

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

K60 MLB

LAST_MODIFIED= Tue Feb 8 14:39:56 2011

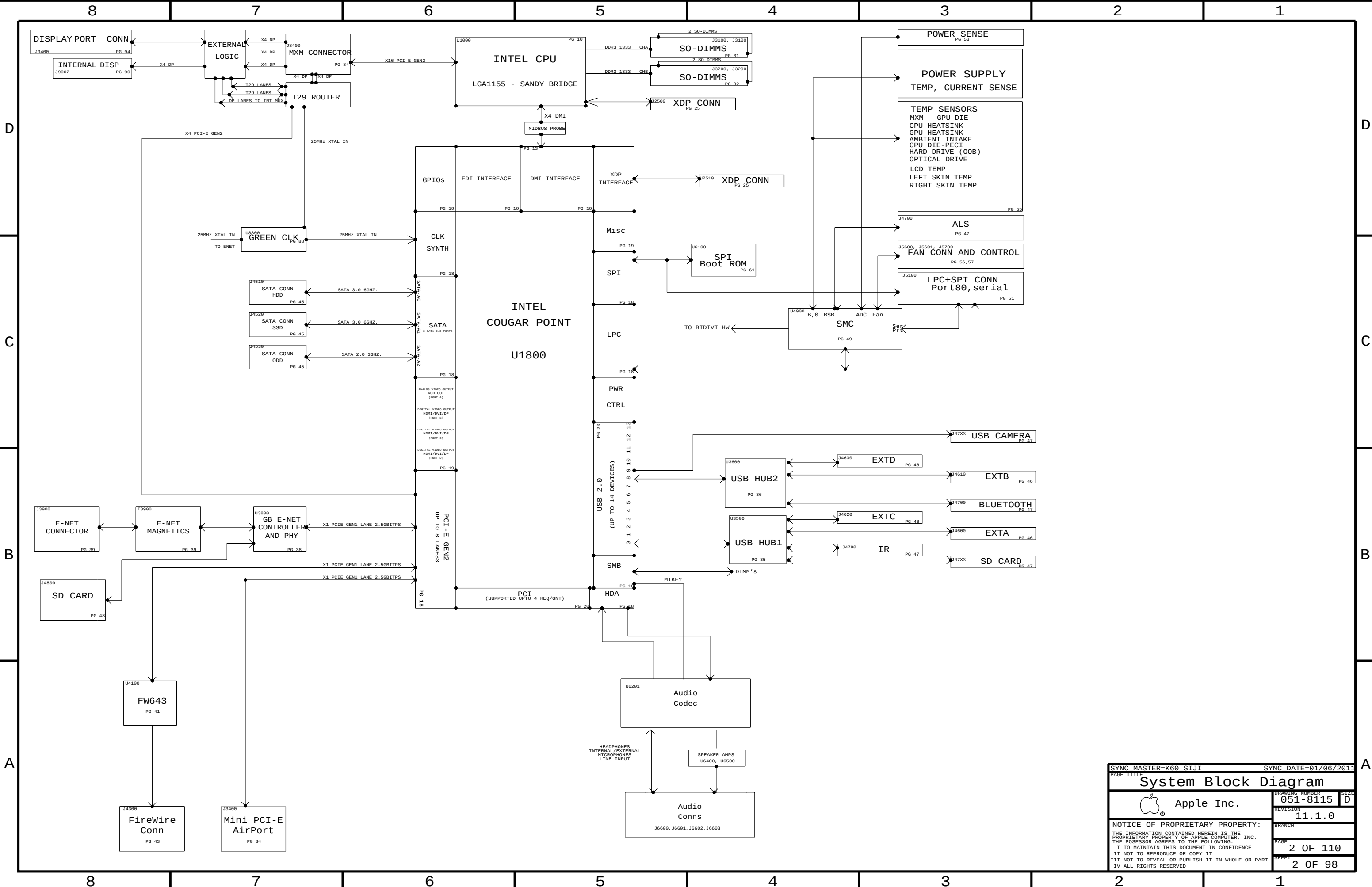
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64	POWER SEQUENCING PGOOD	K62	01/06/2011
65	VREG: PVPCORE_S0_CPU	K60_AARON	N/A
66	VREG: CPU CORE - PHASES 1-3	K62	01/06/2011
67	VREG:AXG PHASE/CORE - CAPS	K60_AARON	N/A
68	1V05 REGULATOR	K62	01/06/2011
69	CPU VCCSA REGULATOR	K62	11/15/2010
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74	12V_S0 & 12V_S5 switch	K60_JERRY	01/06/2011
75	MXM PCIE, DP & Power	K62	01/06/2011
76	MXM I/O	K60_MASTER	N/A
77	MXM PCIE CAPS	K62	01/06/2011
78	DP ALIAS AND CONTROL	K60_AARON	07/18/2010
79	GREEN CLOCK	K62	01/06/2011
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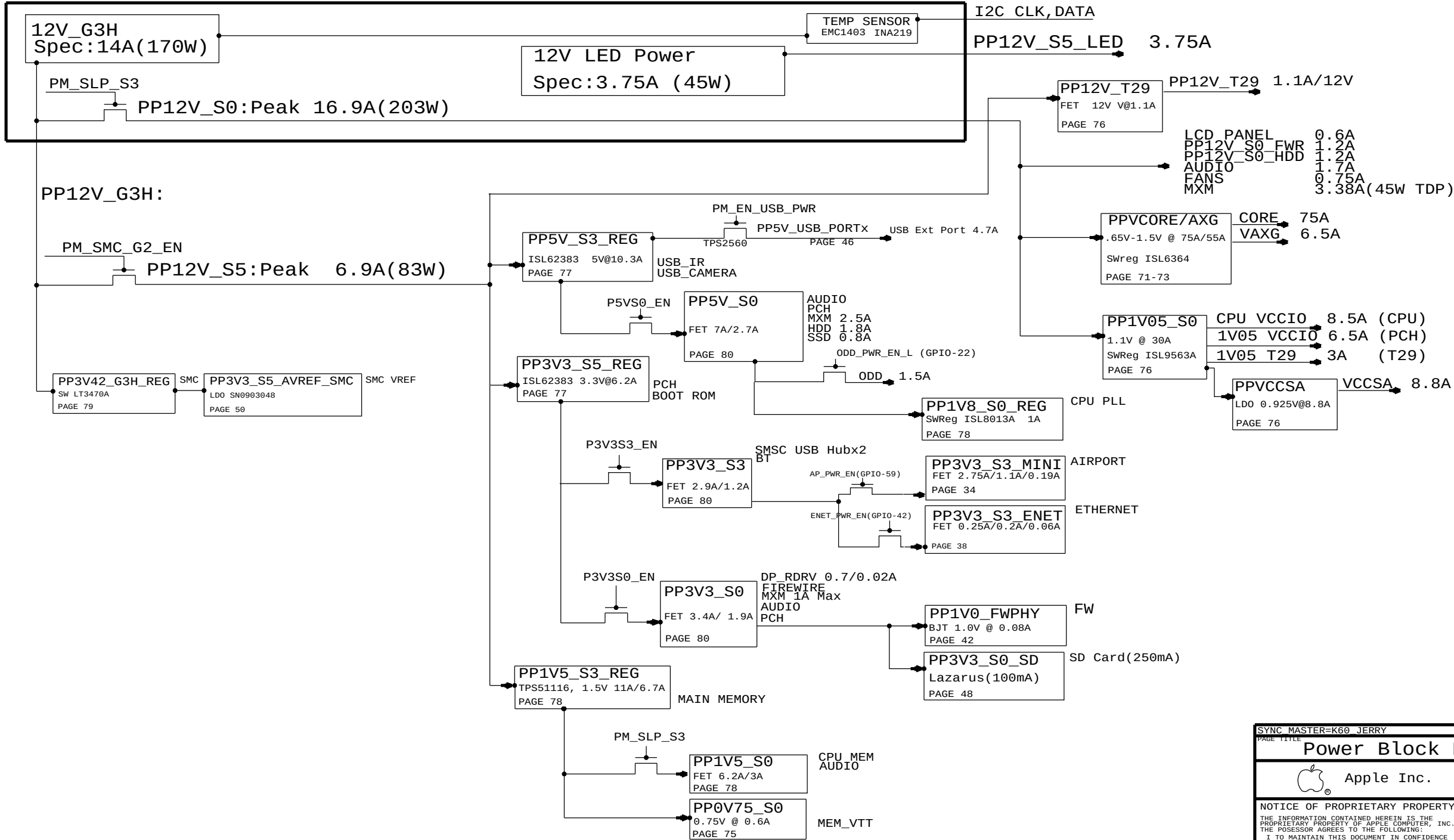
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			2011-02-08

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AC/DC POWER SUPPLY (Spec:215W)



SYNC MASTER=K60 JERRY		SYNC DATE=01/06/2011	
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-0801	PCBA, MLB, DEV, K60	DEVELOPMENT, DEV_GROUP
639-1767	PCBA, MLB, K60, 2.5G, 4C, PRQ, P2_ODD	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, YES_DBG
639-1820	PCBA, MLB, K60, 2.7G, 4C, PRQ, P2_ODD	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, YES_DBG
639-1821	PCBA, MLB, K60, 2.8G, 4C, PRQ, P2_ODD	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, YES_DBG
639-2160	PCBA, MLB, K60, 2.5G, 4C, PRQ, P2_ODD, NO_DBG	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, NO_DBG
639-2159	PCBA, MLB, K60, 2.7G, 4C, PRQ, P2_ODD, NO_DBG	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, NO_DBG
639-2161	PCBA, MLB, K60, 2.8G, 4C, PRQ, P2_ODD, NO_DBG	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, NO_DBG
639-2118	PCBA, MLB, K60, 2.5G, 4C, PRQ, P1_ODD	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, YES_DBG
639-2122	PCBA, MLB, K60, 2.7G, 4C, PRQ, P1_ODD	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, YES_DBG
639-2119	PCBA, MLB, K60, 2.8G, 4C, PRQ, P1_ODD	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, YES_DBG
639-2132	PCBA, MLB, K60, 2.5G, 4C, PRQ, P1_ODD, NO_DBG	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, NO_DBG
639-2133	PCBA, MLB, K60, 2.7G, 4C, PRQ, P1_ODD, NO_DBG	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, NO_DBG
639-2134	PCBA, MLB, K60, 2.8G, 4C, PRQ, P1_ODD, NO_DBG	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, NO_DBG

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC1	COMMON, ALTERNATE, MXM, FCIM, CPU_1V5_SENSE, CPU_VCCSA_SENSE, 1V05_PCH_SENSE, HUB_USX2061, PRODUCTION, VAXG, SSD
BASIC2	AP, BT, IR, T29
DEV_GROUP	VREFMRGN_A, VREFMRGN_B, DIMM_1V5_SENSE
YES_DBG	XDP, XDP_CONN, XDP_CPU_BPM, MOJOMUX: YES, LPCPLUS: YES
NO_DBG	MOJOMUX: NO, LPCPLUS: NO

CPU SOCKET & ILM SUB-BOMS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0071	1	SOCKET, LGA1155, CPU- LF	U1000	CRITICAL	TYCO_SOCKET
604-1474	1	ASSY, PURCHASED, ILM, TYCO	ILM	CRITICAL	TYCO_SOCKET
511S0073	1	SOCKET, LGA1155, CPU- LF	U1000	CRITICAL	MOLEX_SOCKET
604-1161	1	ASSY, PURCHASED, ILM, MOLEX	ILM	CRITICAL	MOLEX_SOCKET

BOM NUMBER	BOM NAME	BOM OPTIONS
085-2452	SUB ASSY,CPU SOCKET,K60,TYCO	TYCO_SOCKET
085-2453	SUB ASSY,CPU SOCKET,K60,MOLEX	MOLEX_SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-2452	1	TYCO CPU SOCKET AND ILM	SKT_ILM	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-2453	085-2452		SKT_ILM	MOLEX ALTERNATE

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
	337S4688	1	IC, COUGAR POINT, SL34F, B082Z68, PRQ, B3	U1800	CRITICAL	
	353S3955	1	IC, P13VEDP212, X2 DP MUX, QFN	U9390	CRITICAL	
	338S0753	1	IC, FW643, 1394B PCIE, PHY/LINK	U4100	CRITICAL	
	825-7122	1	MLB LABEL, 48.6X4.8	X14	CRITICAL	
	343S0534	1	IC, BCM57765, ENET&SD, 8X8	U3900	CRITICAL	
RAW: 335S0807	341T0184	1	FLASH, EFI BOOTROM, K60/K62	U6100	CRITICAL	
RAW: 335S0539	341T0328	1	SFLASH ENET 2MBIT, CIV	U3990	CRITICAL	
	338S0945	1	T29 ROUTER, IC ASSP	U9700	CRITICAL	T29
RAW: 335S0550	341T0257	1	IC, T29, SERIAL EEPROM	U9790	CRITICAL	T29
RAW: 337S3997	341T0326	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29
RAW: 335S0709	341T0330	1	IC, MXM SYS ROM, 24C02	U8570	CRITICAL	
RAW: 338S0878	341T0185	1	IC, SMC, K60	U4900	CRITICAL	K60

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754043	1	SNB, SR00S, PRQ, D2, 2.5, 65W, +1, 6M, LGA	CPU	CRITICAL	2P5GHZ_SNB_CPU_PRQ
33754062	1	SNB, SR009, PRQ, D2, 2.7, 65W, +1, 6M, LGA	CPU	CRITICAL	2P7GHZ_SNB_CPU_PRQ
33754061	1	SNB, SR00E, PRQ, D2, 2.8, 65W, +1, 8M, LGA	CPU	CRITICAL	2P8GHZ_SNB_CPU_PRQ

K60 PARTS

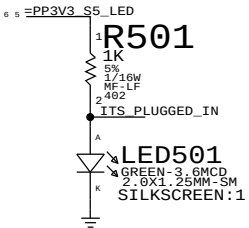
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951-8115	1	SCH, MLB, K60	SCH1		K60
820-2641	1	PCBF, MLB, K60	MLB1		K60

K60 ALTERNATE PARTS

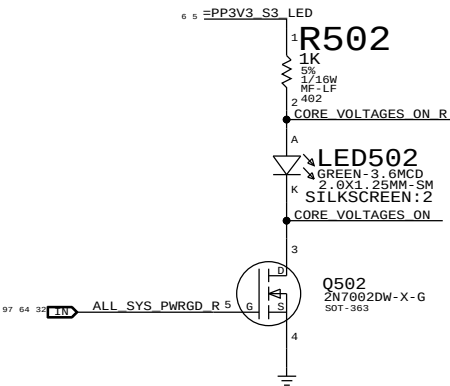
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12850298	12850293			330UF
37150679	37150652			PTN DIODE
37750187	37750066			USB DIODE
37650972	37650612			ROHM TRA-BJT

SYNCR MASTER=K60 AARON		SYNCR DATE=N/A	
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BOM Configuration			
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	REVISION	11.1.0	
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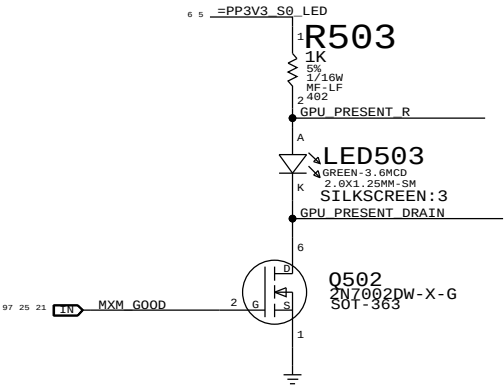
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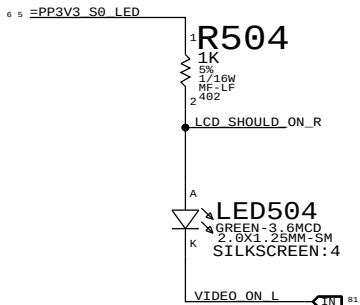
ALL_SYS_PWRGD Led



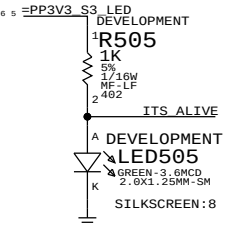
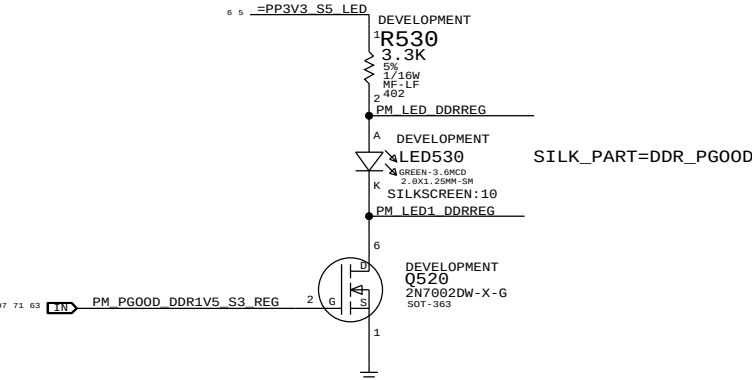
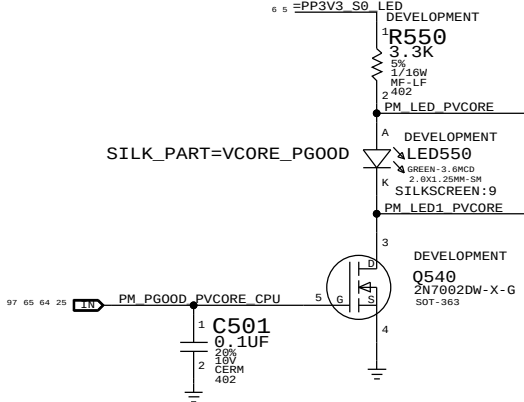
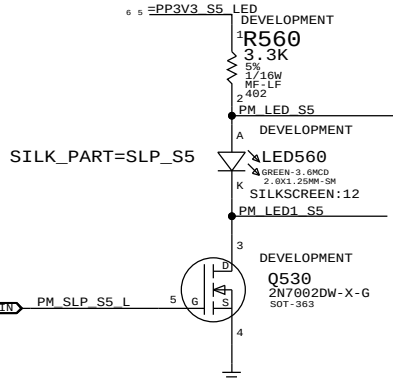
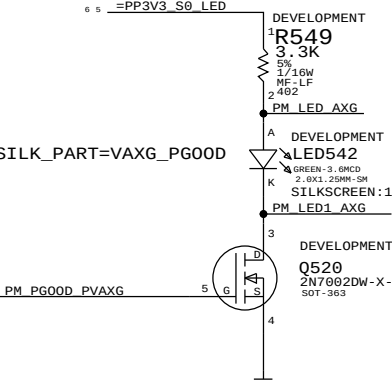
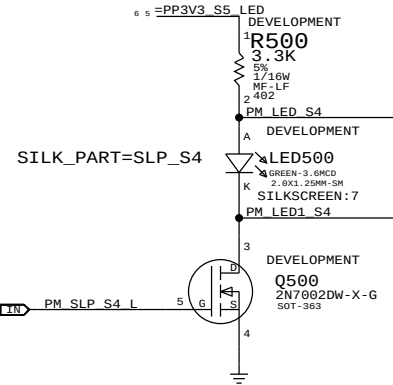
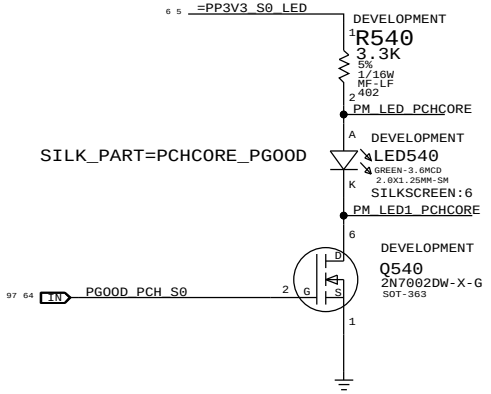
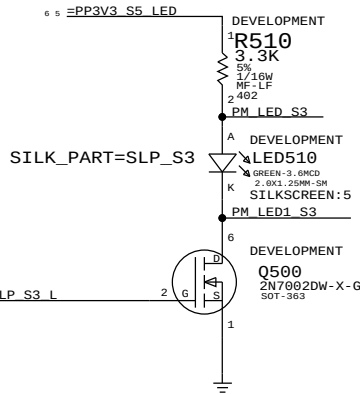
MXM PWR GOOD Led



VIDEO ON Led



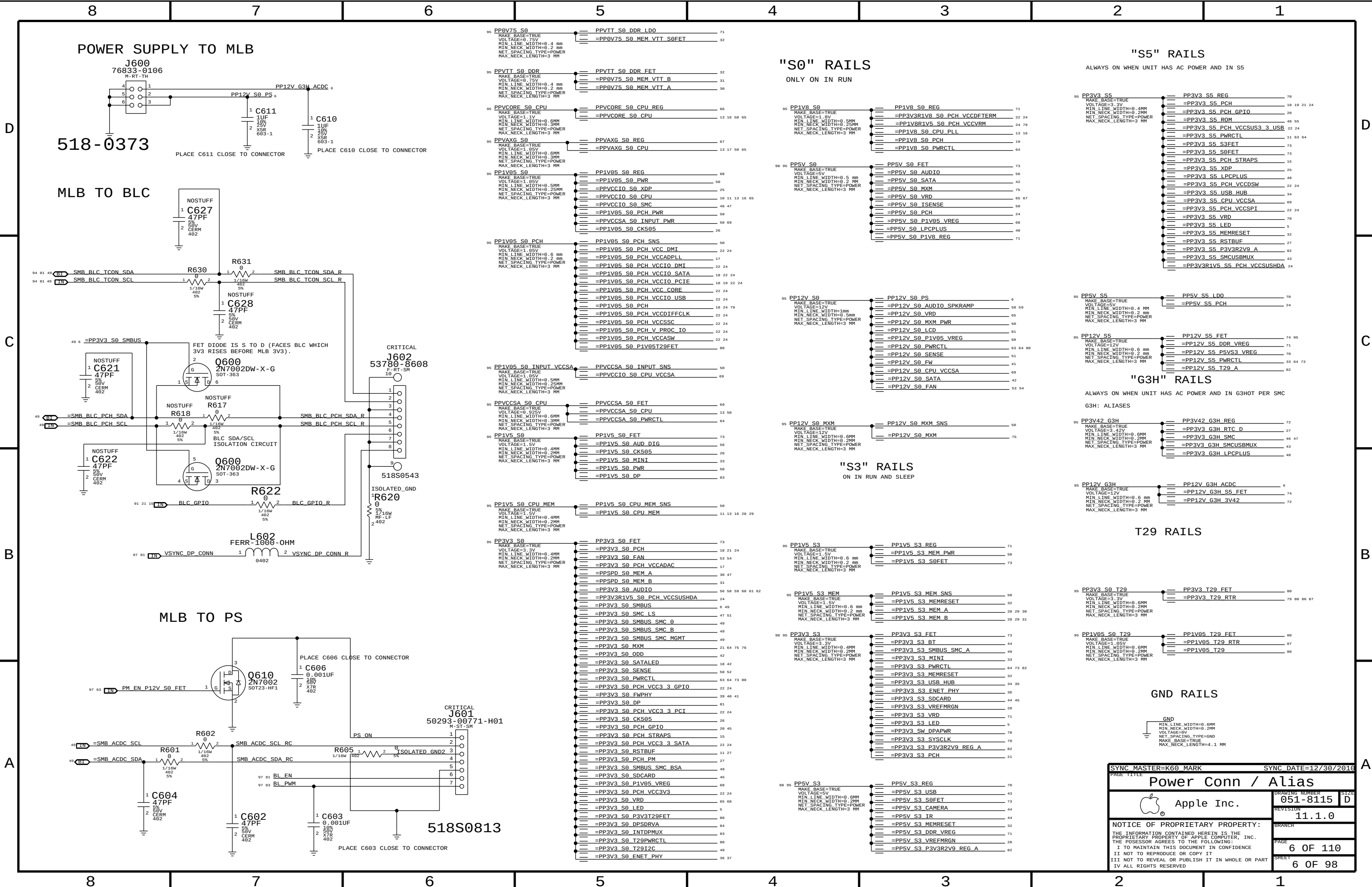
PROTO DEBUG LEDS ARE SHOWN BELOW

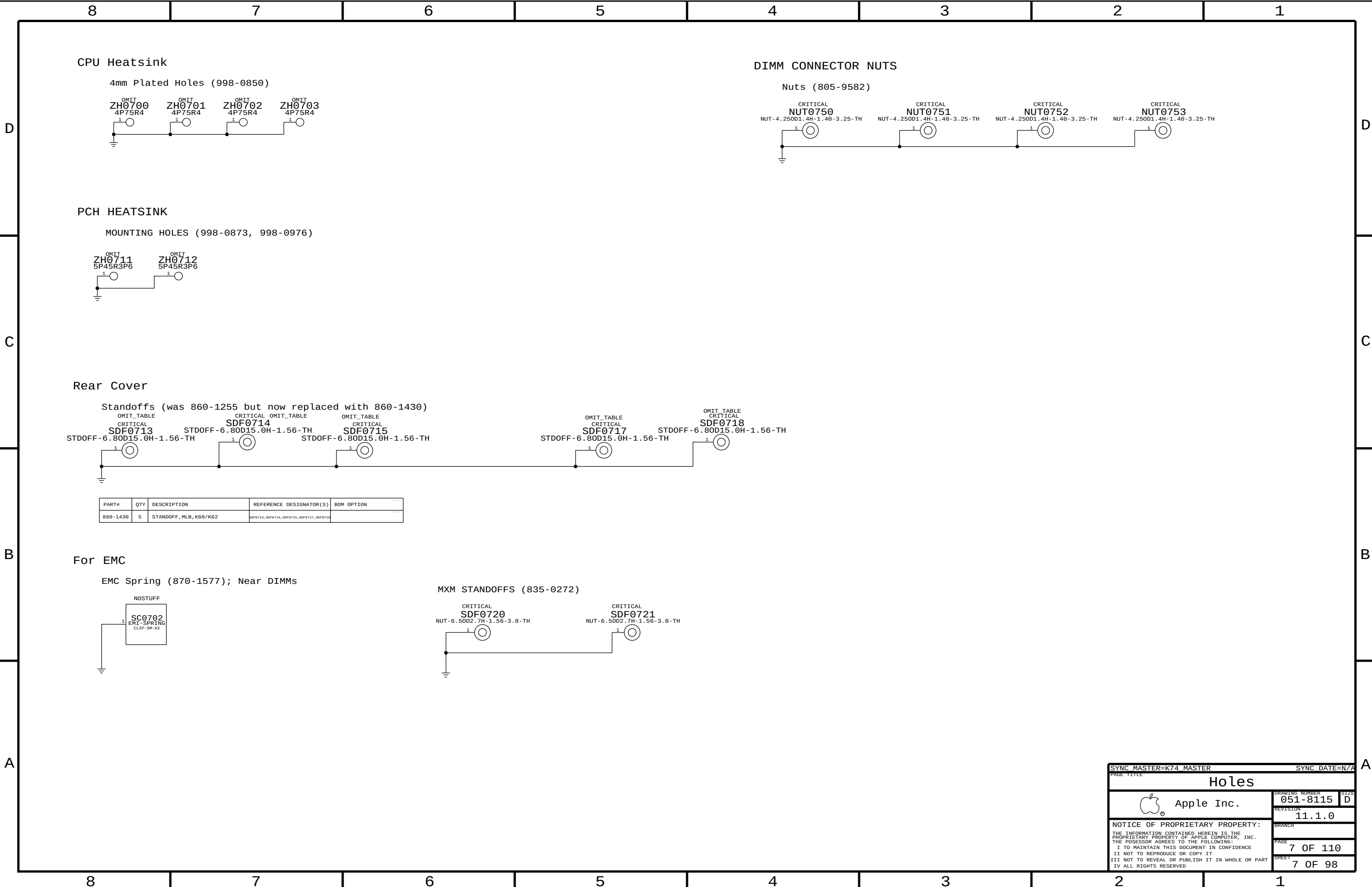


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SYNC DATE=01/06/2011

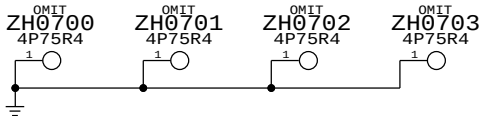
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Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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		PAGE	5 OF 110
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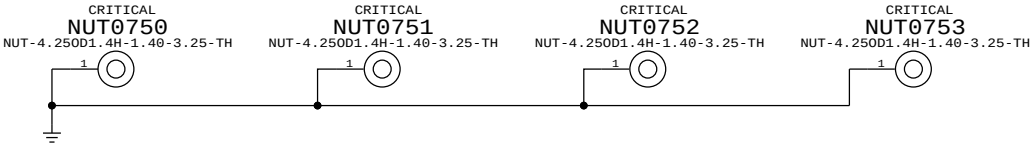
CPU Heatsink

4mm Plated Holes (998-0850)



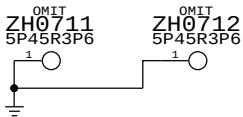
DIMM CONNECTOR NUTS

Nuts (805-9582)



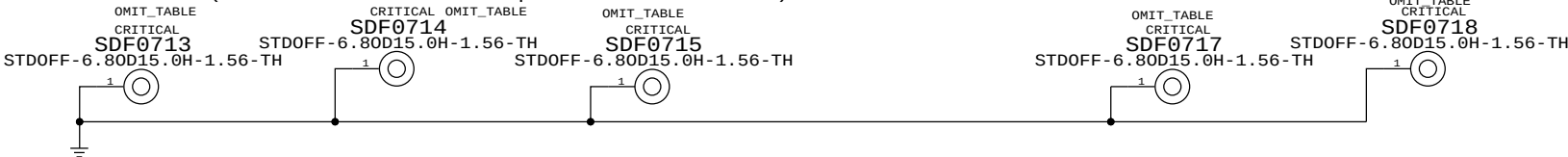
PCH HEATSINK

MOUNTING HOLES (998-0873, 998-0976)



Rear Cover

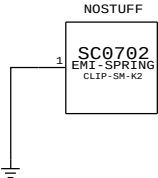
Standoffs (was 860-1255 but now replaced with 860-1430)



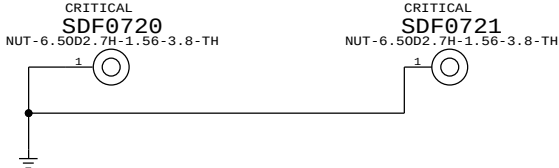
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
860-1430	5	STANDOFF, MLB, K60/K62	SDF0713, SDF0714, SDF0715, SDF0717, SDF0718	

For EMC

EMC Spring (870-1577); Near DIMMs



MXM STANDOFFS (835-0272)



SYNC MASTER=K74 MASTER

SYNC DATE=N/A

PAGE TITLE

Holes

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8	7	6	5	4	3	2	1
UNUSED CPU SIGNALS		NC ON UNUSED PCIE ALIASES		NC ON UNUSED DISPLAY ALIASES		NC ON UNUSED FDI ALIASES	
19 TP_CPU_RSVD<16..1> == NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CPU_RSVD<46..19> == NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		18 TP_PCIE_CLK100M_PE5P == NC_PCIE_CLK100M_PE5P MAKE_BASE=TRUE NO_TEST=TRUE 18 TP_PCIE_CLK100M_PE5N == NC_PCIE_CLK100M_PE5N MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE6P == NC_PCIE_CLK100M_PE6P MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE6N == NC_PCIE_CLK100M_PE6N MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE7P == NC_PCIE_CLK100M_PE7P MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE7N == NC_PCIE_CLK100M_PE7N MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_RX_N<3..0> == NC_PE_RXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_RX_P<3..0> == NC_PE_RXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_TX_N<3..0> == NC_PE_TXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_TX_P<3..0> == NC_PE_TXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_D2R_PERN4 == NC_PCIE_D2R_PERN4 MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_D2R_PERP4 == NC_PCIE_D2R_PERP4 MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_R2D_PETN4 == NC_PCIE_R2D_PETN4 MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_R2D_PETP4 == NC_PCIE_R2D_PETP4 MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_CLK100M_PE4P == NC_PCIE_CLK100M_PE4P MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_CLK100M_PE4N == NC_PCIE_CLK100M_PE4N MAKE_BASE=TRUE NO_TEST=TRUE		19 TP_CRT_IG_DDC_CLK == NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_DDC_DATA == NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_RED == NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_GREEN == NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_BLUE == NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_HSYNC == NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_VSYNC == NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_MLN<3..0> == NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_MLP<3..0> == NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_AUX_N == NC_DP_IG_B_AUXN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_AUX_P == NC_DP_IG_B_AUXP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_HPD == NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_DDC_CLK == NC_DP_IG_B_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_DDC_DATA == NC_DP_IG_B_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_MLN<3..0> == NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_MLP<3..0> == NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_AUX_N == NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_AUX_P == NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_HPD == NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_CTRL_CLK == NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_CTRL_DATA == NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_MLN<3..0> == NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_MLP<3..0> == NC_DP_IG_D_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_AUXN == NC_DP_IG_D_AUXN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_AUXP == NC_DP_IG_D_AUXP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_HPD == NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_CTRL_CLK == NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_CTRL_DATA == NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_TVCLKINN == NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_TVCLKINP == NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_STALLN == NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_STALLP == NC_SDVO_STALLP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_INTN == NC_SDVO_INTN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_INTP == NC_SDVO_INTP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_L_BKLTCTL == NC_PCH_L_BKLTCTL MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_L_BKLTEN == NC_PCH_L_BKLTEN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_L_VDD_EN == NC_PCH_L_VDD_EN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_CLKOUT_DPN == NC_PCH_CLKOUT_DPN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_CLKOUT_DPP == NC_PCH_CLKOUT_DPP MAKE_BASE=TRUE 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NC_SATA_F_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SATA_F_R2D_CP == NC_SATA_F_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE	
NC ON UNUSED PCI ALIASES		NC ON UNUSED USB ALIASES				NC ON UNUSED SATA ALIASES	
20 TP_PCI_AD<31..0> == NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_PCI_C_BE_L<3..0> == NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_PCI_PAR == NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_PCI_RESET_L == NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_LPC_DREQ0_L == NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_USB_1N == NC_USB_1N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_1P == NC_USB_1P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_2N == NC_USB_2N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_2P == NC_USB_2P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_3N == NC_USB_3N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_3P == NC_USB_3P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_4N == NC_USB_4N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_4P == NC_USB_4P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_5N == NC_USB_5N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_5P == NC_USB_5P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_6N == NC_USB_6N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_6P == NC_USB_6P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_7N == NC_USB_7N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_7P == NC_USB_7P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_10N == NC_USB_10N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_10P == NC_USB_10P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_11N == NC_USB_11N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_11P == NC_USB_11P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_12N == NC_USB_12N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_12P == NC_USB_12P MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_13N == NC_USB_13N MAKE_BASE=TRUE NO_TEST=TRUE 20 TP_USB_13P == NC_USB_13P MAKE_BASE=TRUE NO_TEST=TRUE					
NC ON UNUSED MEM ALIASES							
12 TP_MEM_A_DQ_CB<7..0> == NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_A_DQS_N<8> == NC_MEM_A_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_A_DQS_P<8> == NC_MEM_A_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_B_DQ_CB<7..0> == NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_B_DQS_N<8> == NC_MEM_B_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_B_DQS_P<8> == NC_MEM_B_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE							
NC ON UNUSED MISC ALIASES							
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8	7	6	5	4	3	2	1

SYNC_MASTER=K62

SYNC_DATE=01/06/2011

UNUSED SIGNAL ALIAS

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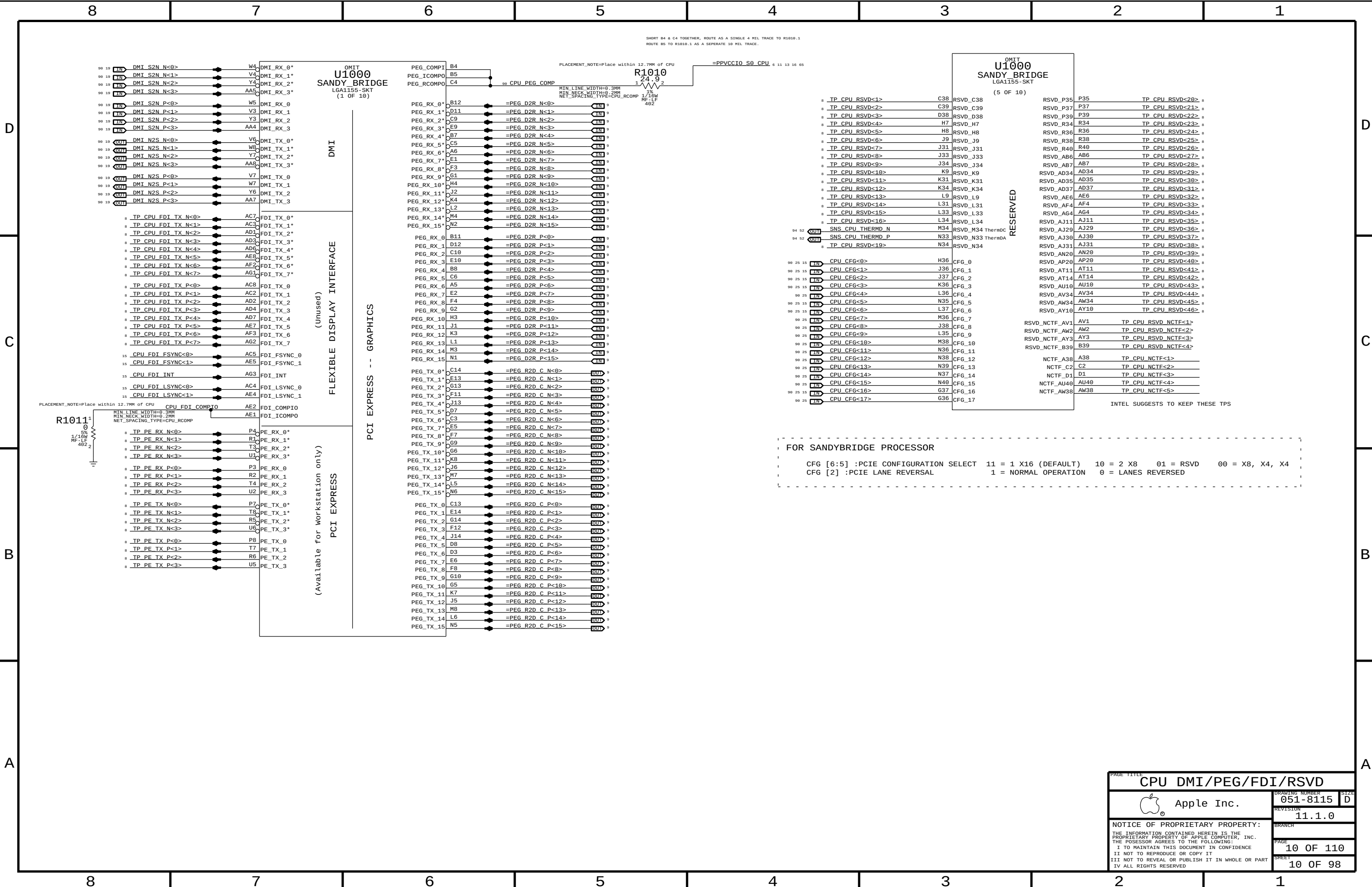
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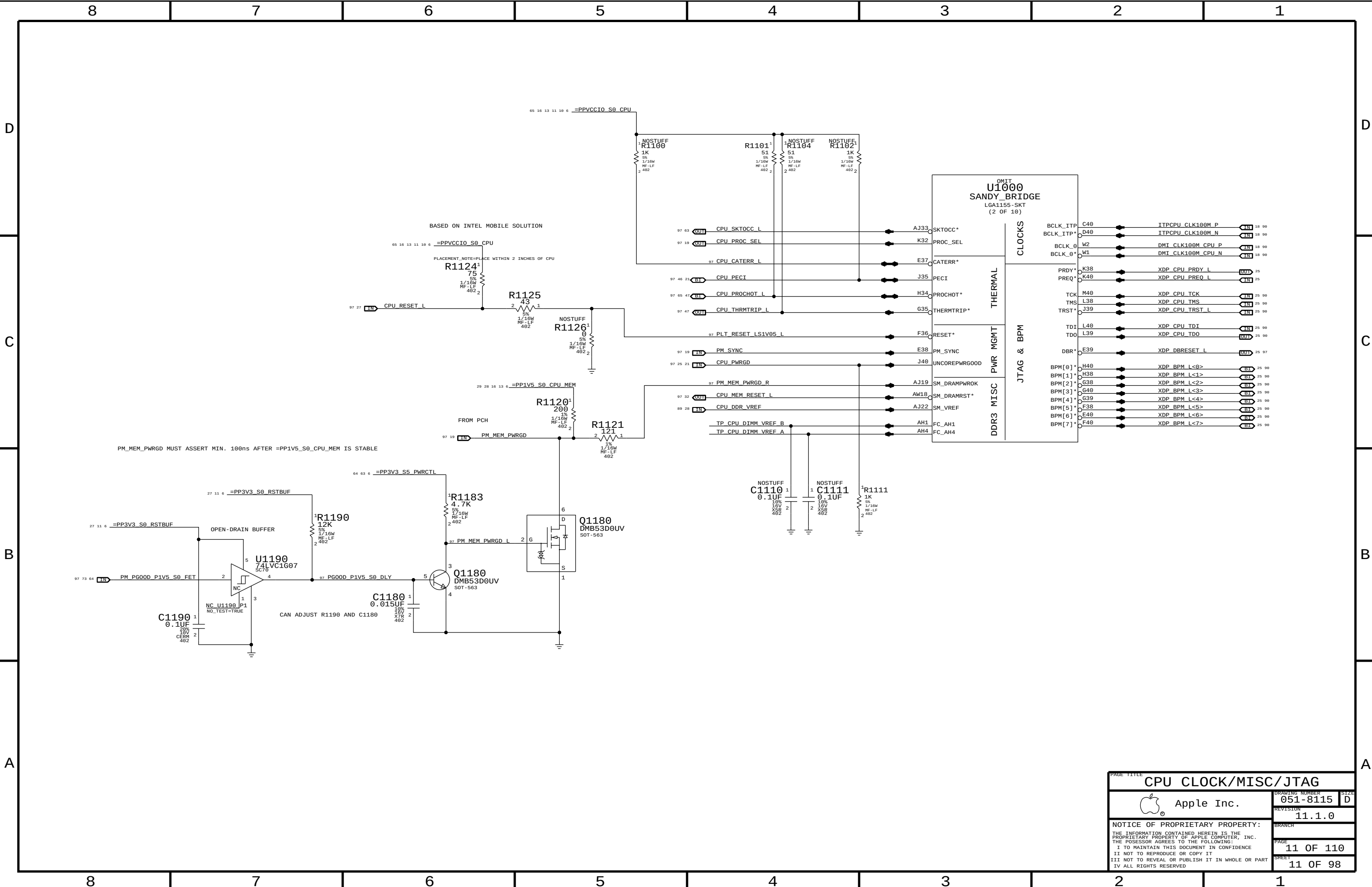
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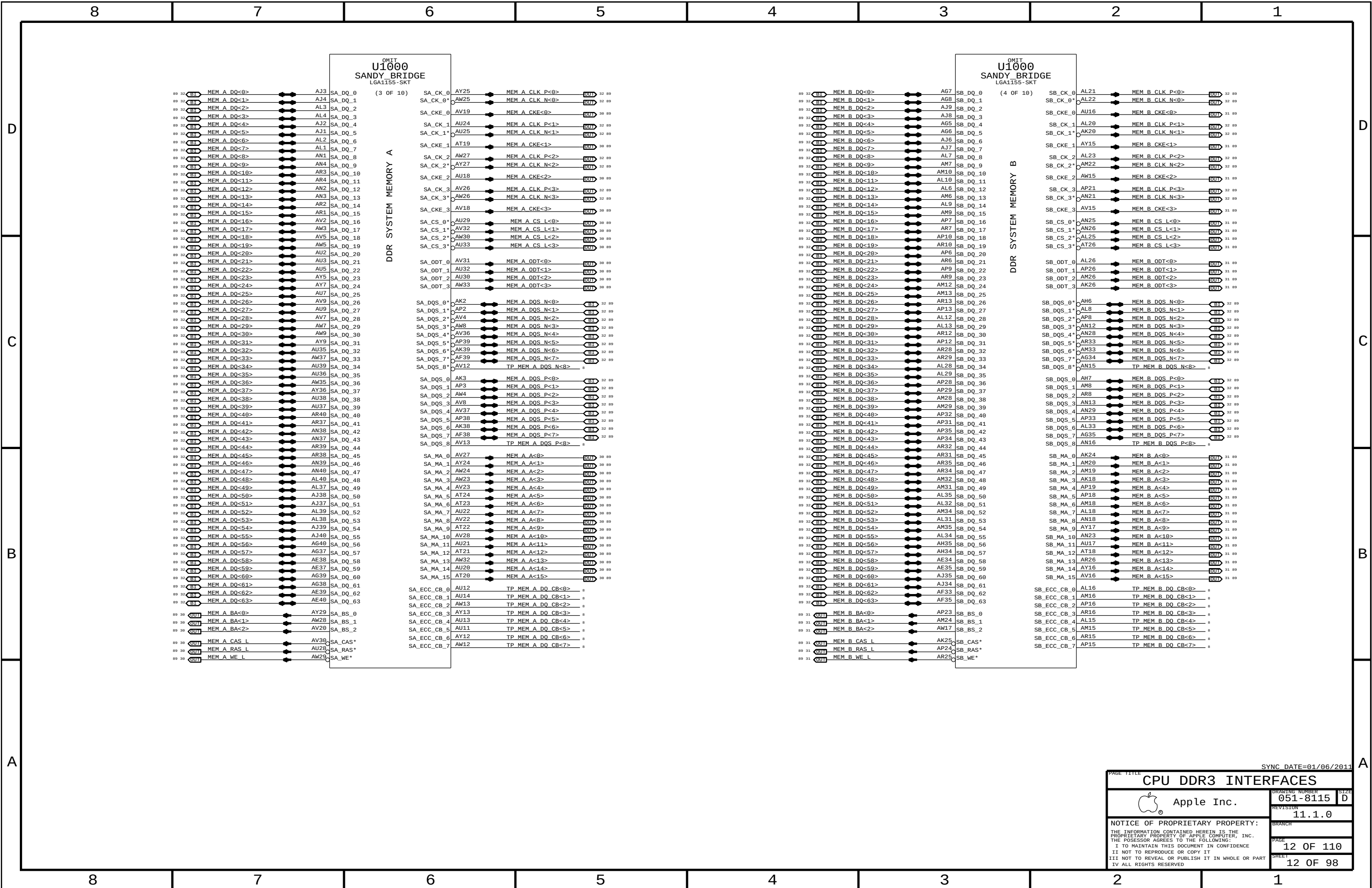
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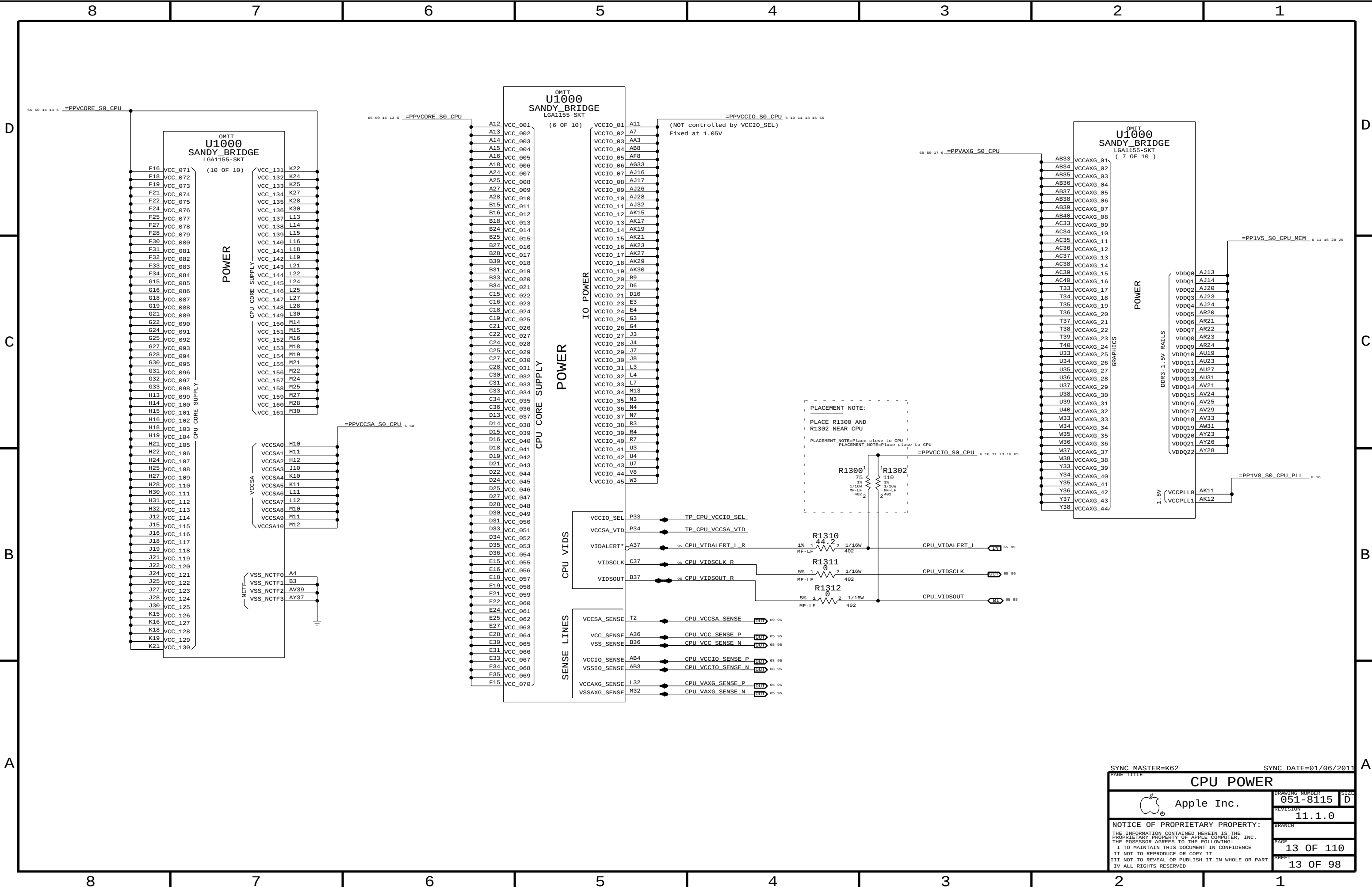


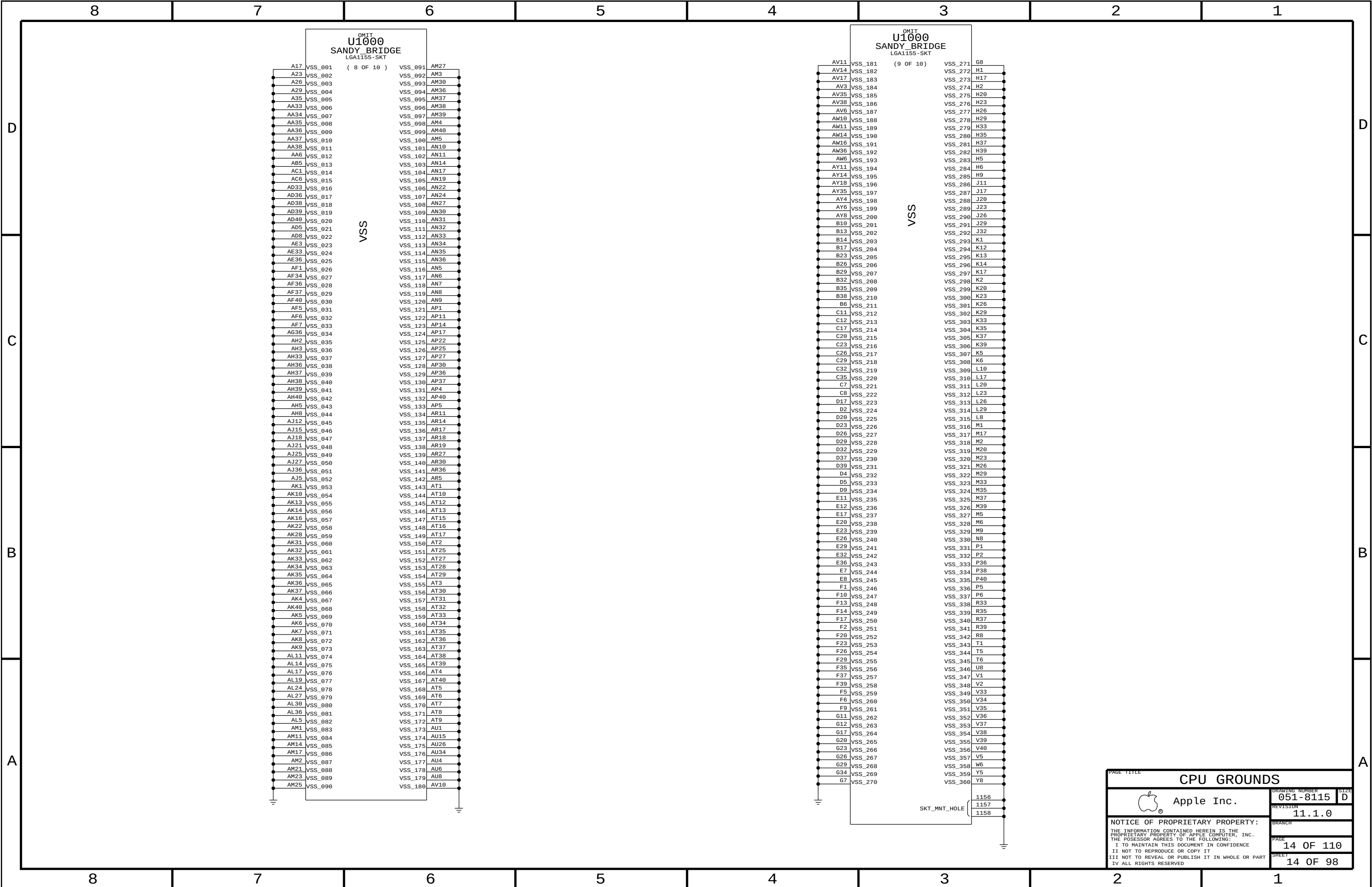
FOR SANDYBRIDGE PROCESSOR

CFG [6:5] :PCIE CONFIGURATION SELECT 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [2] :PCIE LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

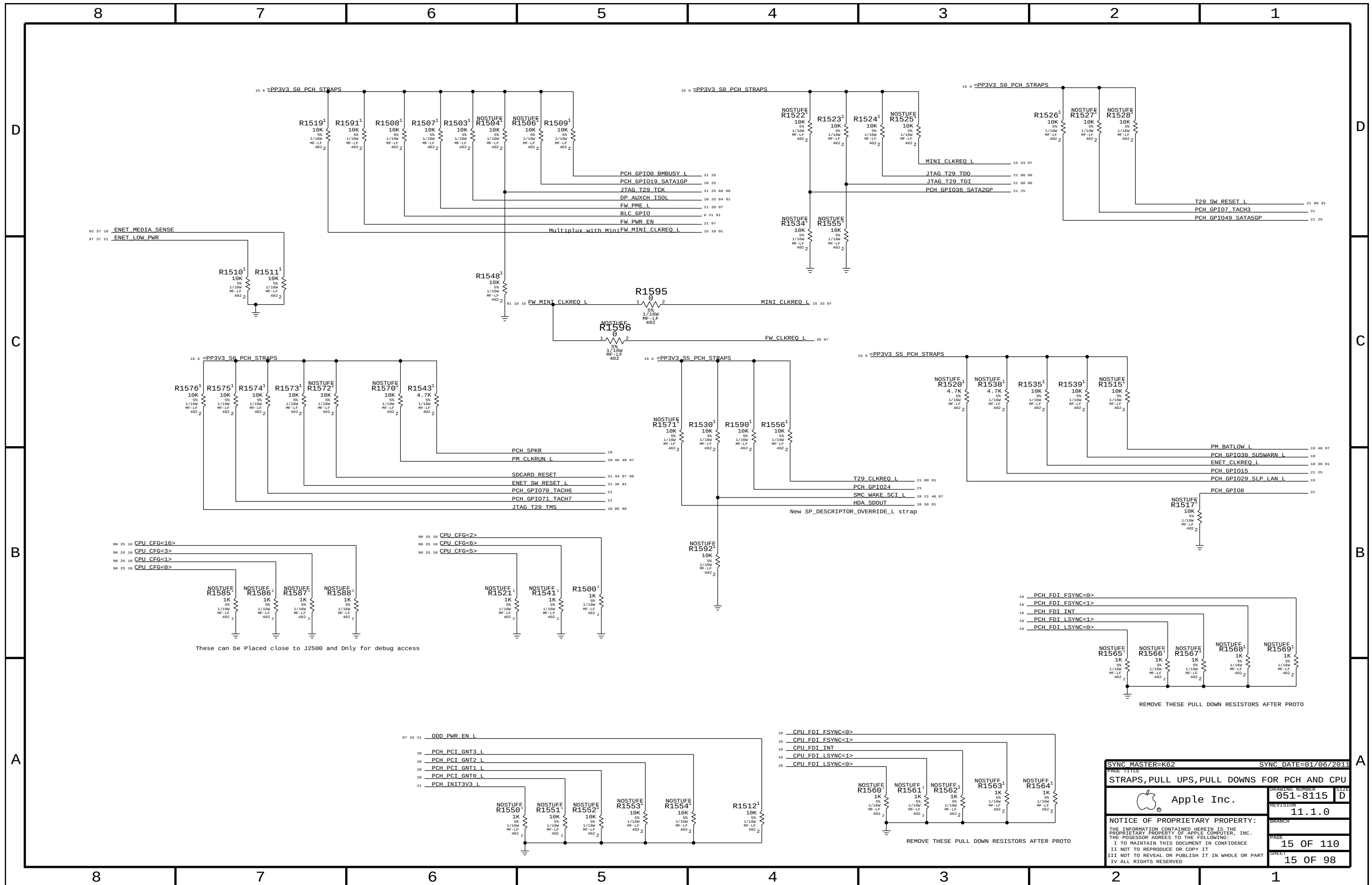


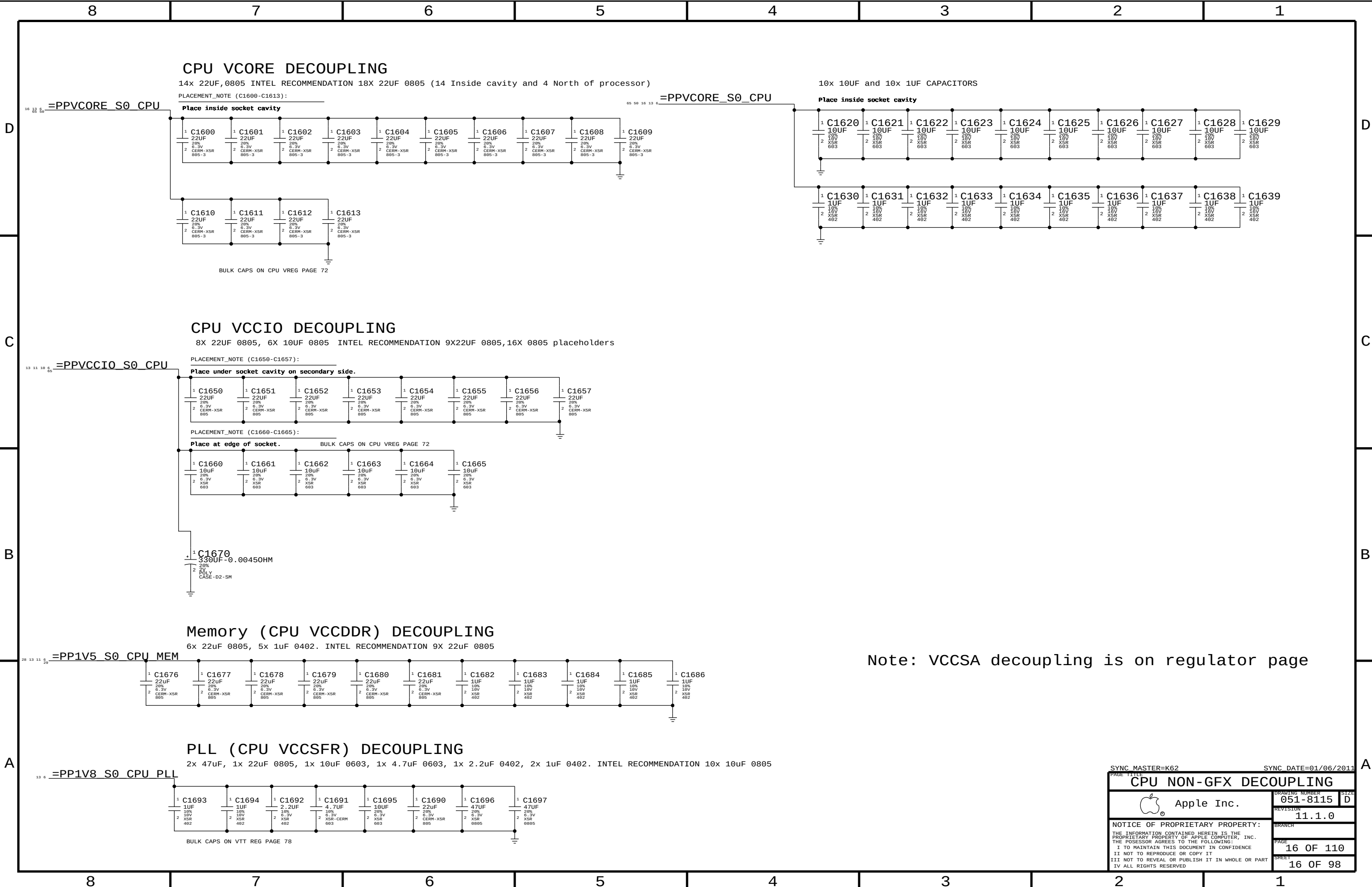






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CPU GROUNDS			
 Apple Inc.		DRAWING NUMBER	051-8115
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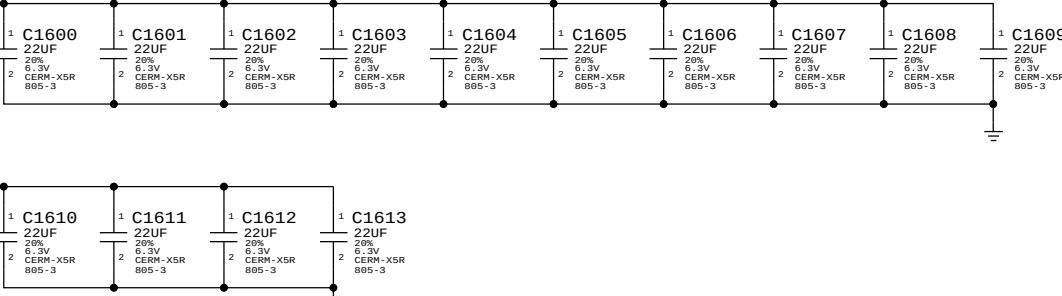


CPU Vcore DECOUPLING

14x 22uF,0805 INTEL RECOMMENDATION 18X 22uF 0805 (14 Inside cavity and 4 North of processor)

PLACEMENT_NOTE (C1600-C1613):

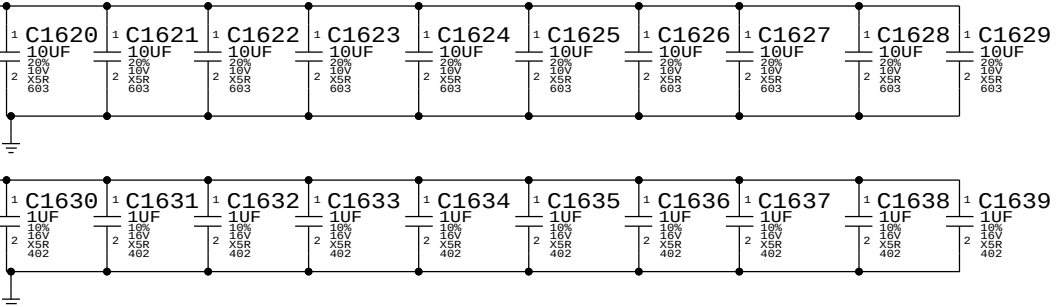
Place inside socket cavity



BULK CAPS ON CPU VREG PAGE 72

10x 10uF and 10x 1uF CAPACITORS

Place inside socket cavity

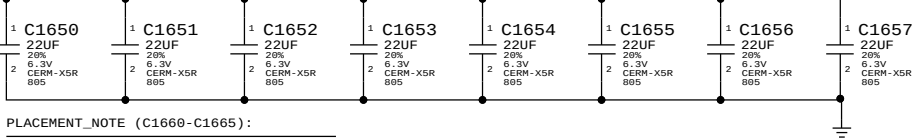


CPU Vccio DECOUPLING

8X 22uF 0805, 6X 10uF 0805 INTEL RECOMMENDATION 9X22uF 0805,16X 0805 placeholders

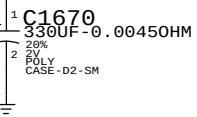
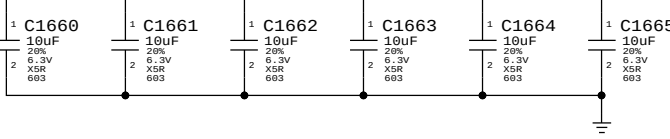
PLACEMENT_NOTE (C1650-C1657):

Place under socket cavity on secondary side.



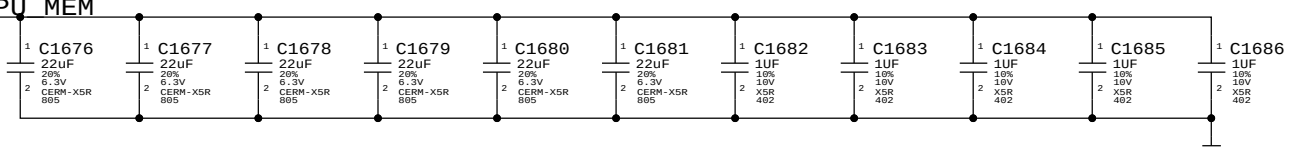
PLACEMENT_NOTE (C1660-C1665):

Place at edge of socket.



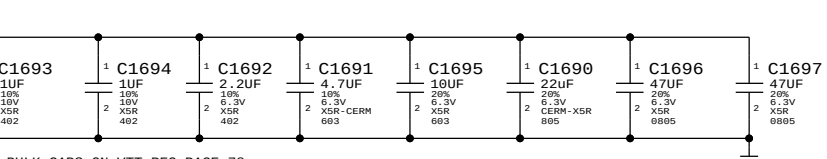
Memory (CPU VCCDDR) DECOUPLING

6x 22uF 0805, 5x 1uF 0402. INTEL RECOMMENDATION 9X 22uF 0805



PLL (CPU VCCSFR) DECOUPLING

2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 10x 10uF 0805



BULK CAPS ON VTT REG PAGE 78

Note: VCCSA decoupling is on regulator page

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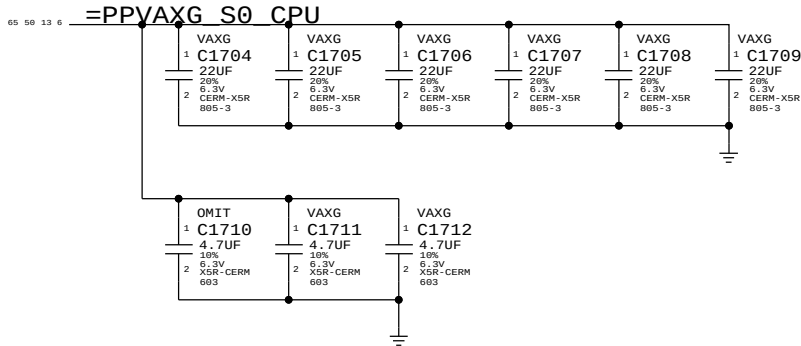
CPU NON-GFX DECOUPLING		
 Apple Inc.	DRAWING NUMBER	051-8115
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VAXG DECOUPLING

INTEL RECOMMENDATION 6X22UF 0805,3X 4.7UF

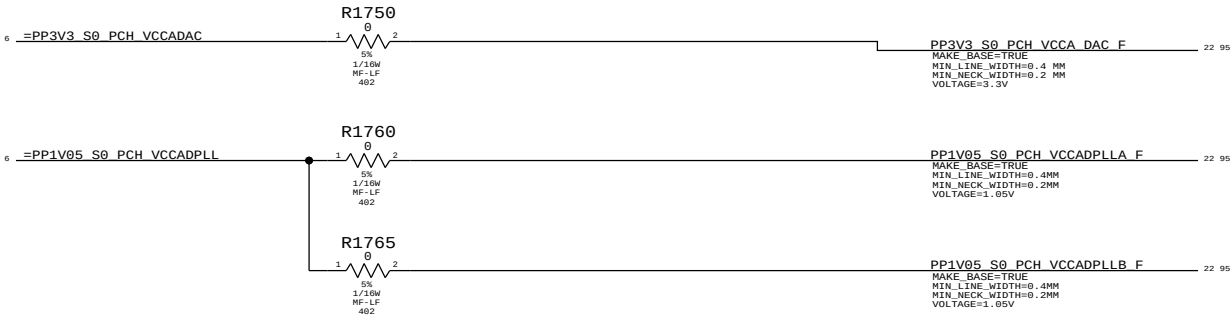
PLACEMENT_NOTE (C1704-C1709):

Place inside socket cavity



BULK CAPS ON CPU VREG PAGE 73

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
138S0586	1	CAP, 4.7UF, 10%, 6.3V, 0603	C1710	VAXG
113S0022	1	RES, 0 OHM, 5%, 0603	C1710	NO_VAXG



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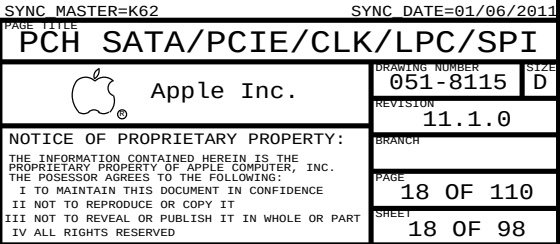
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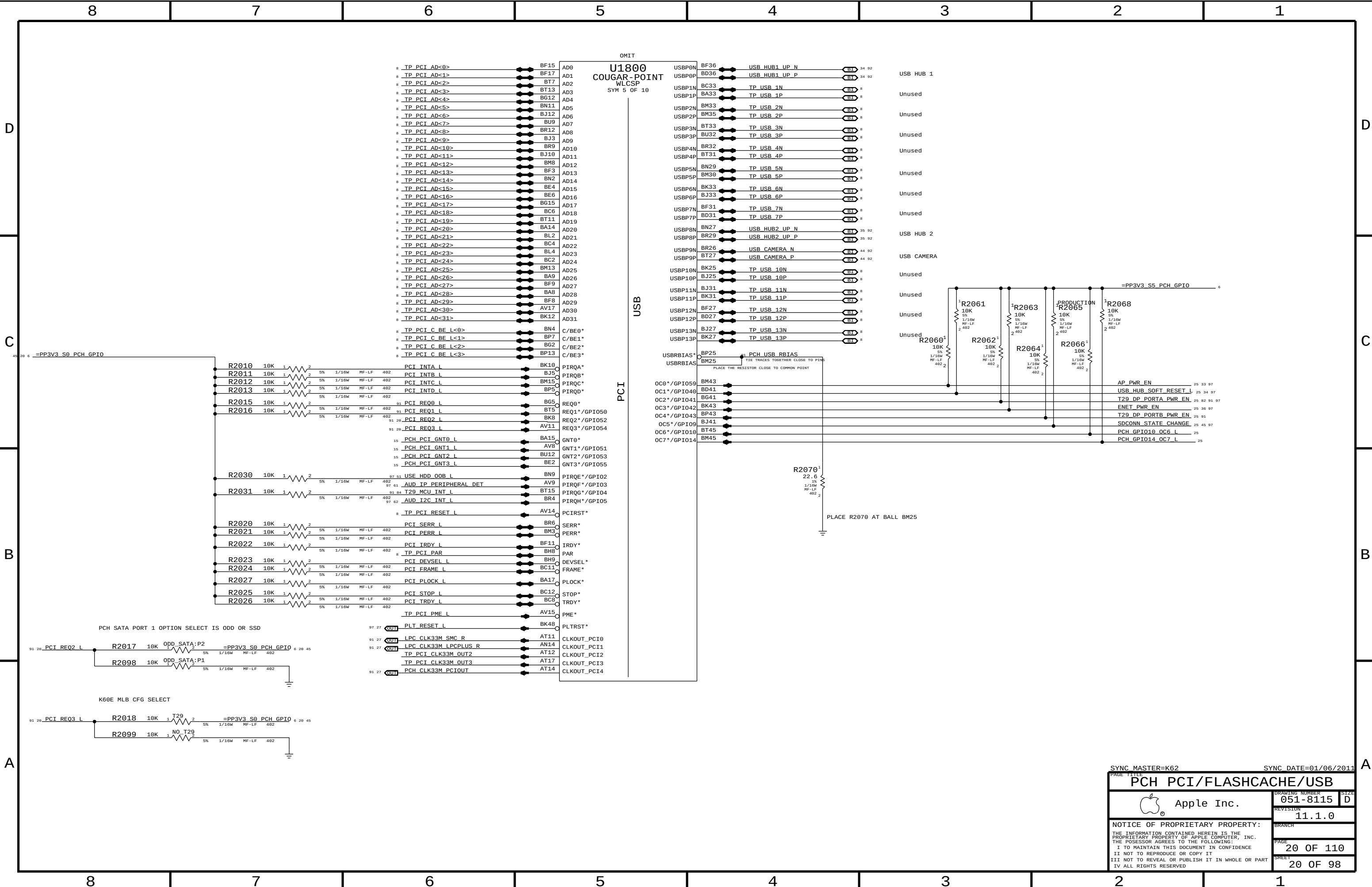
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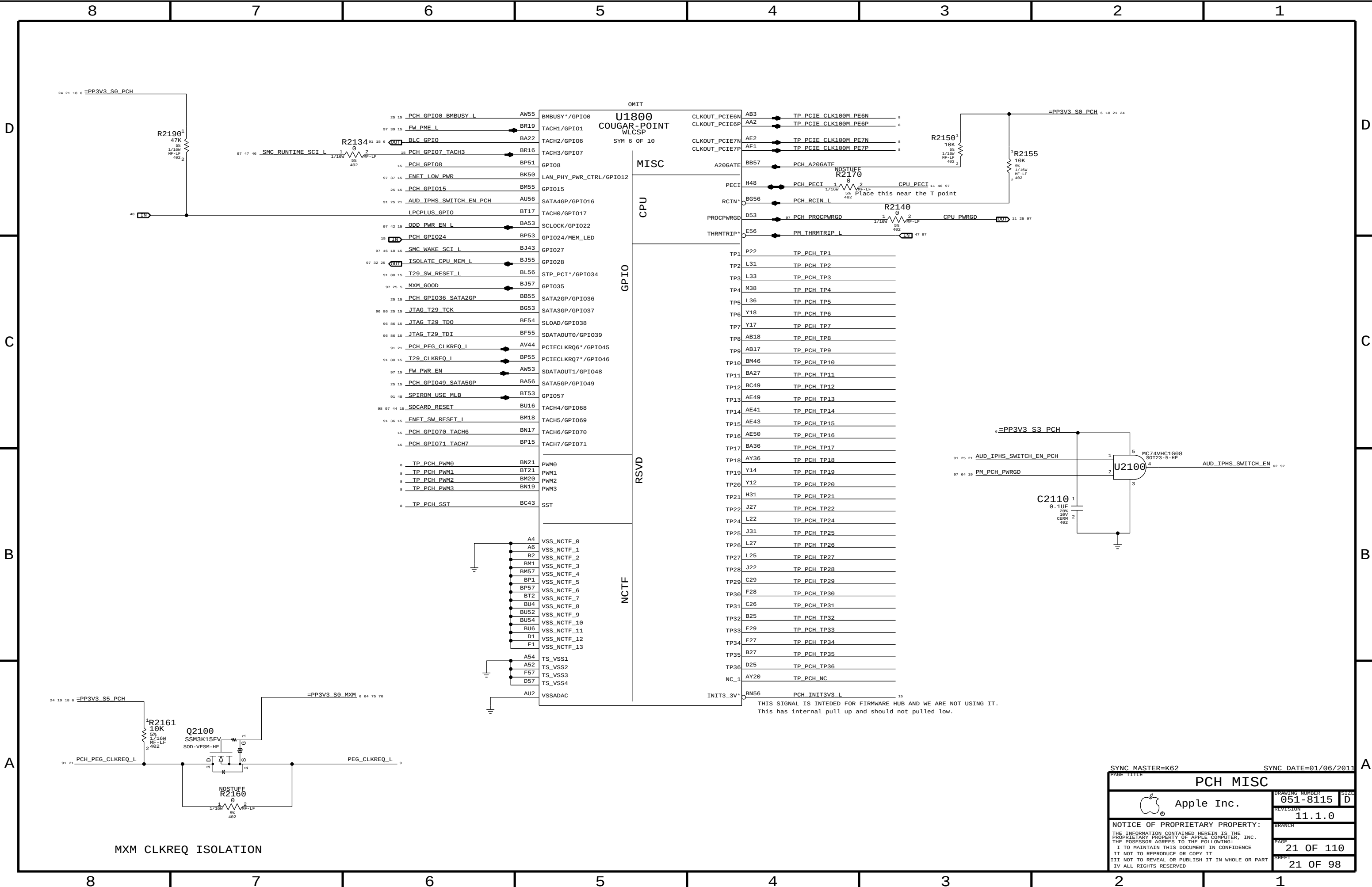
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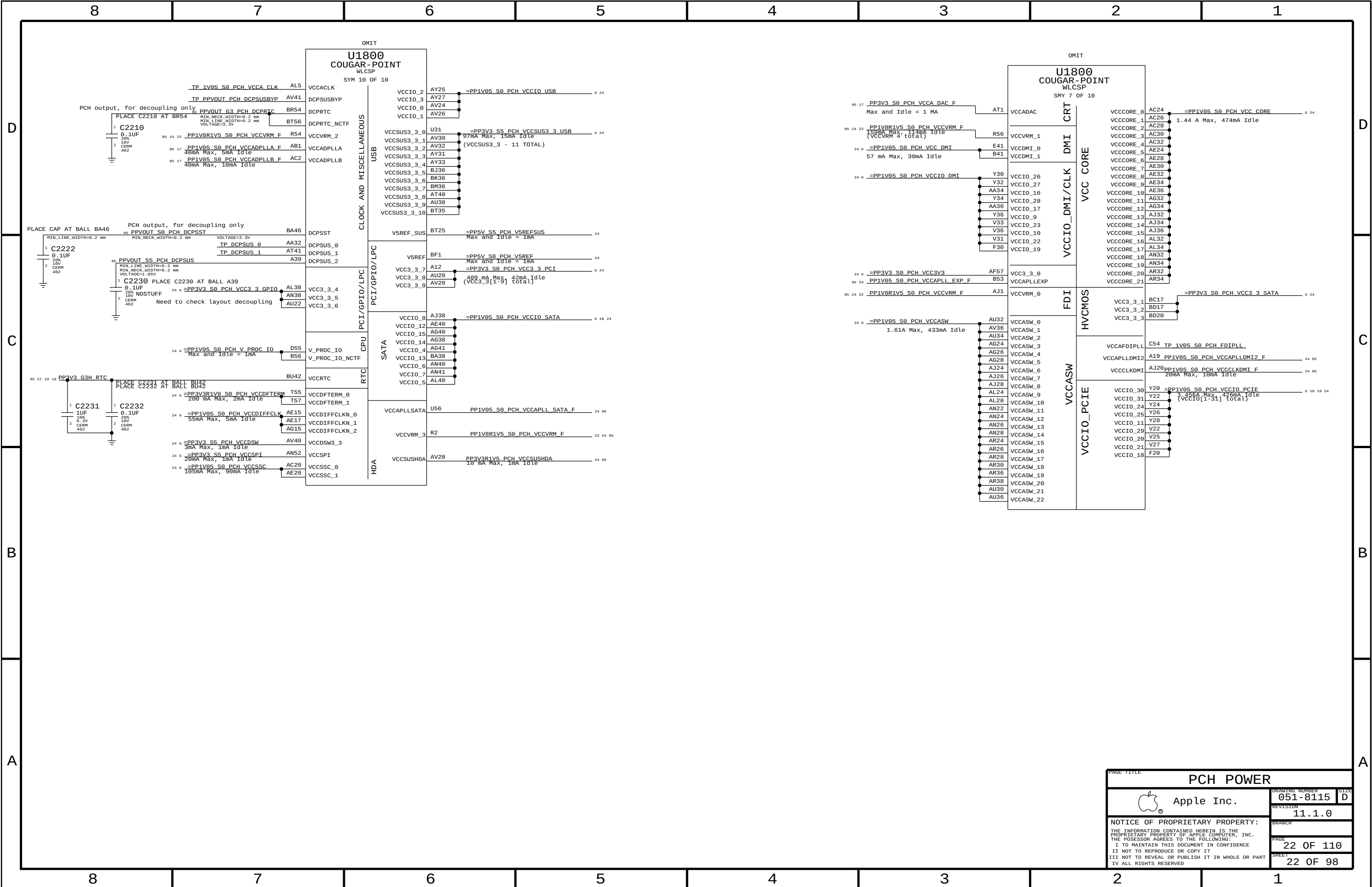
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17 OF 98

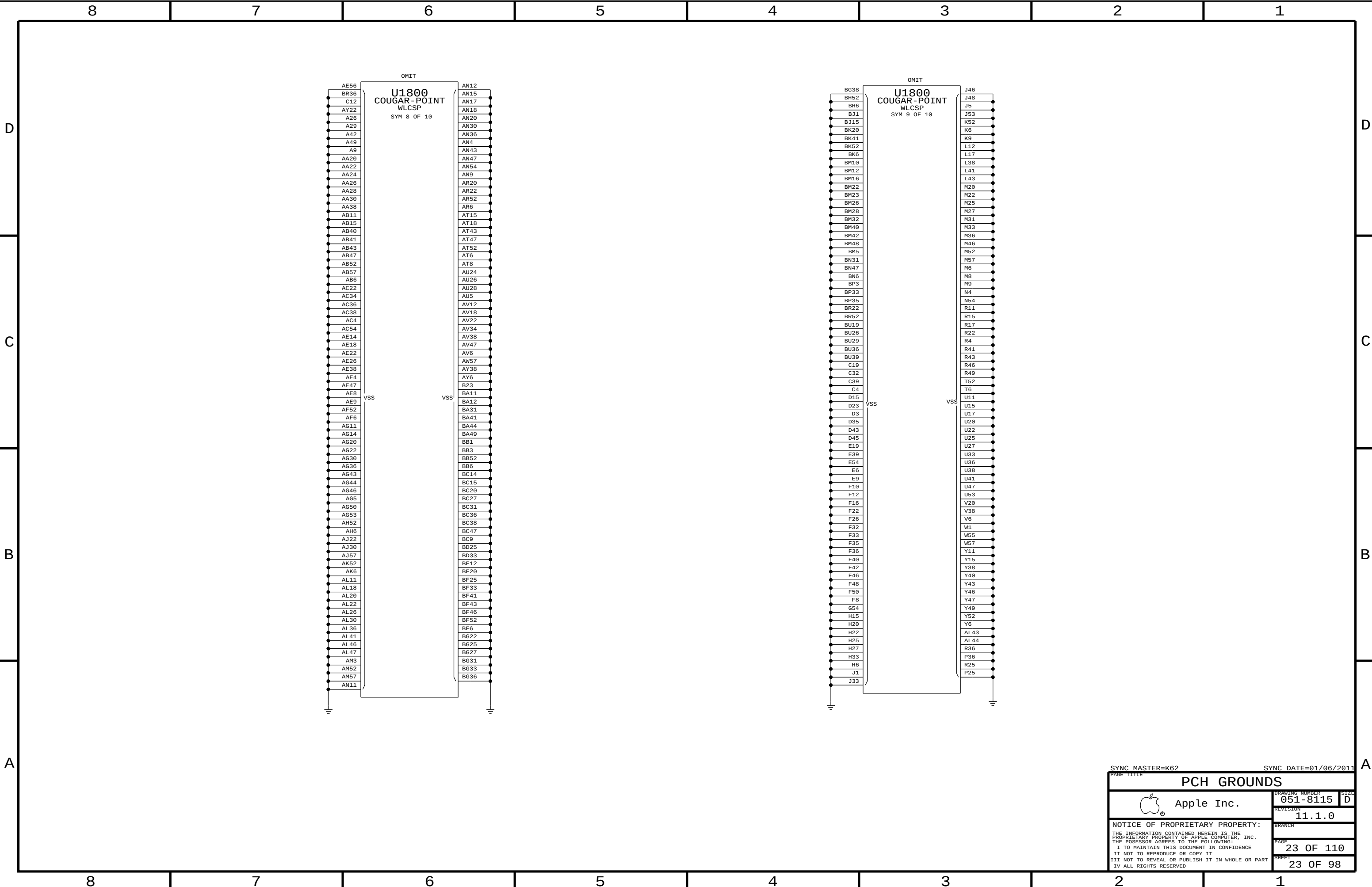
SIZE
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SYNC MASTER=K62

SYNC DATE=01/06/2011

PAGE TITLE		PCH GROUNDS	
 Apple Inc.		DRAWING NUMBER	051-8115
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		BRANCH	
		PAGE	23 OF 110
		SHEET	23 OF 98

D

C

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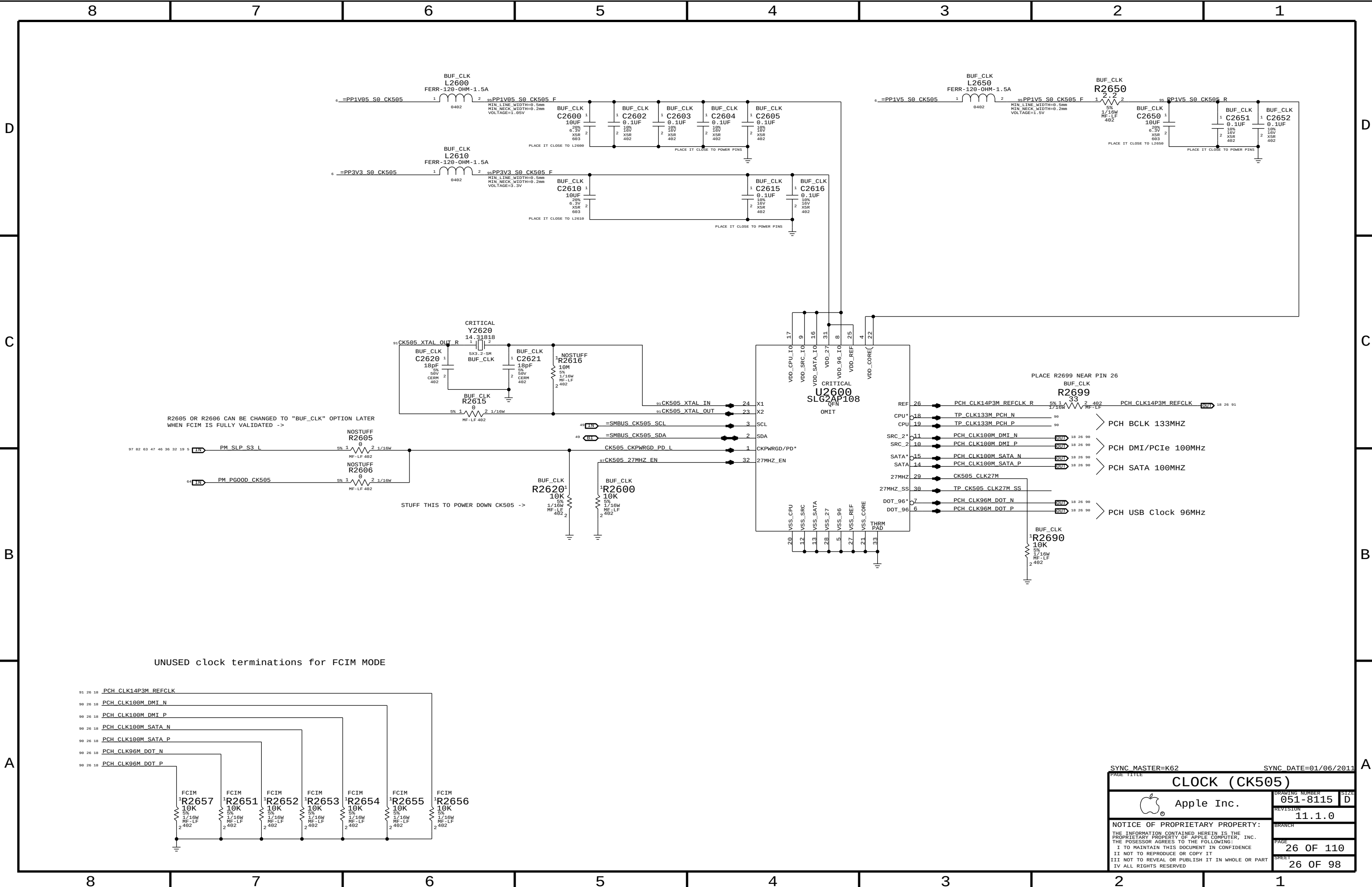
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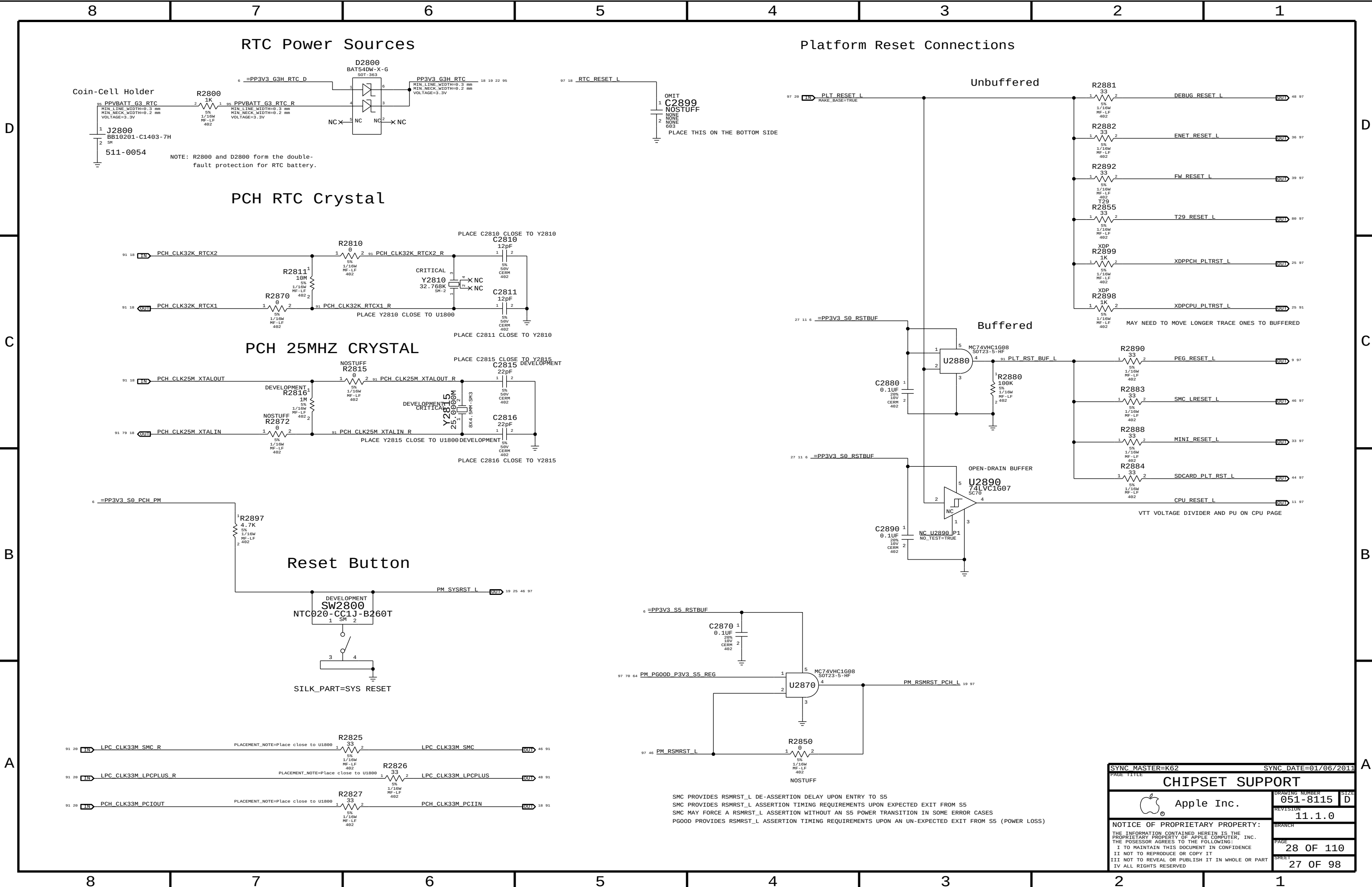
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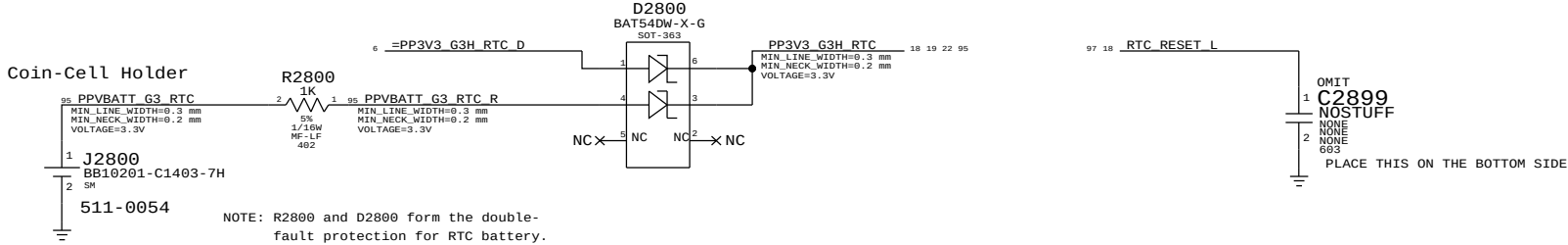
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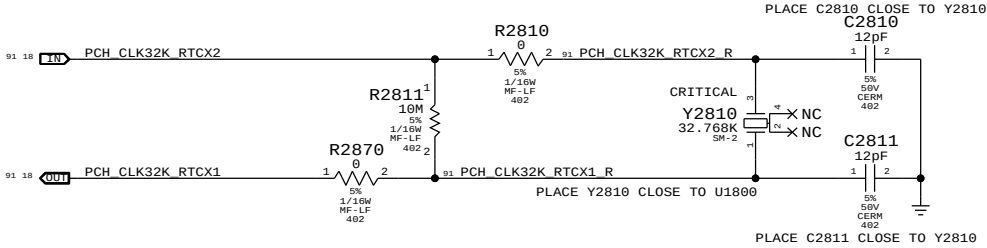
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CLOCK (CK505)		DRAWING NUMBER	051-8115
Apple Inc.		REVISION	11.1.0
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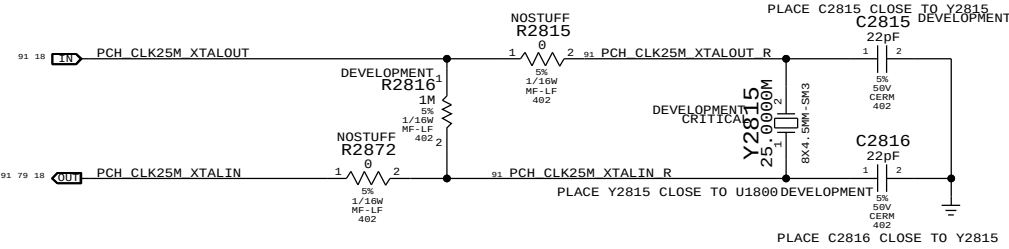
RTC Power Sources



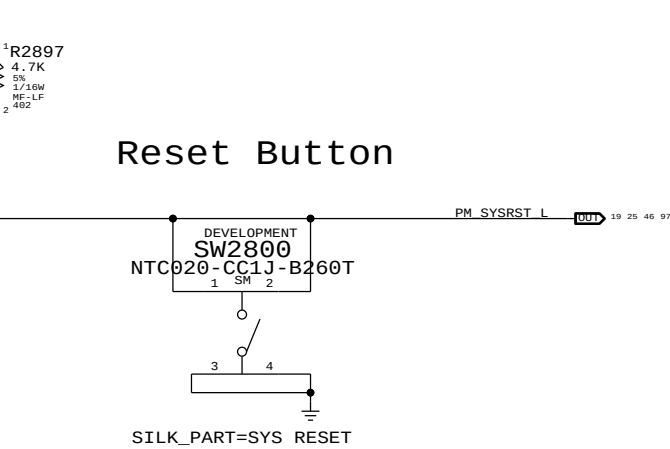
PCH RTC Crystal



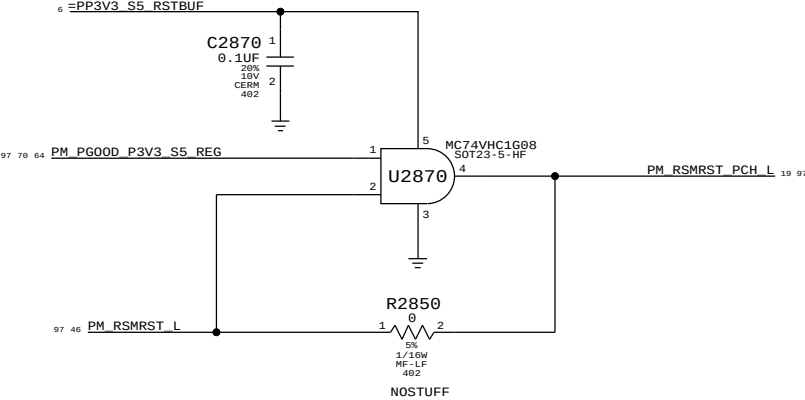
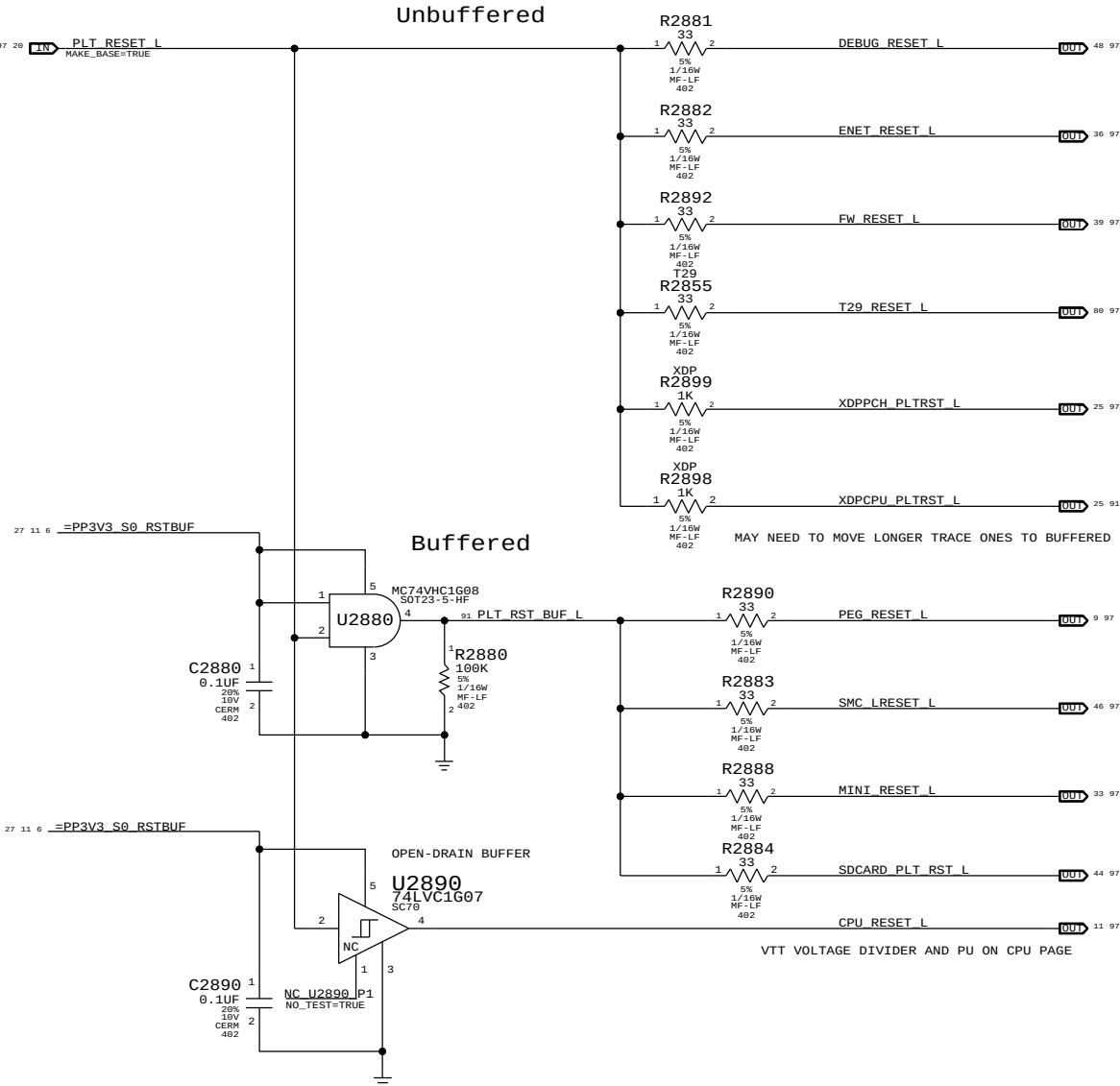
PCH 25MHZ CRYSTAL



Reset Button

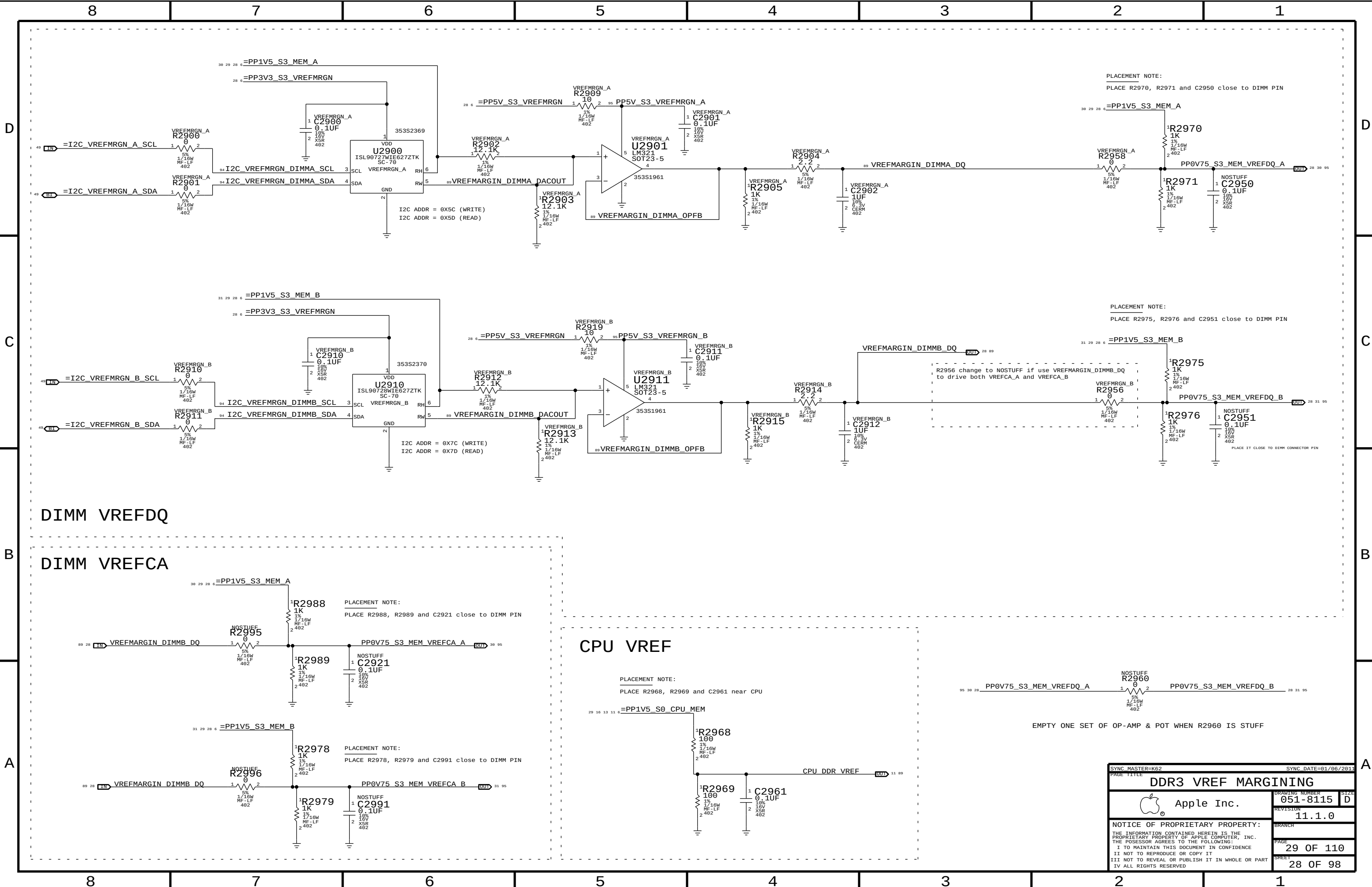


Platform Reset Connections

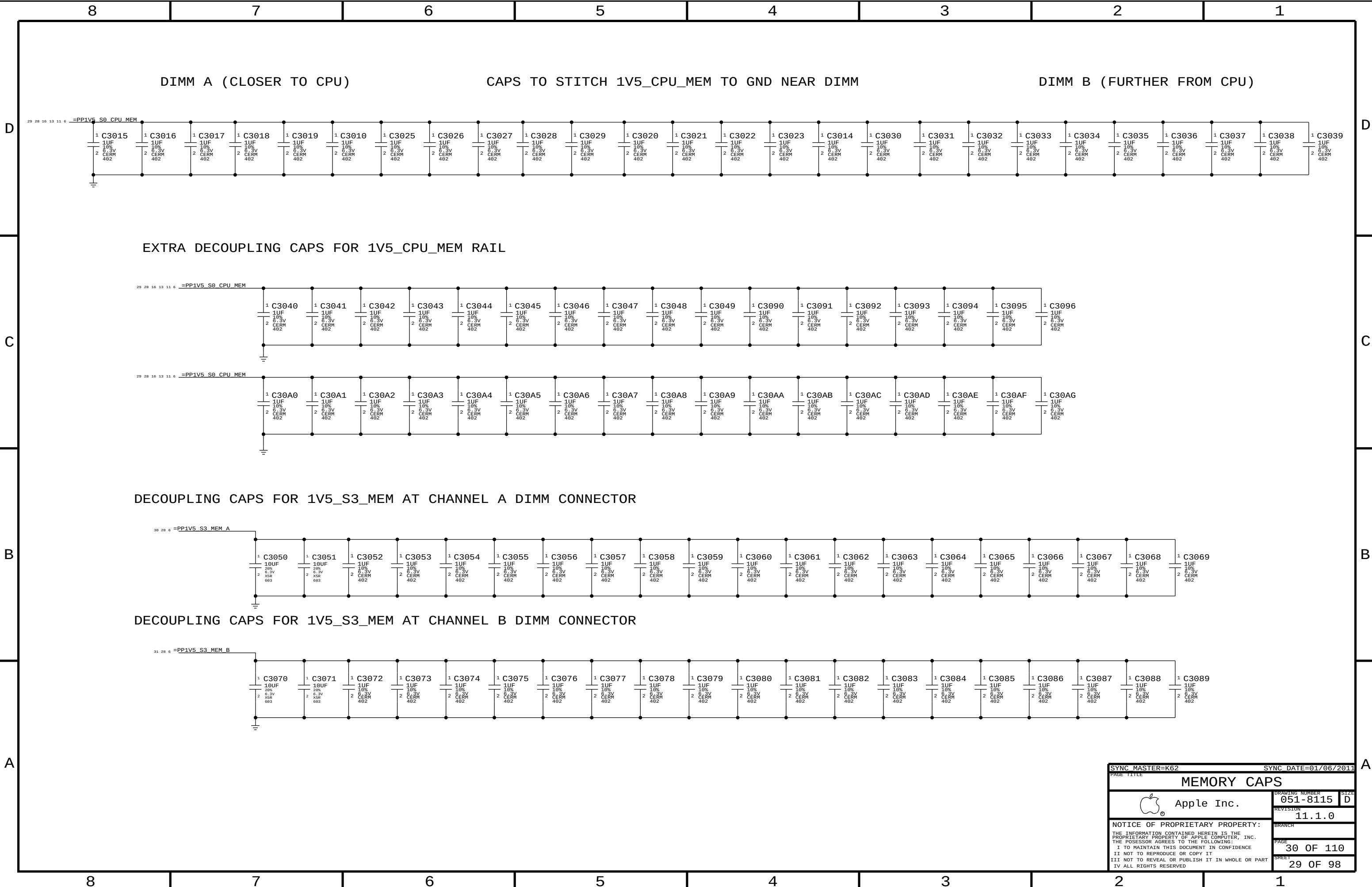


SMC PROVIDES RSMRST_L DE-ASSERTION DELAY UPON ENTRY TO S5
SMC PROVIDES RSMRST_L ASSERTION TIMING REQUIREMENTS UPON EXPECTED EXIT FROM S5
SMC MAY FORCE A RSMRST_L ASSERTION WITHOUT AN S5 POWER TRANSITION IN SOME ERROR CASES
PGOOD PROVIDES RSMRST_L ASSERTION TIMING REQUIREMENTS UPON AN UN-EXPECTED EXIT FROM S5 (POWER LOSS)

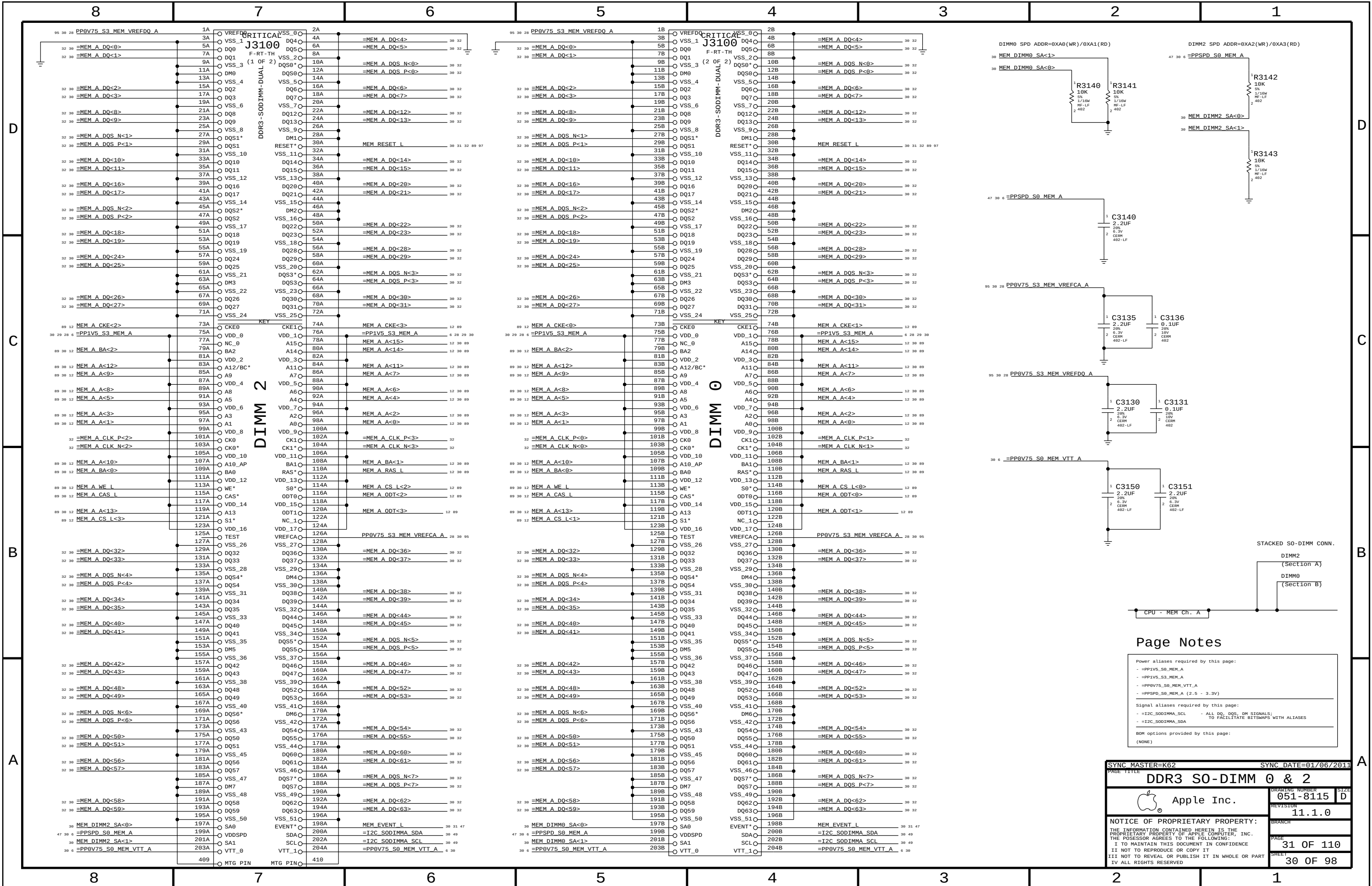
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		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		DDR3 VREF MARGINING	
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		REVISION	11.1.0
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PAGE TITLE		MEMORY CAPS	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMMA_SCL - ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITMAPS WITH ALIASES
- =I2C_S0DIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC MASTER=K62

SYNC DATE=01/06/2011

PAGE TITLE

DDR3 SO-DIMM 0 & 2

 Apple Inc.

DRAWING NUMBER

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SIZE

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REVISION

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PAGE

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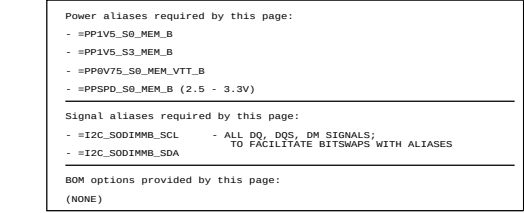
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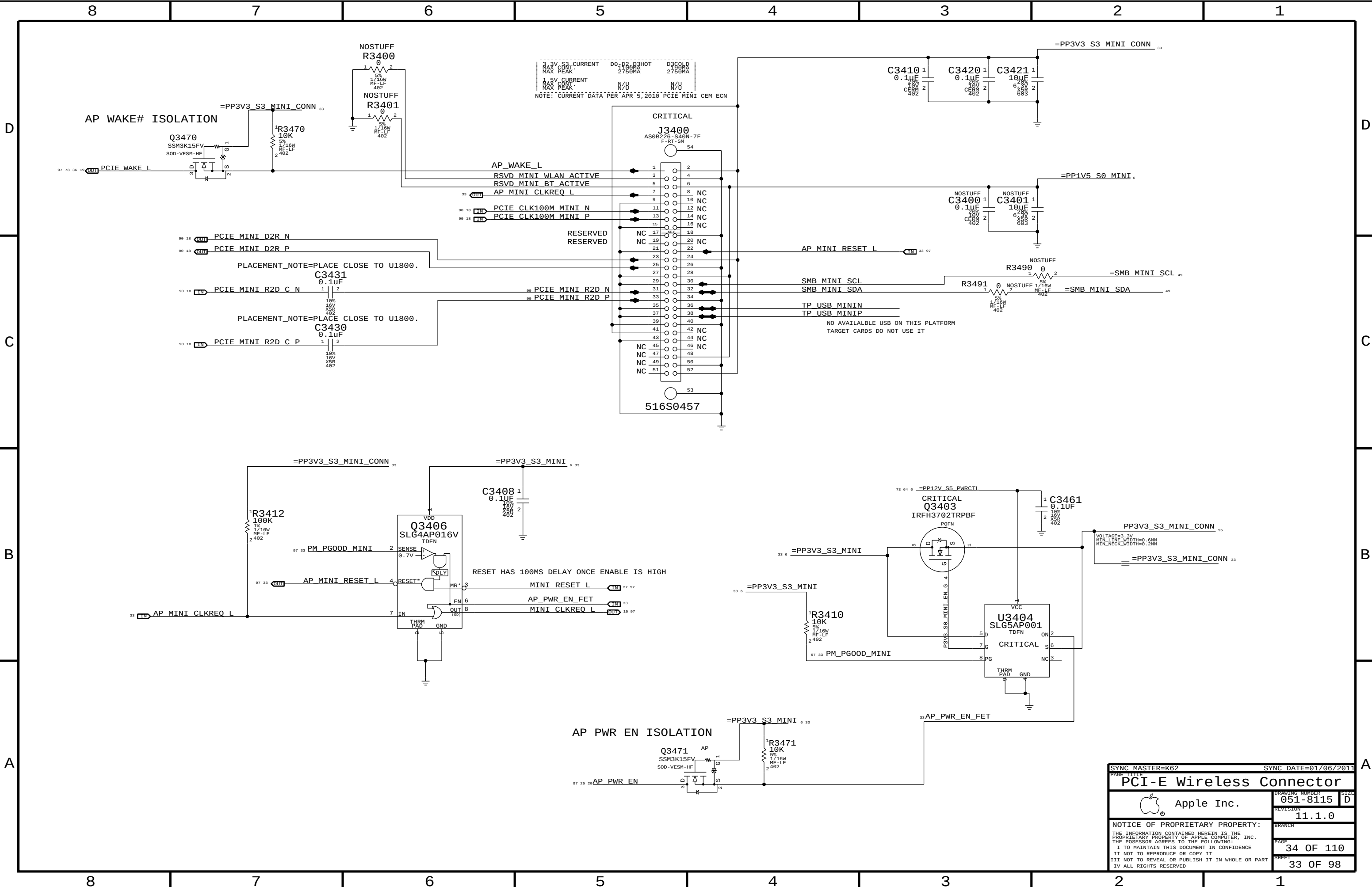
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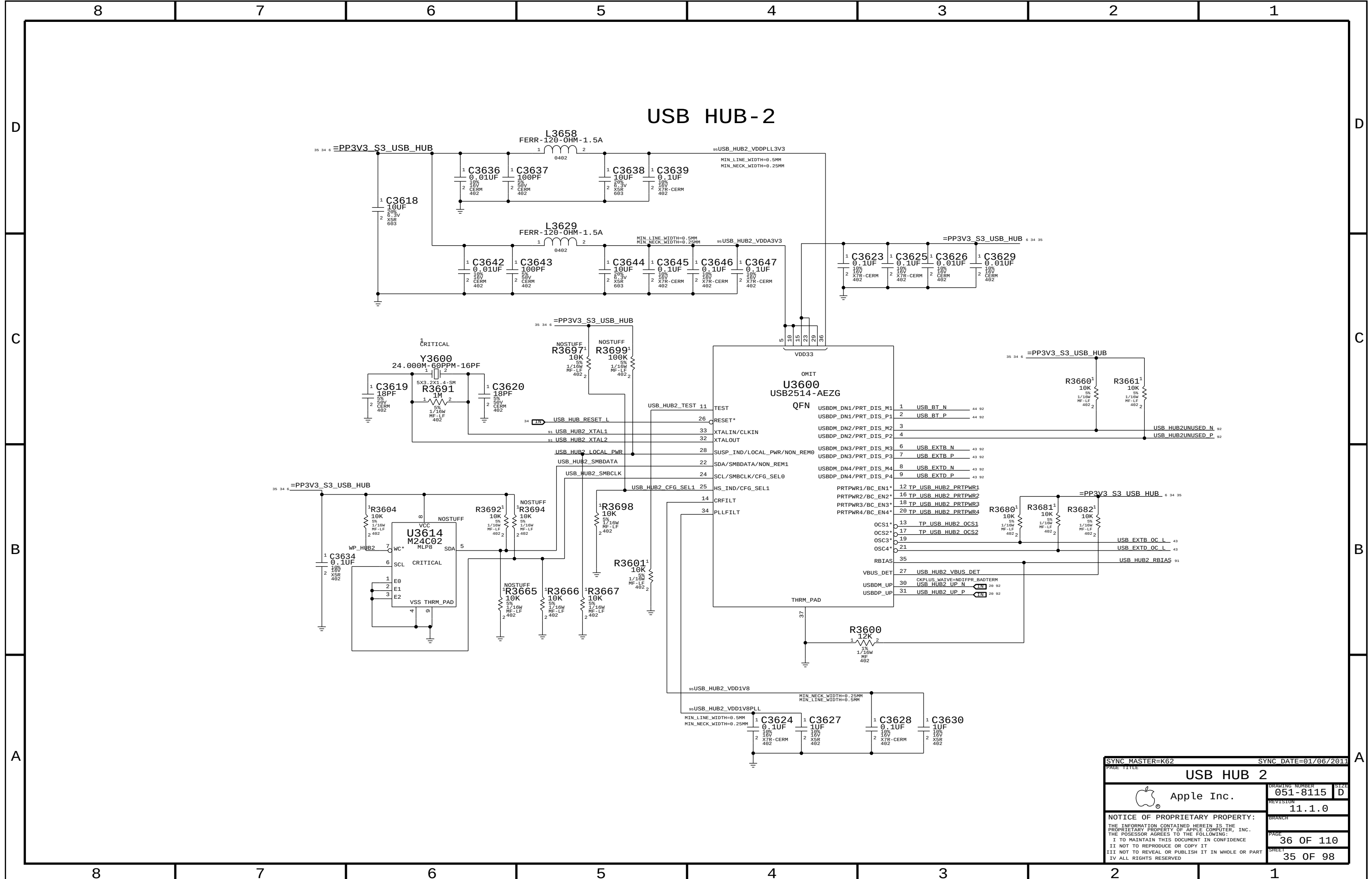
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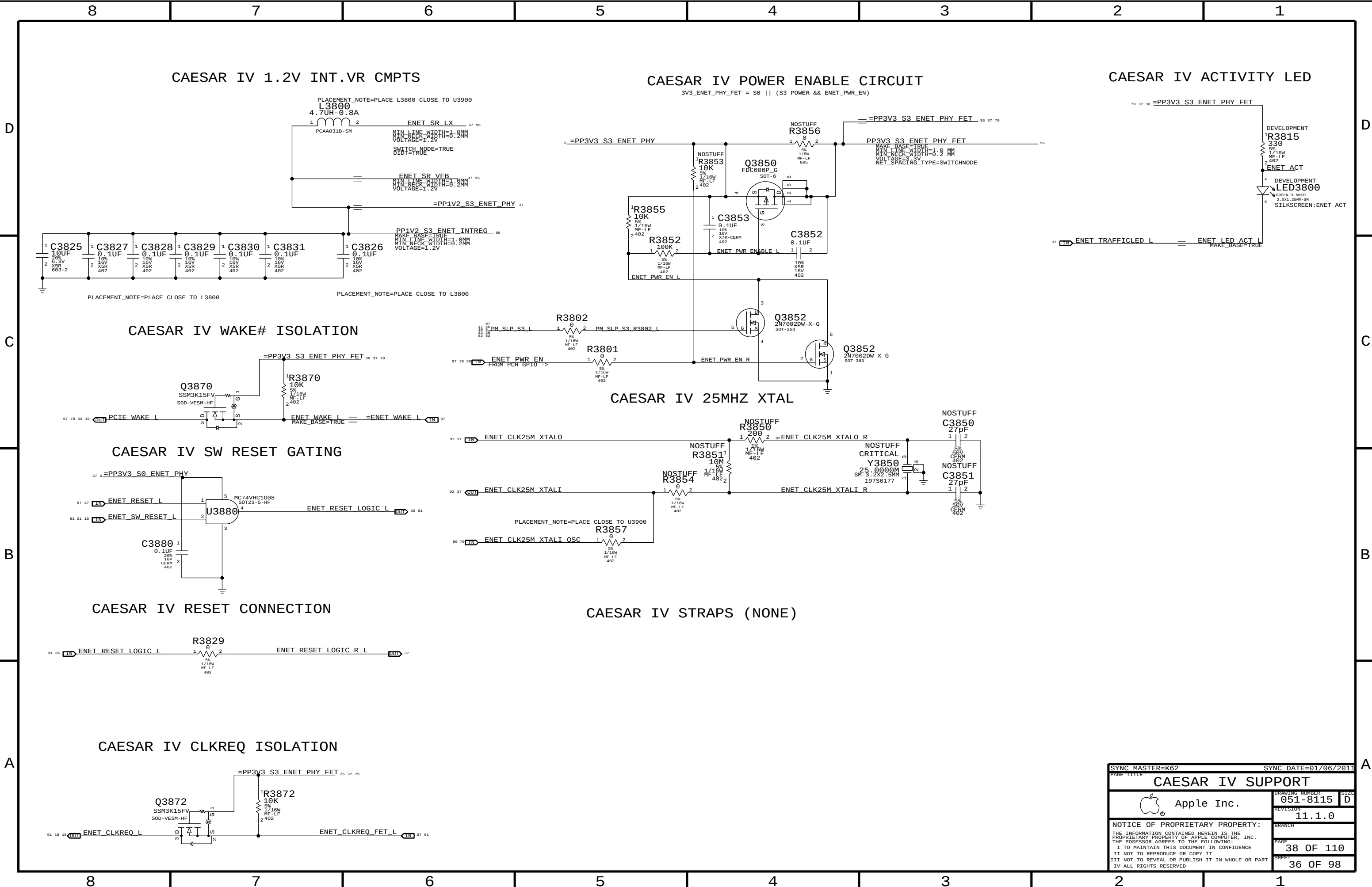




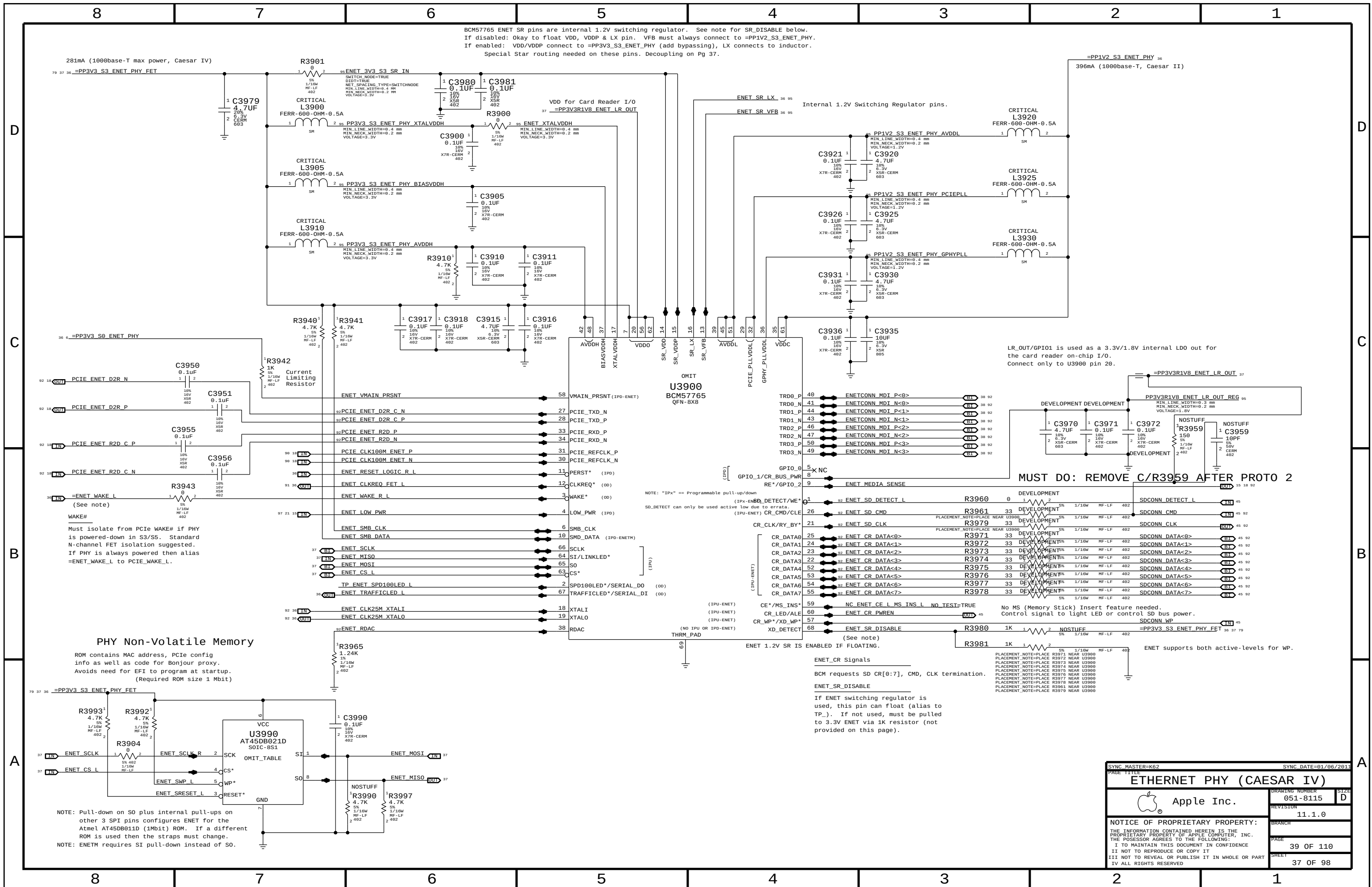
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PAGE TITLE		PCI-E Wireless Connector	
Apple Inc.		DRAWING NUMBER	051-8115
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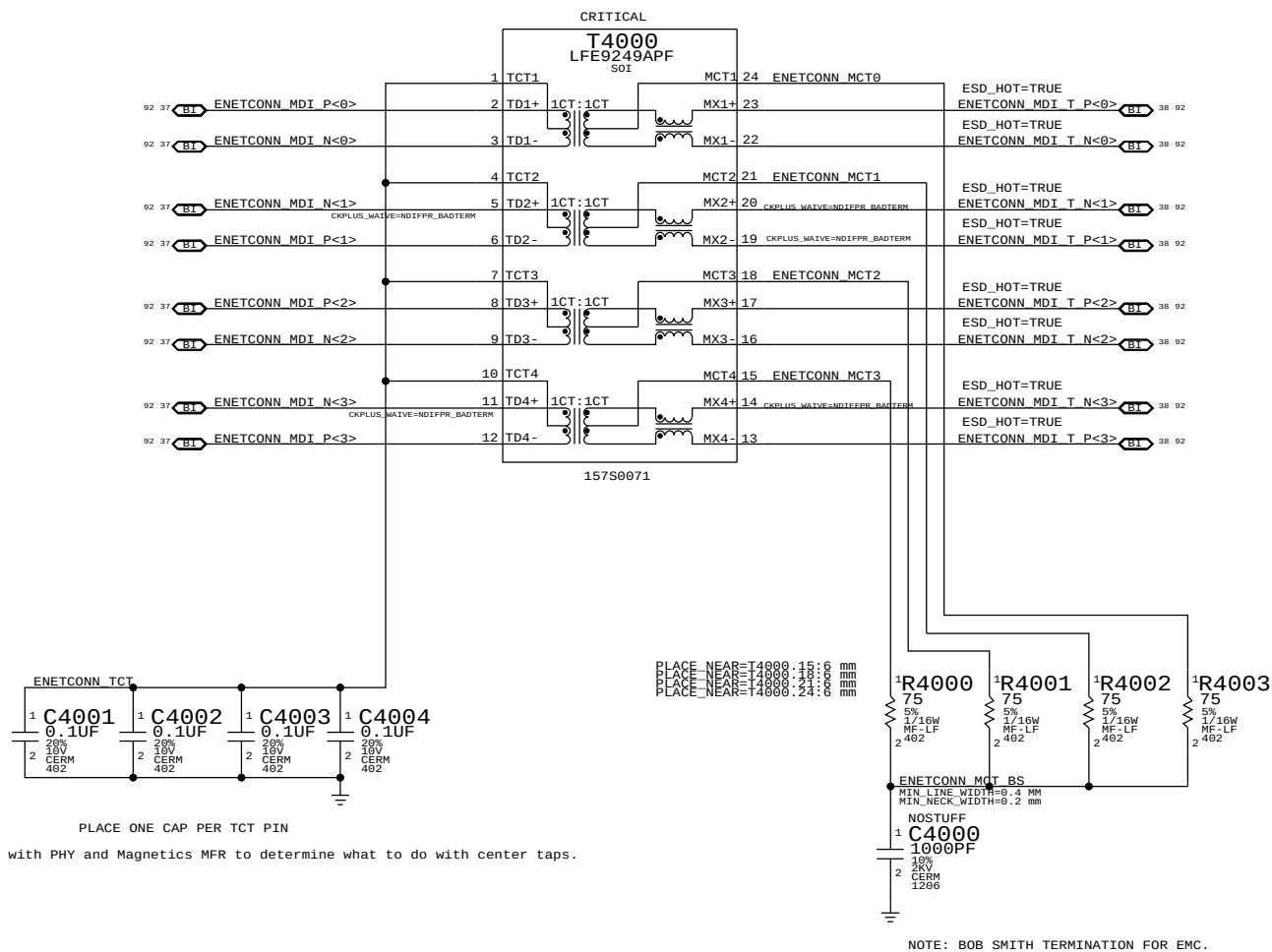
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PAGE TITLE			
USB HUB 2			
 Apple Inc.		DRAWING NUMBER	051-8115
		SIZE	D
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		BRANCH	
		PAGE	36 OF 110
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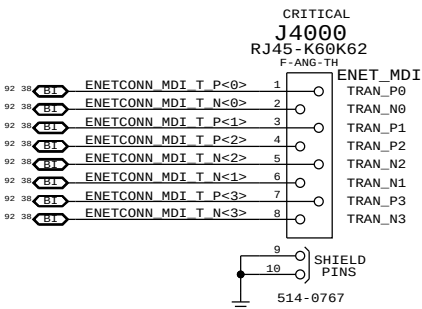
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PAGE TITLE			
CAESAR IV SUPPORT			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-8115		D
		REVISION	
		11.1.0	
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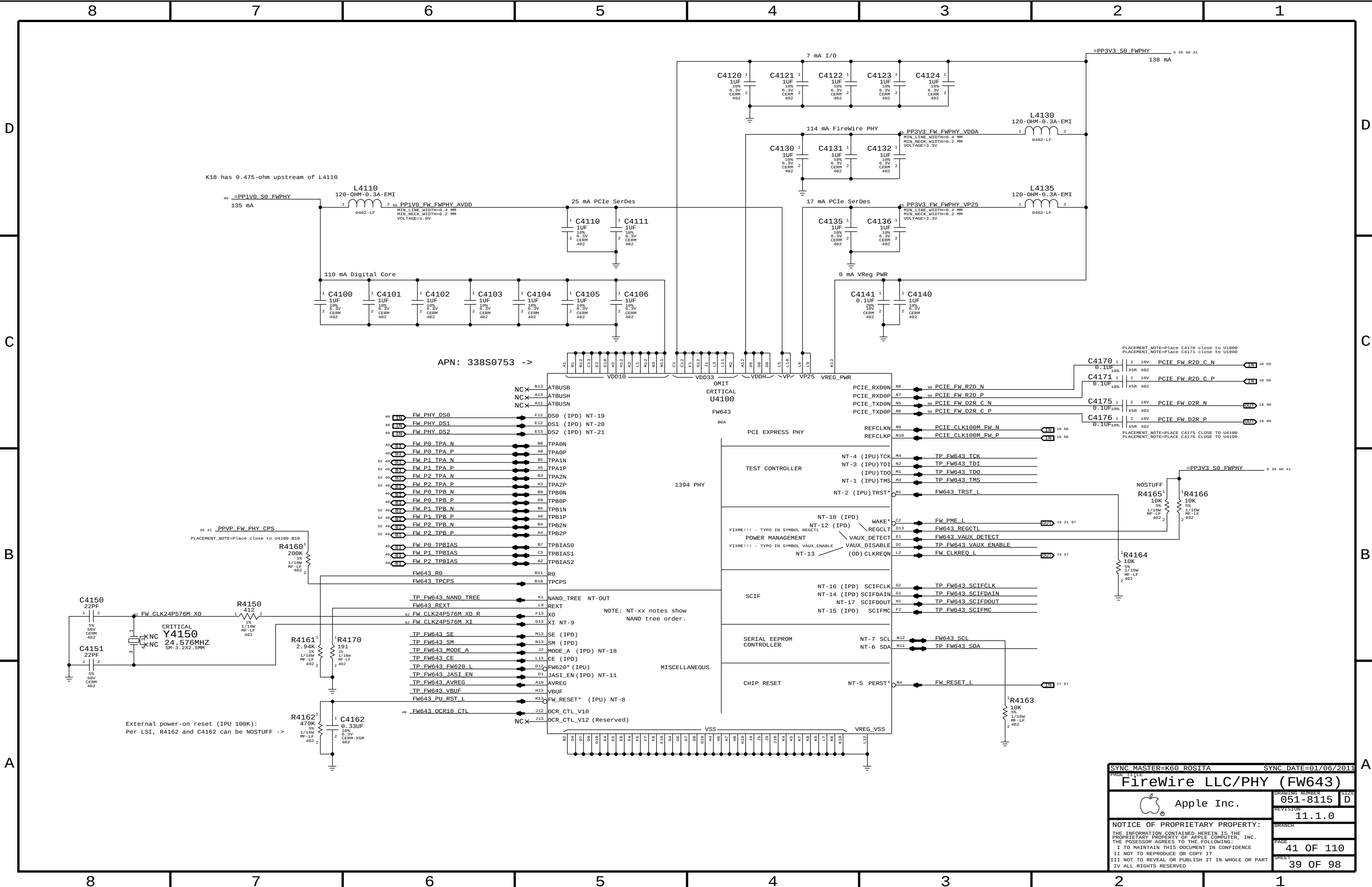
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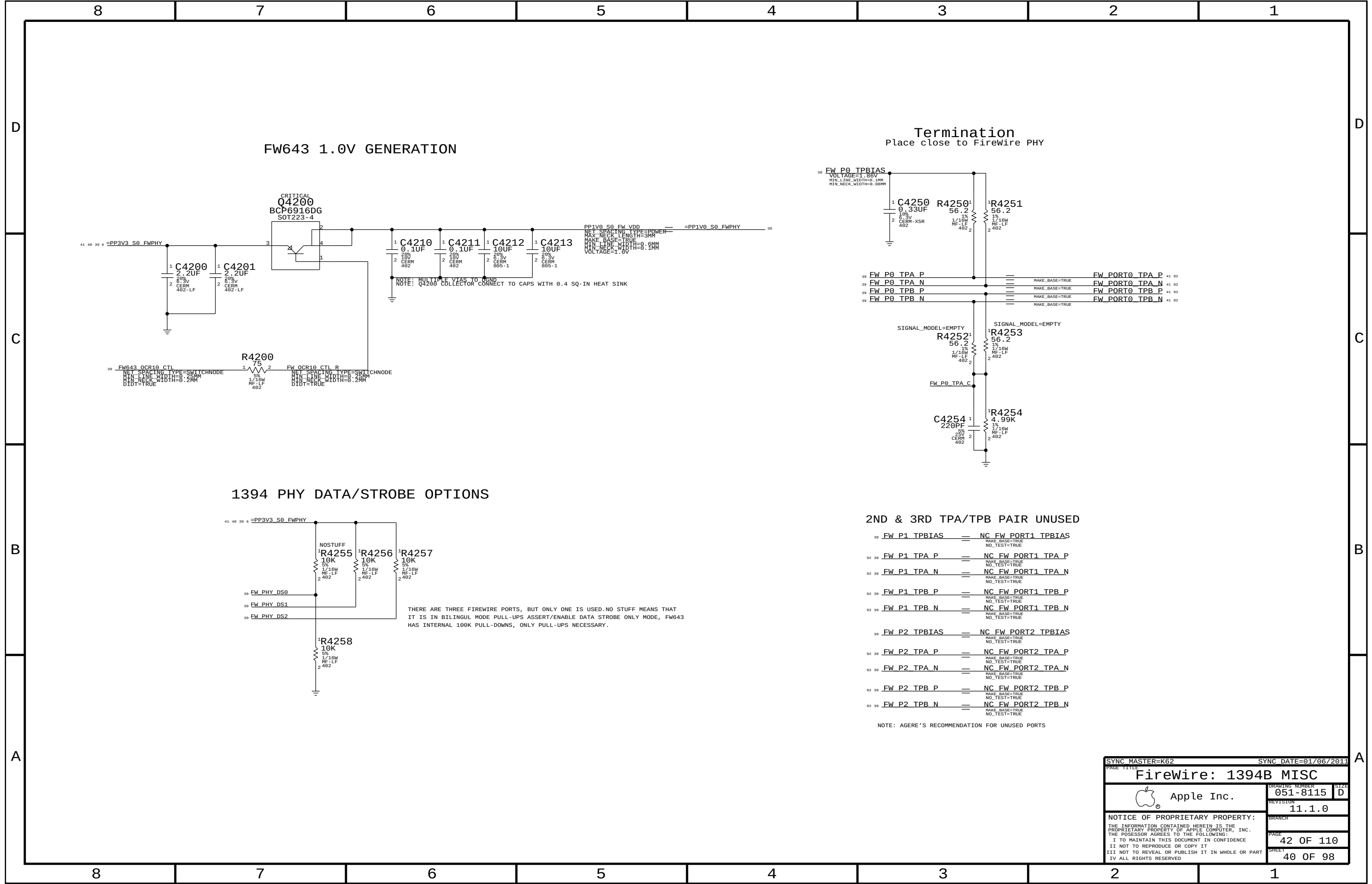


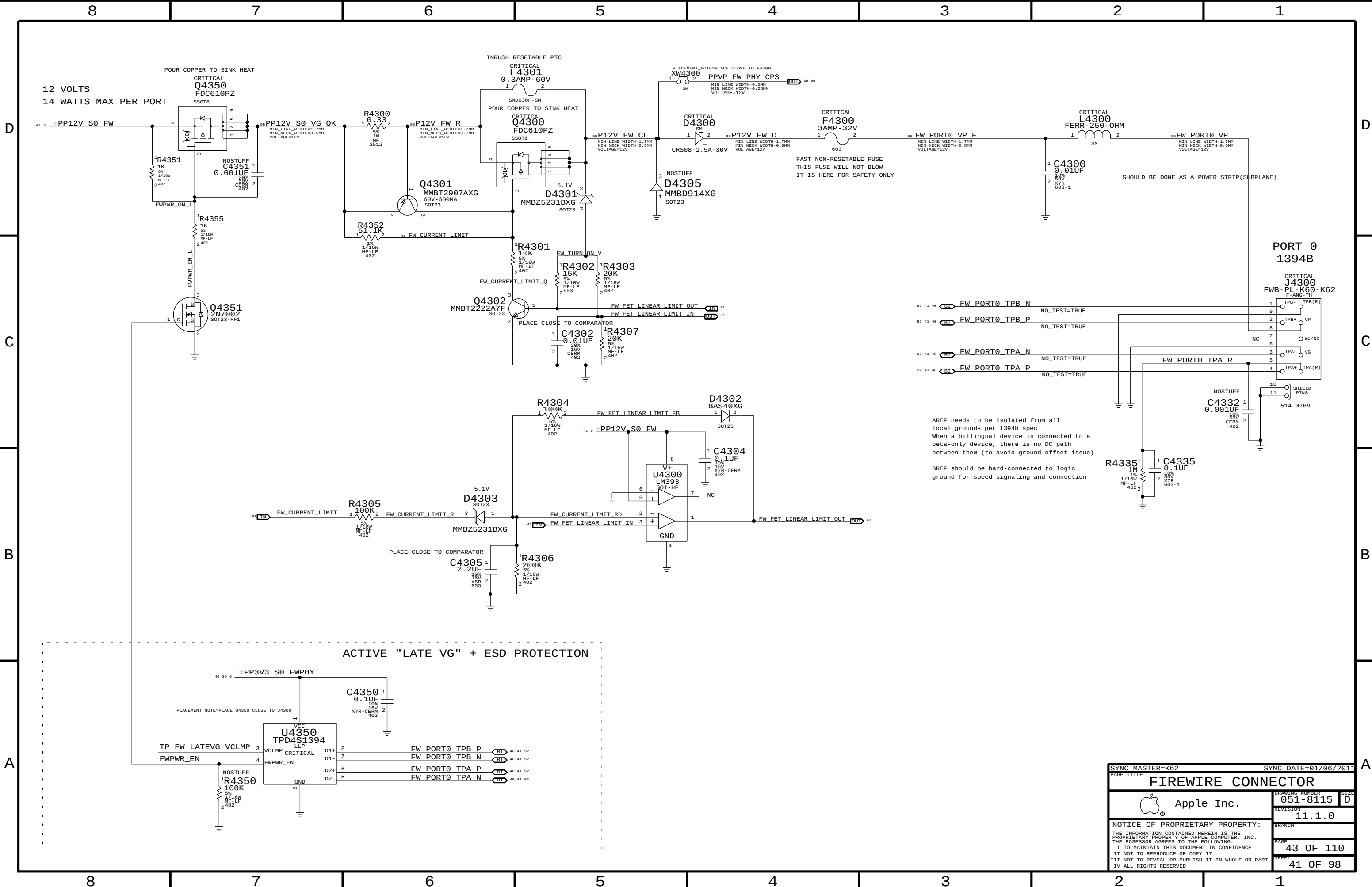
NO PAIR AND PIN POLARITY SWAPS

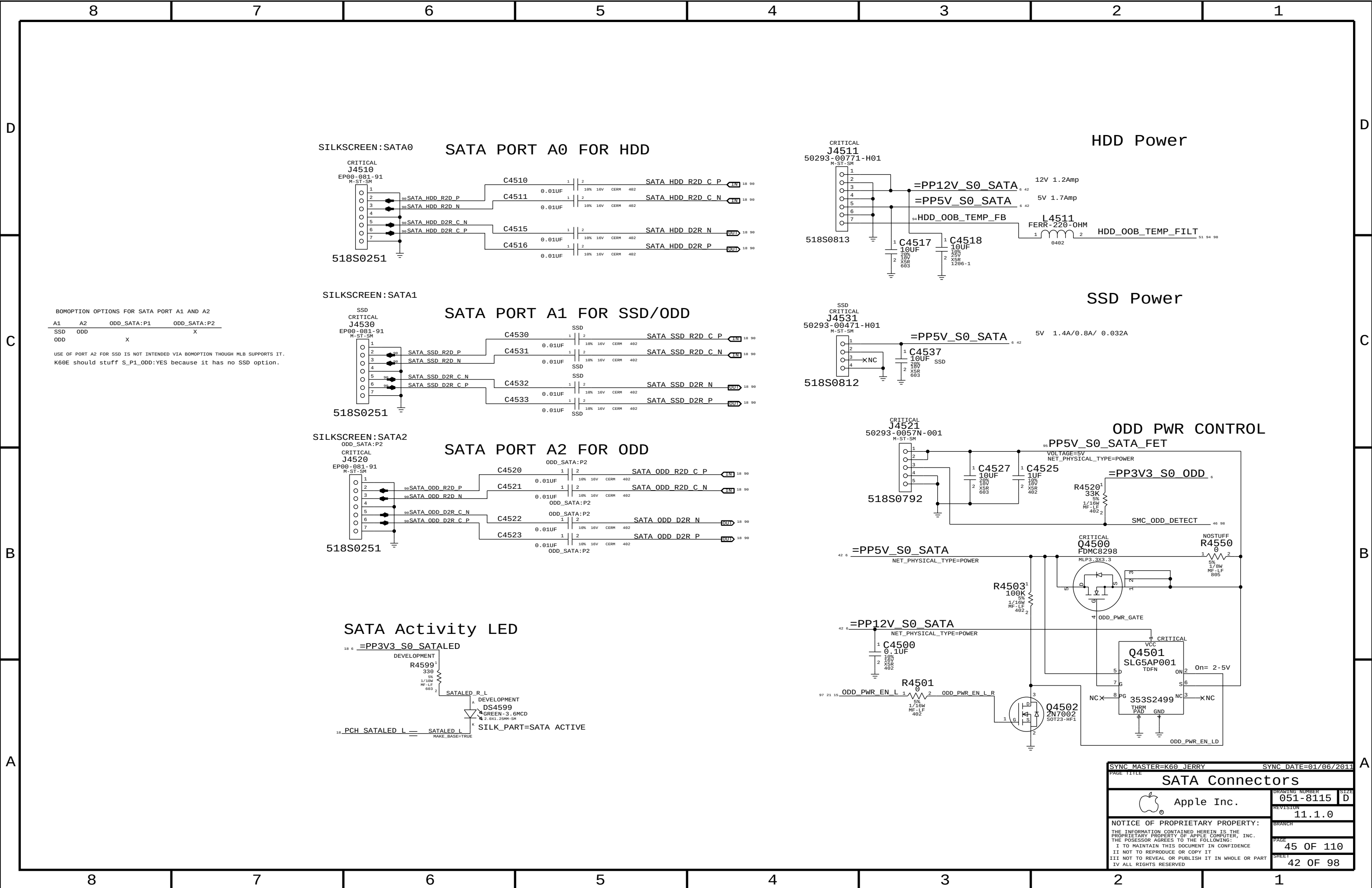


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PAGE TITLE		Ethernet Connector	
Apple Inc.		DRAWING NUMBER	051-8115
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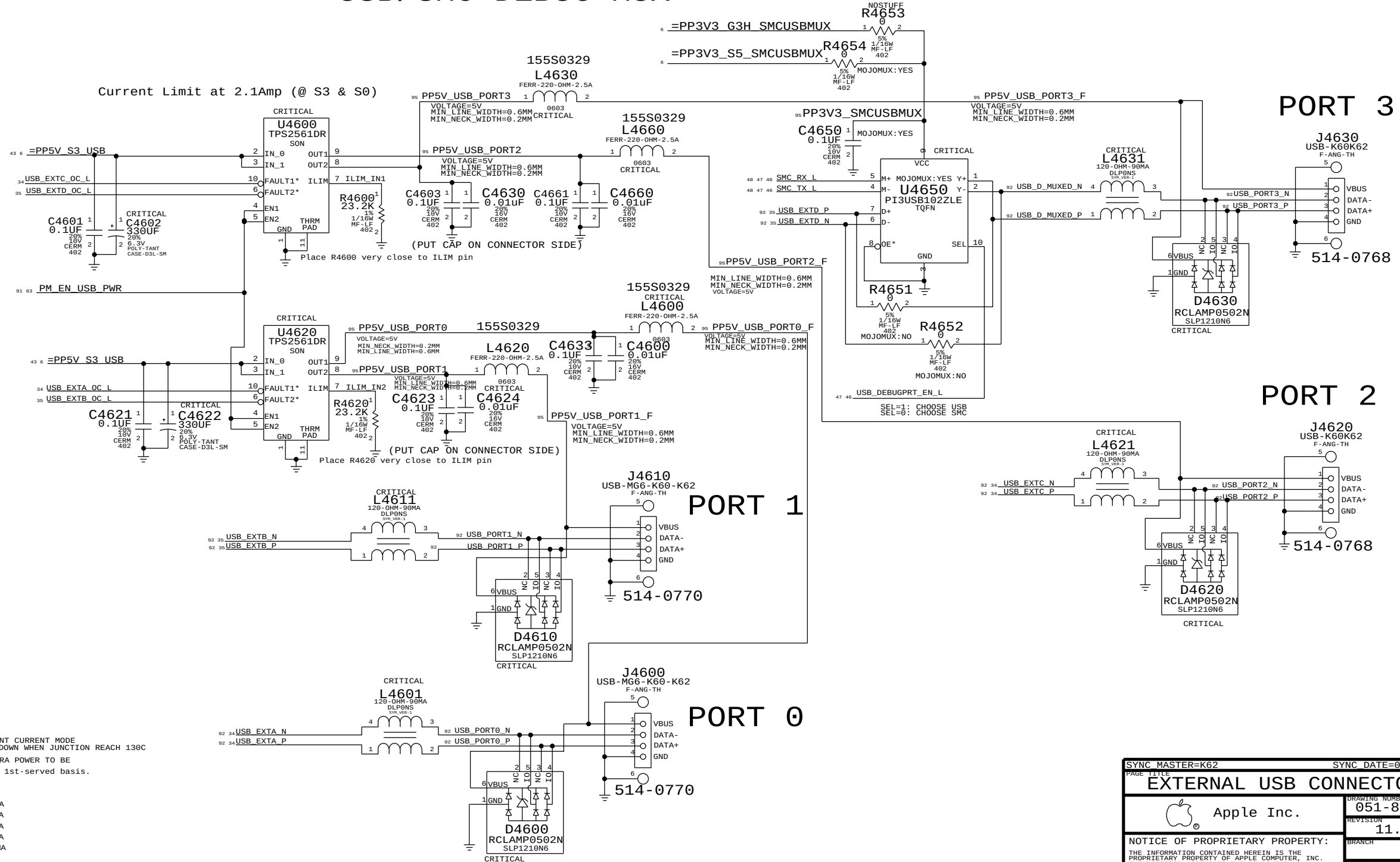
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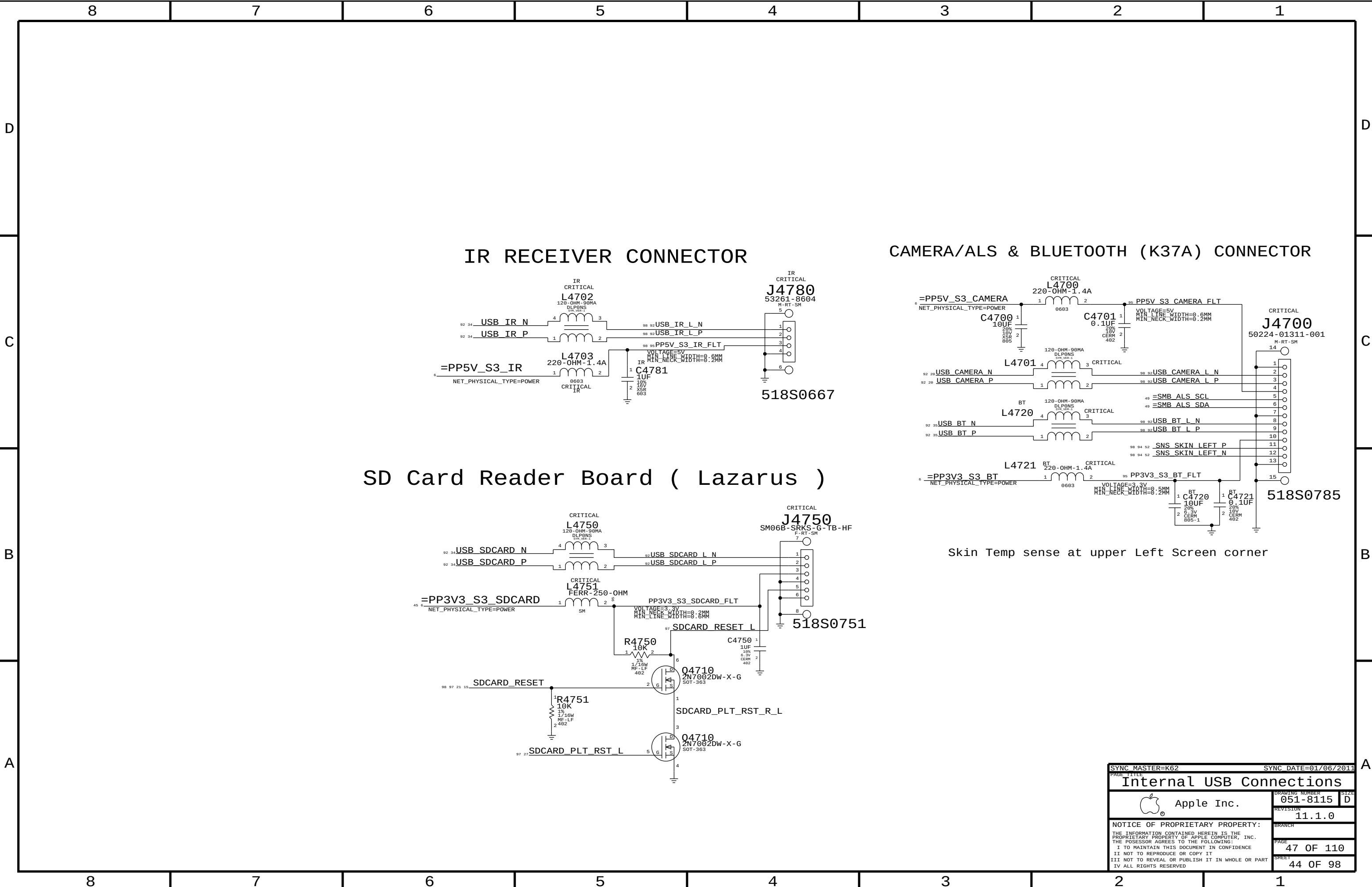
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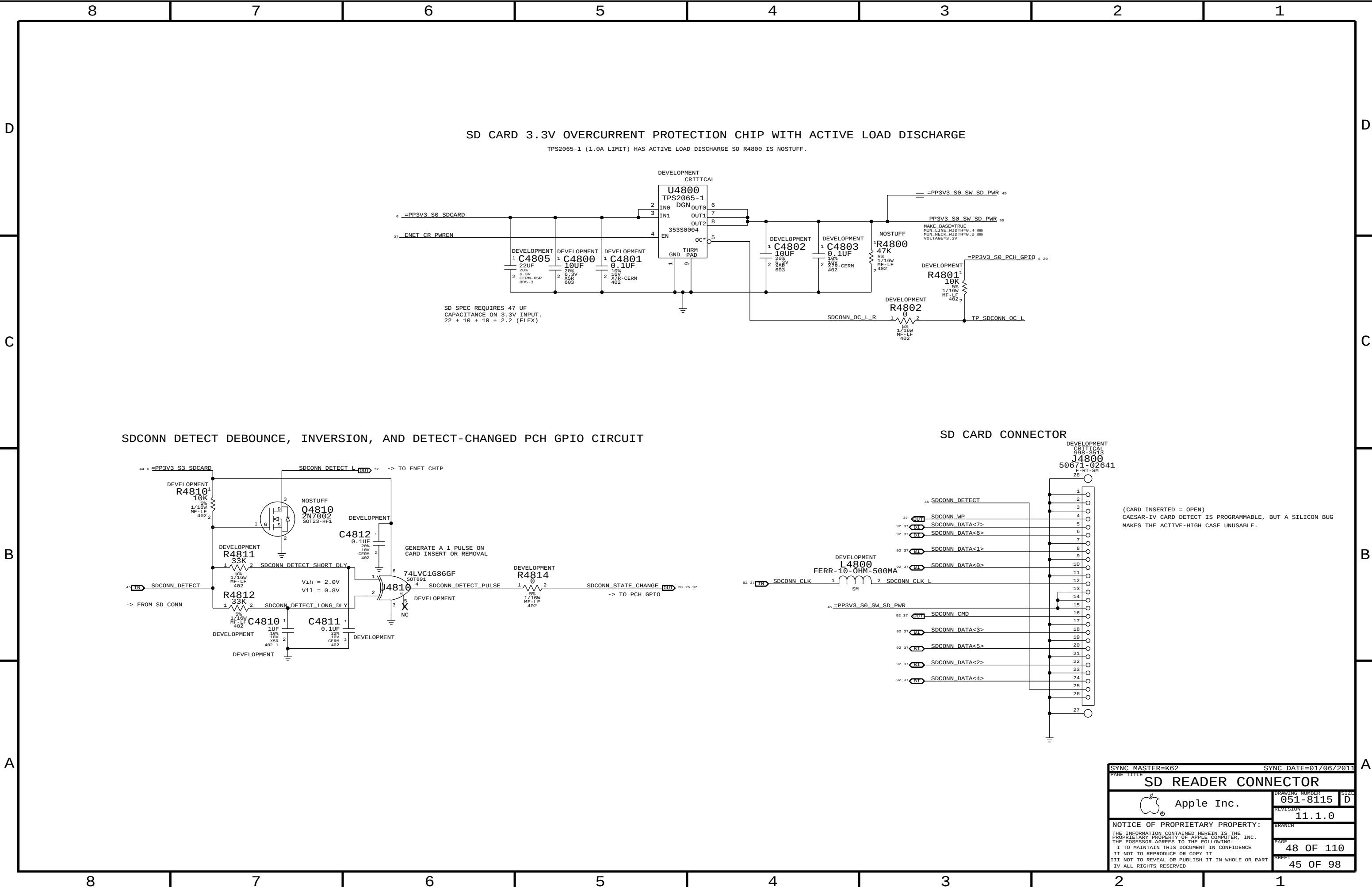
USB/SMC DEBUG MUX

ADDED AT EVT & SWITCH TO S5 RAIL

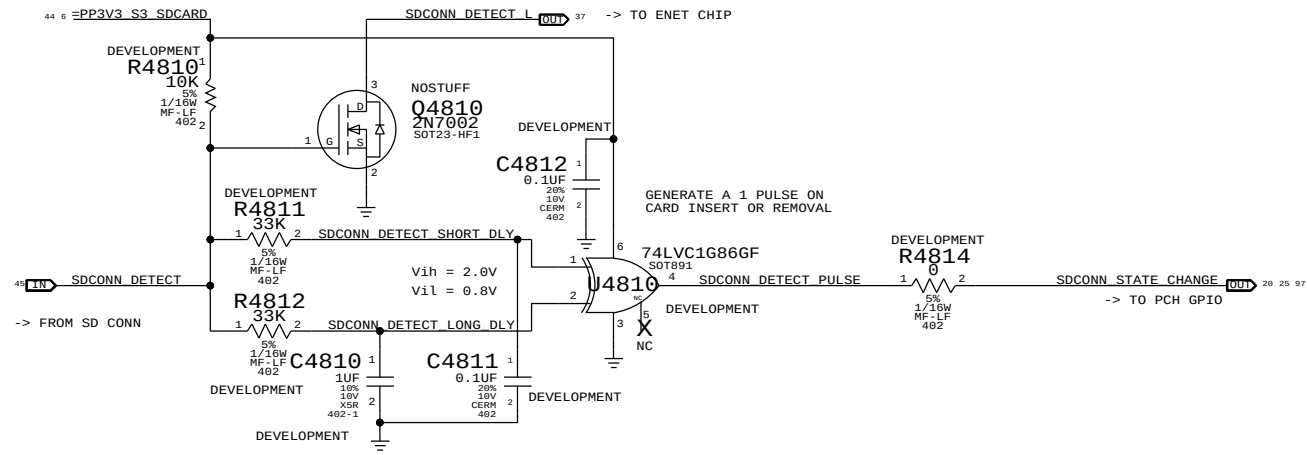


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PAGE TITLE			
EXTERNAL USB CONNECTORS			
 Apple Inc.		DRAWING NUMBER	051-8115
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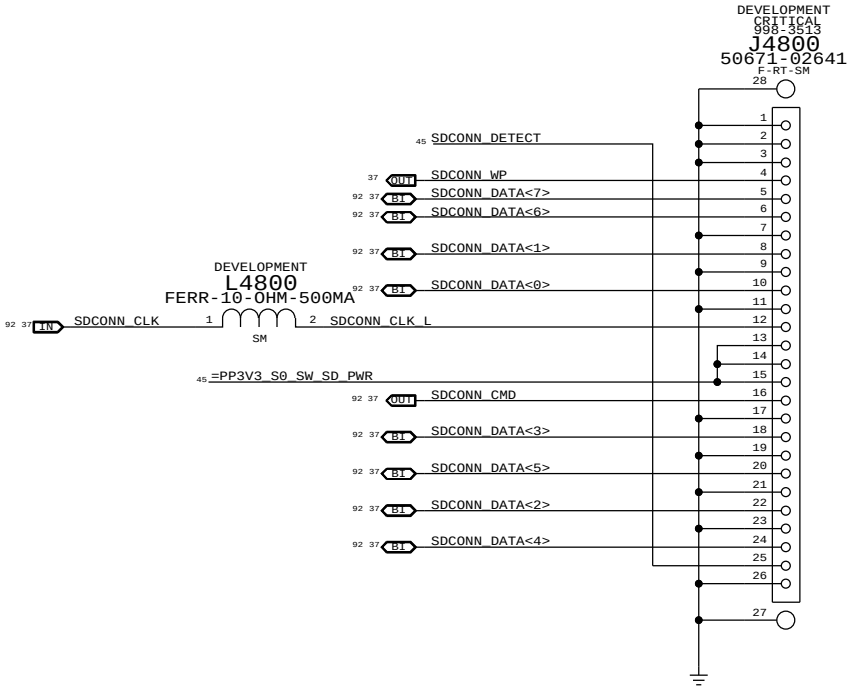




SDCONN DETECT DEBOUNCE, INVERSION, AND DETECT-CHANGED PCH GPIO CIRCUIT



SD CARD CONNECTOR



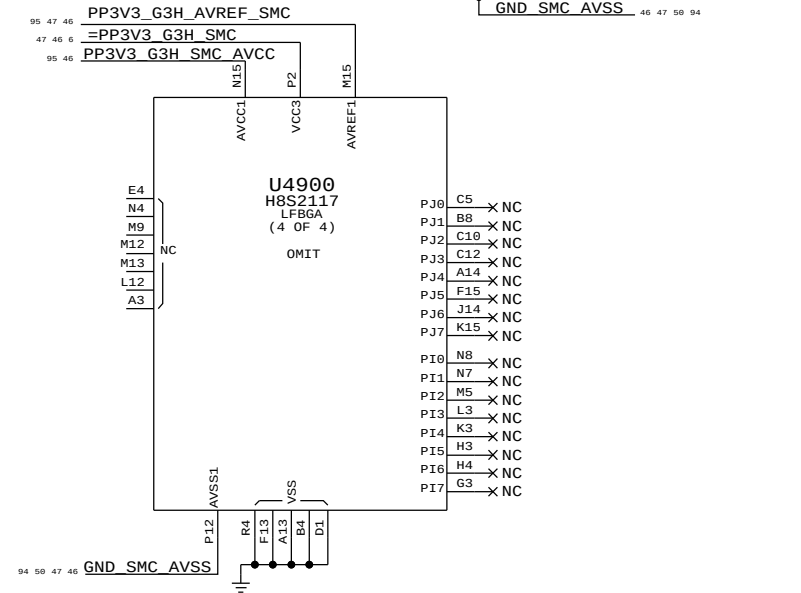
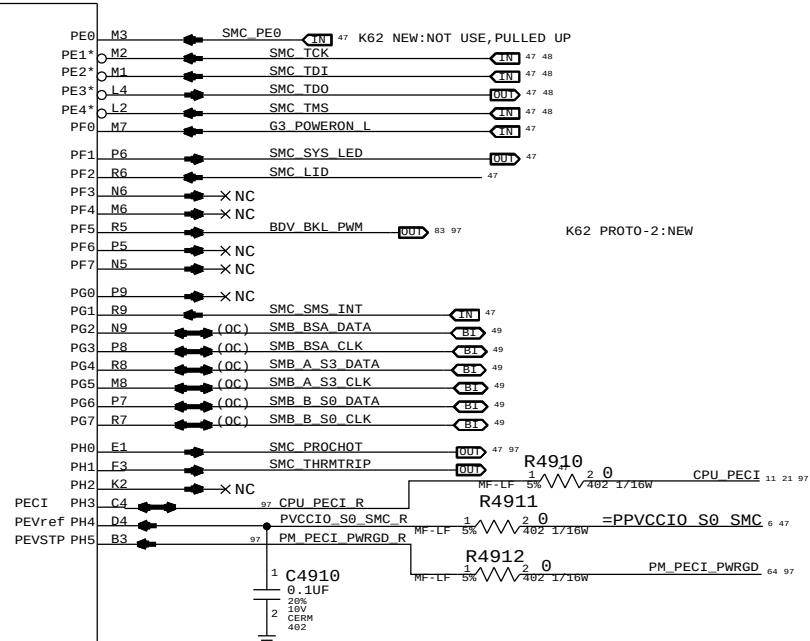
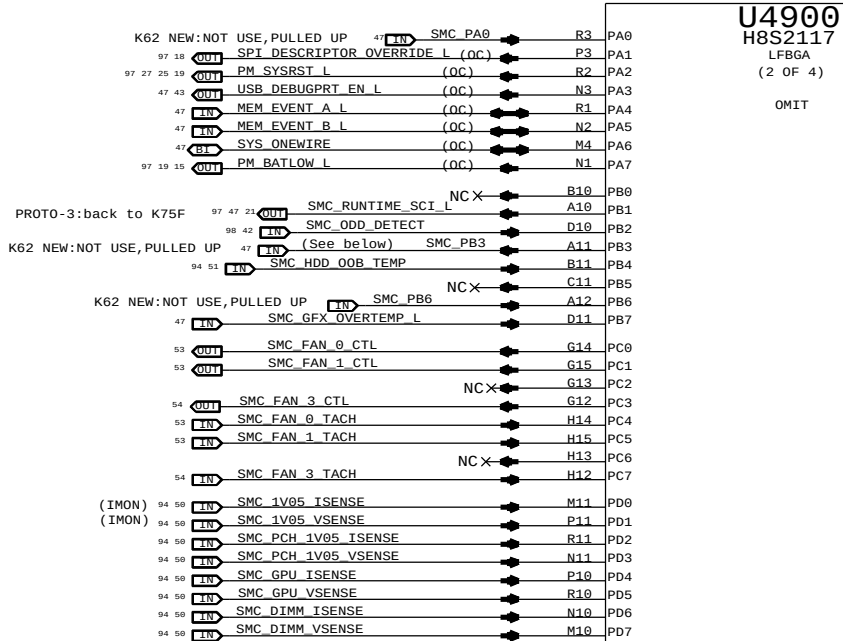
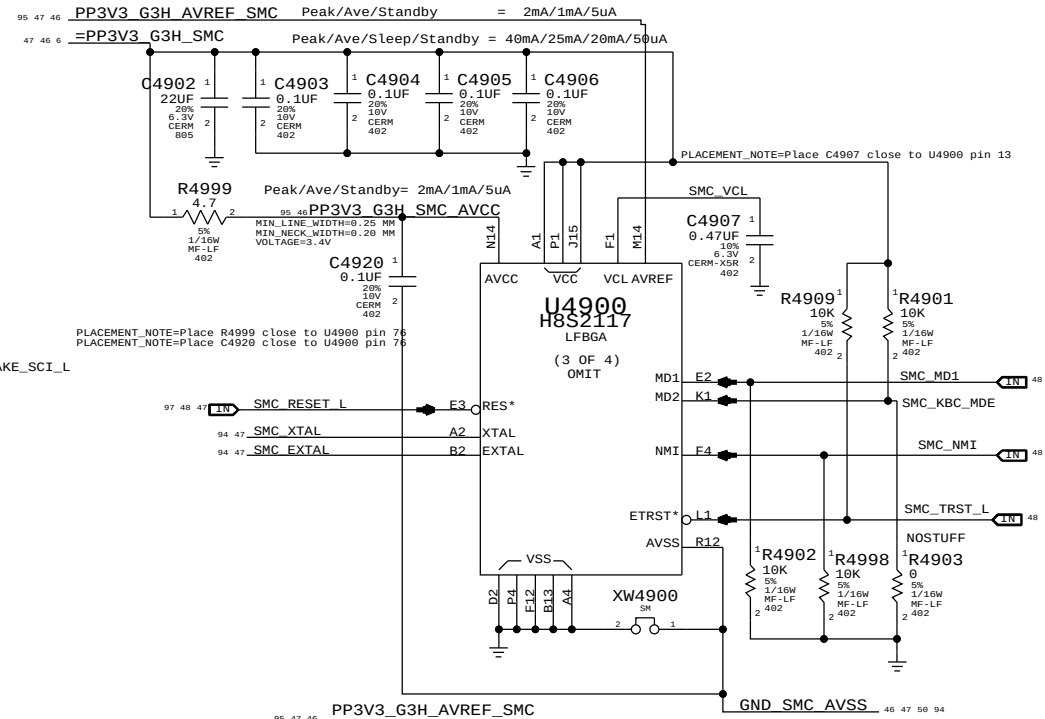
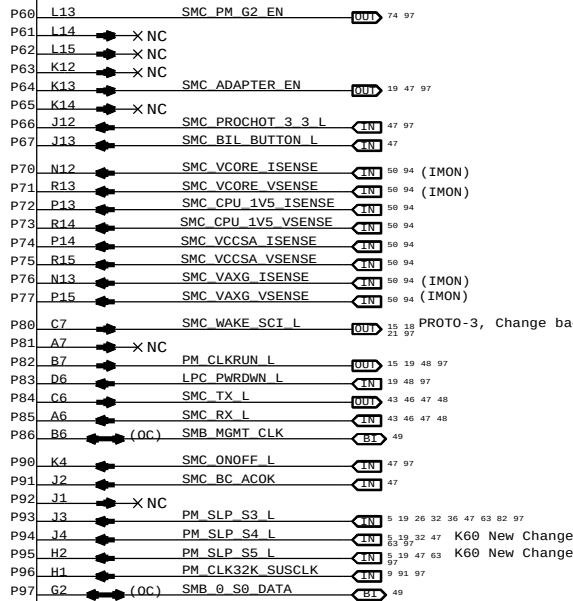
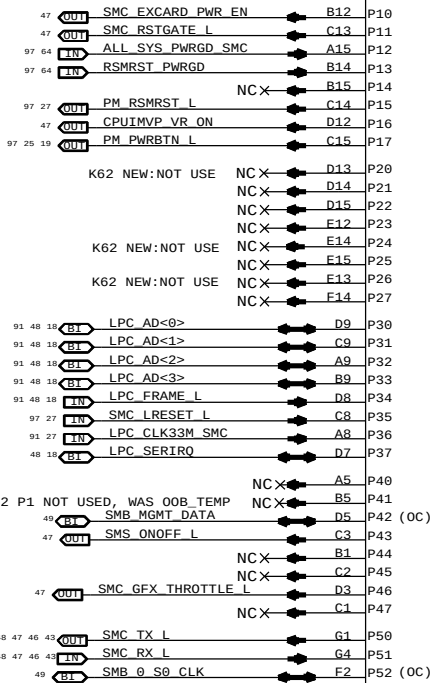
(CARD INSERTED = OPEN)
CAESAR-IV CARD DETECT IS PROGRAMMABLE, BUT A SILICON BUG
MAKES THE ACTIVE-HIGH CASE UNUSABLE.

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
SD READER CONNECTOR			
 Apple Inc.		DRAWING NUMBER	051-8115
		SIZE	D
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		BRANCH	
		PAGE	48 OF 110
		SHEET	45 OF 98

NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

338S0878

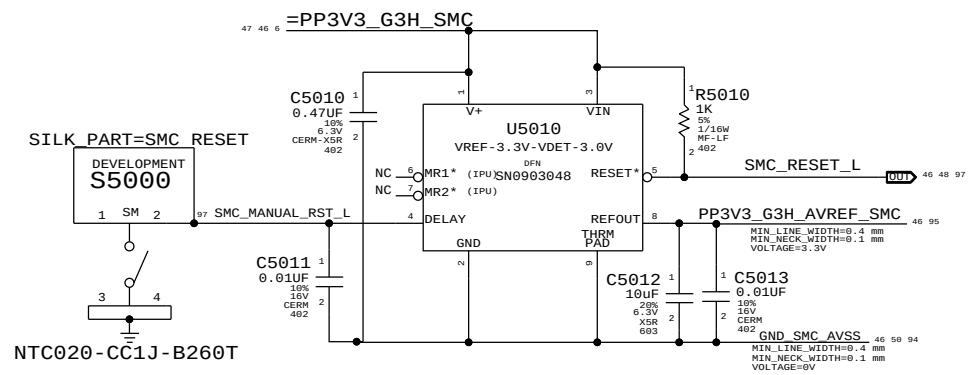
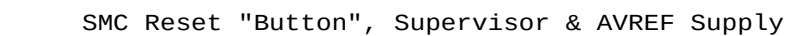
U4900
H8S2117
LFBGA
(1 OF 4)
OMIT



SMC PB3: SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay (was ISENSE_CAL_EN)

SMC PG1: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

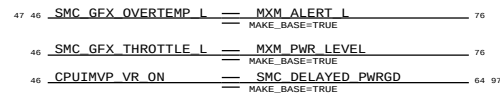
SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		SMC	
Apple Inc.		DRAWING NUMBER	051-8115
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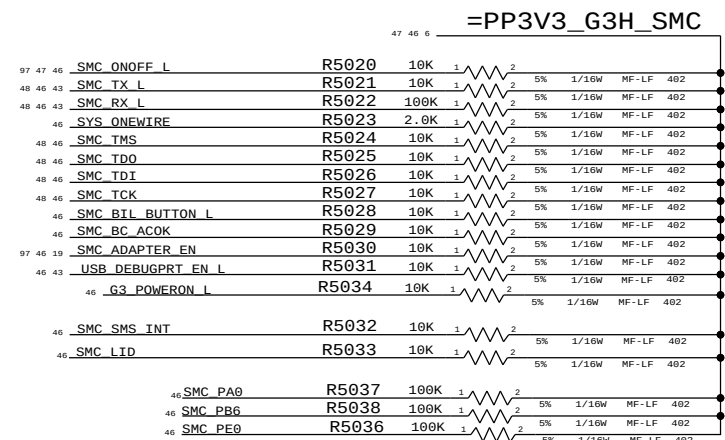
MR1* and MR2* must both be low to cause manual reset.
Used on mobiles to support SMC reset via keyboard.

NOTE: Internal pull-ups are to VIN, not V+.

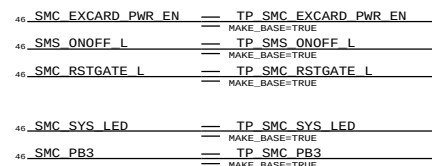
MISC. SIGNAL ALIASES



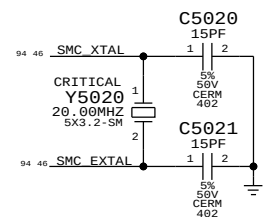
UNUSED PORT 7 ANALOG SENSORS



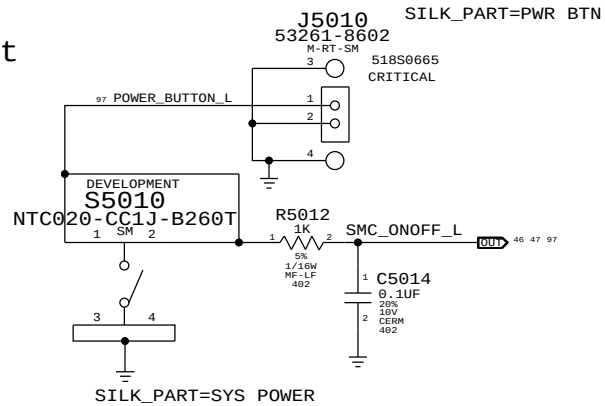
UNUSED TP/NC ALIASES



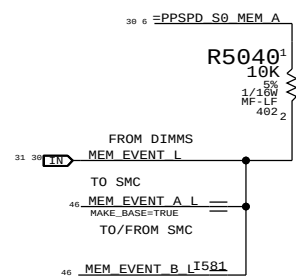
SMC Crystal Circuit



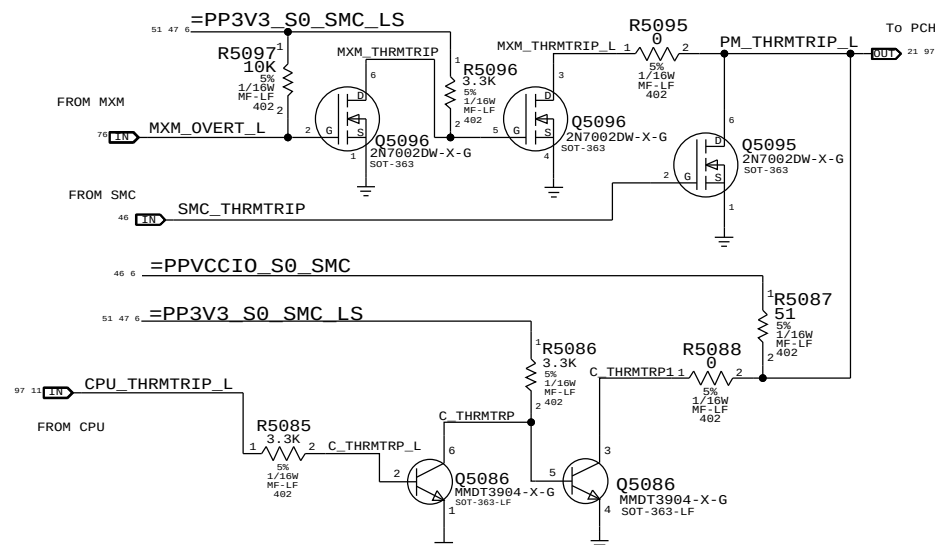
POWER BUTTON



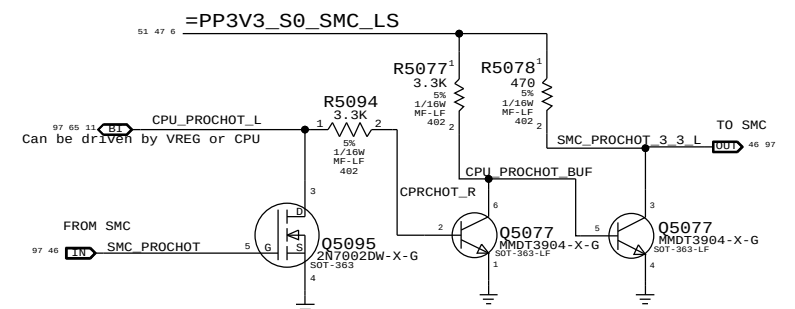
MEM_EVENT

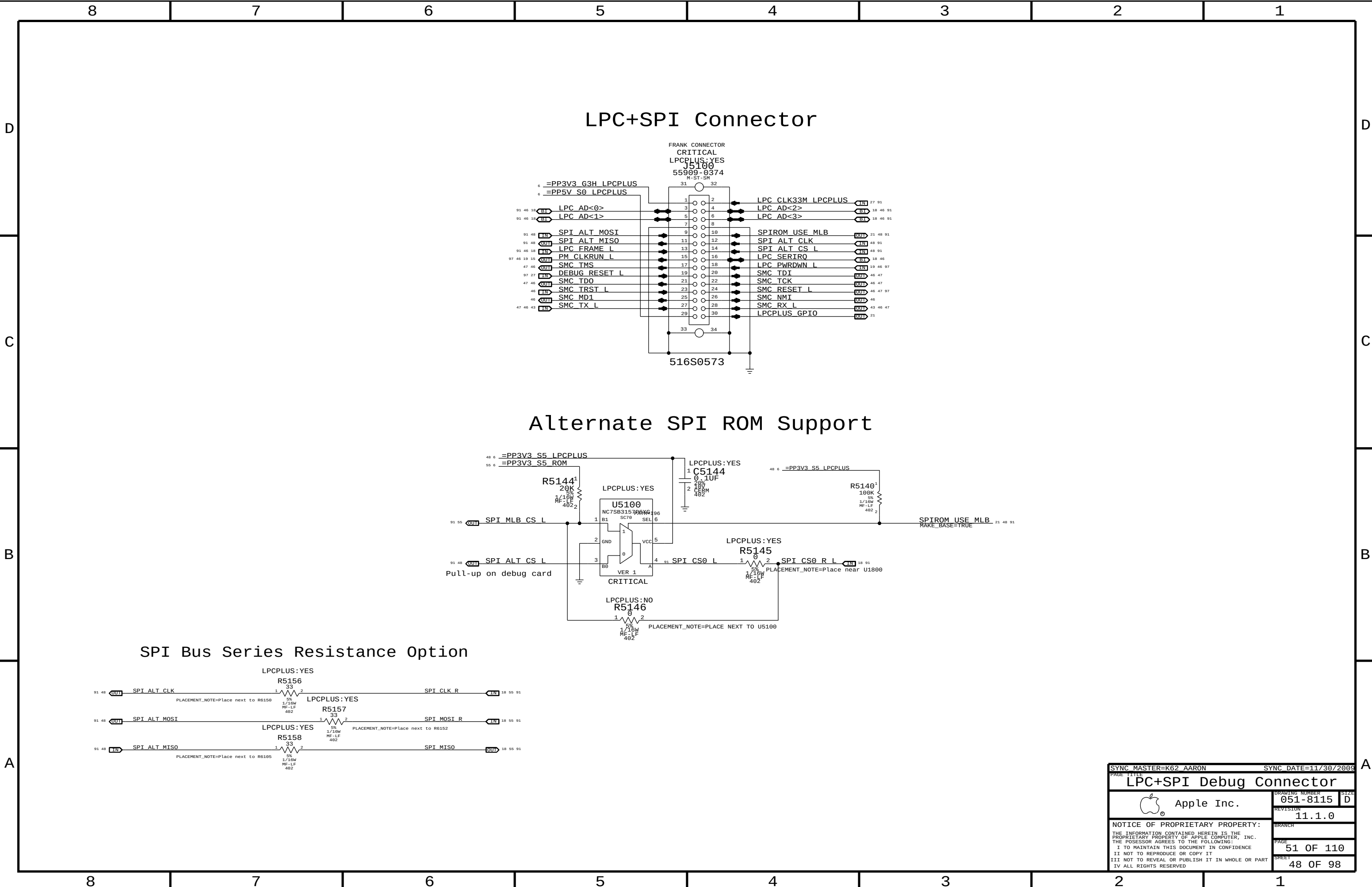


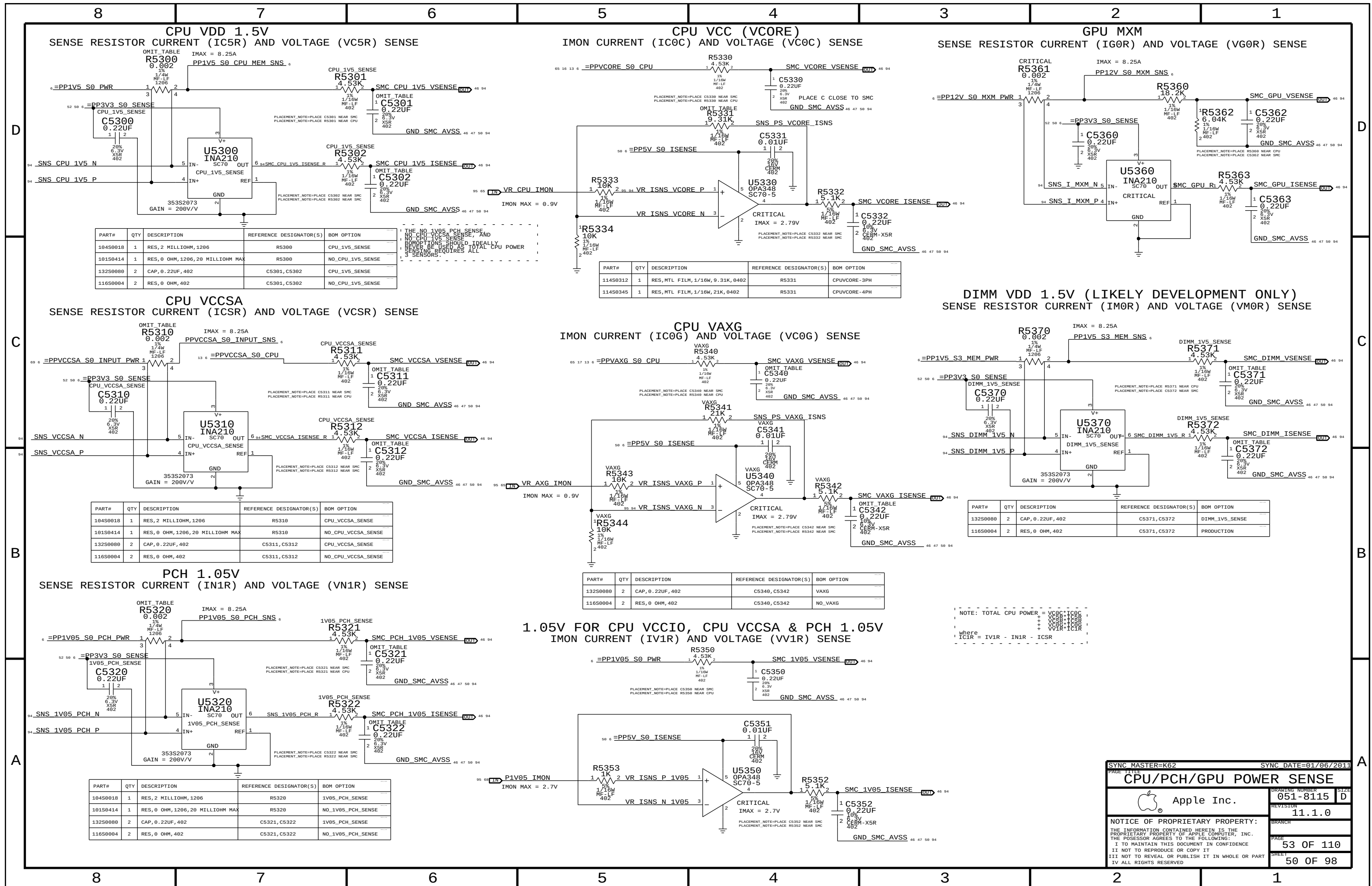
SMC & MXM THERMTRIP LEVEL SHIFTING

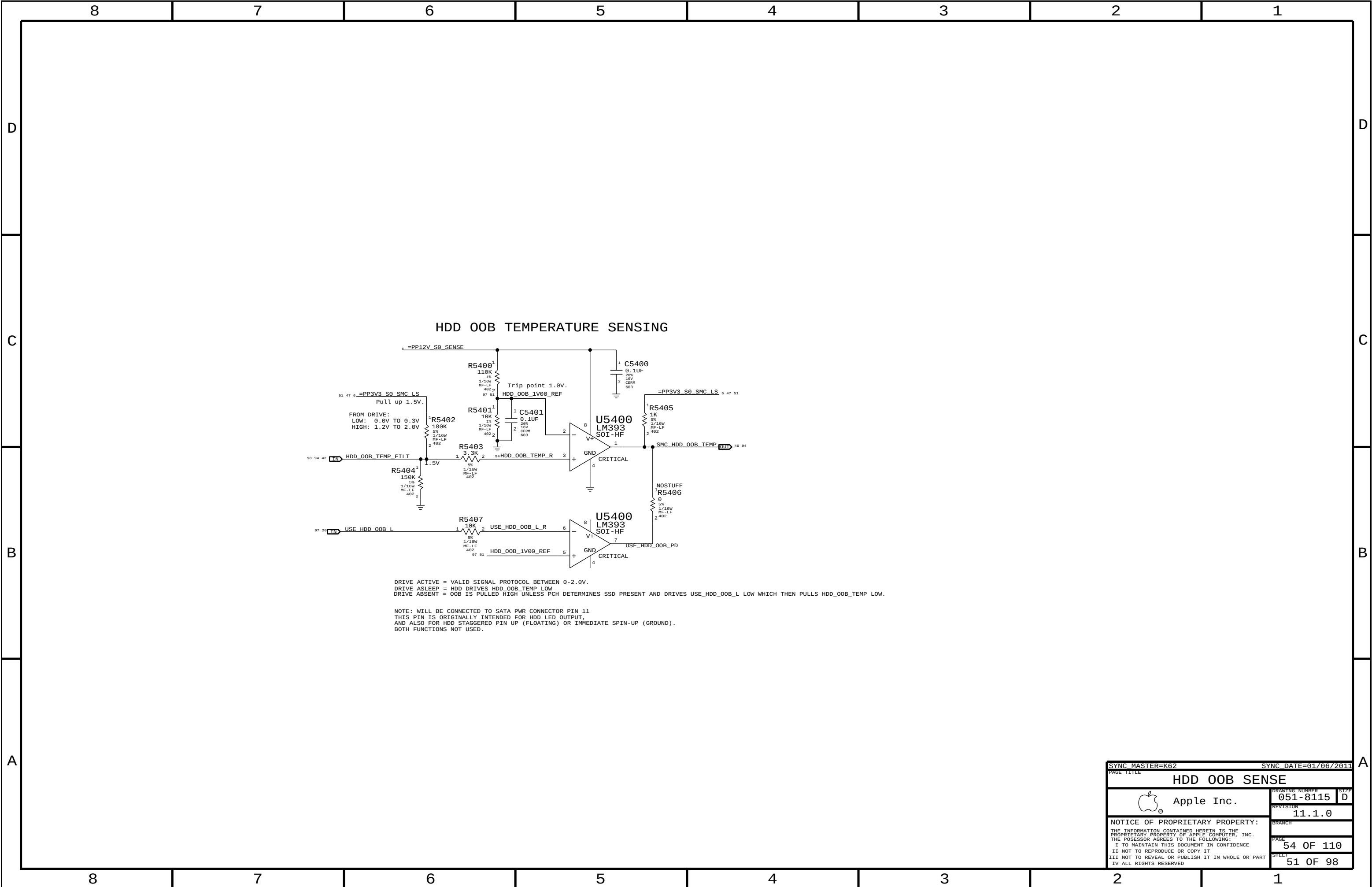


SMC PROCHOT 3.3V LEVEL SHIFTING

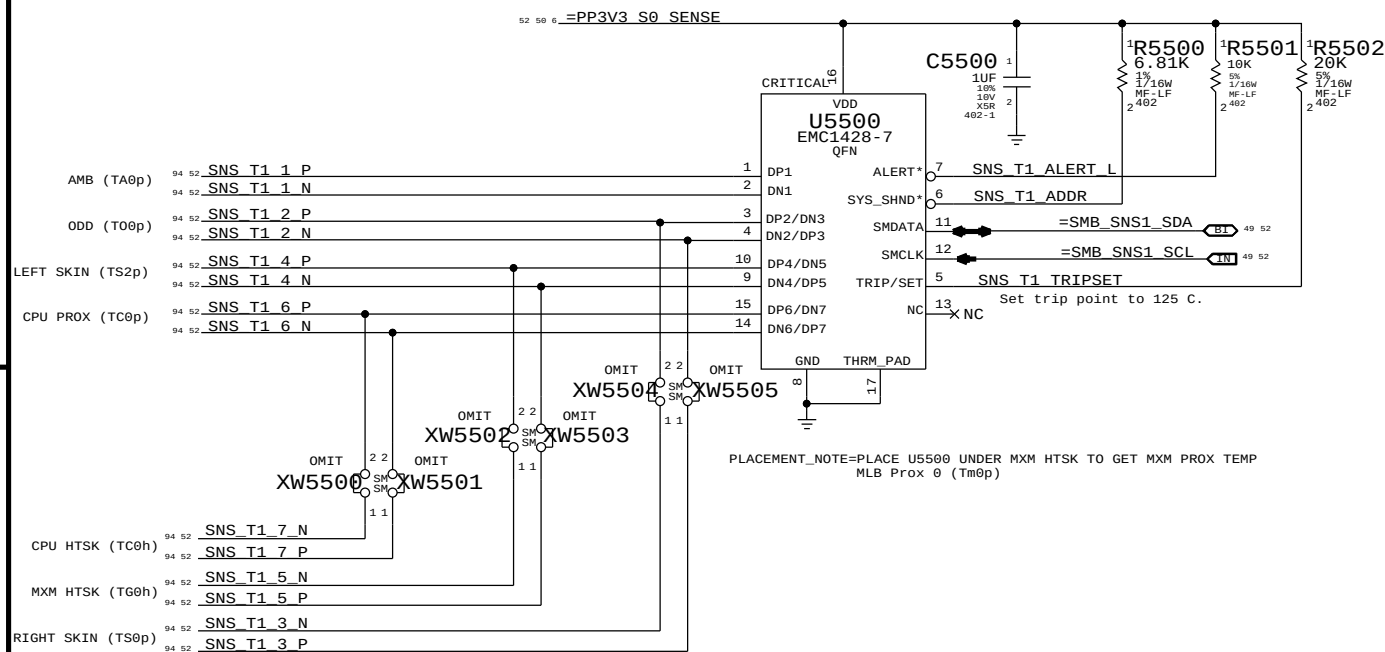






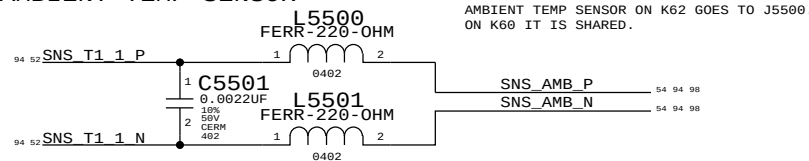


SNS T1: PRODUCTION TEMP SENSOR IC

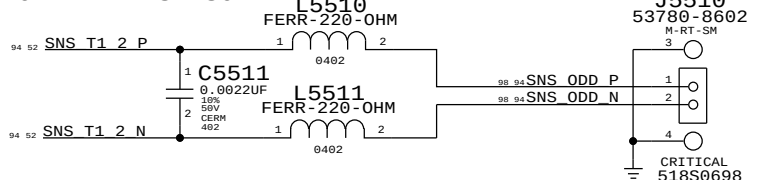


EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

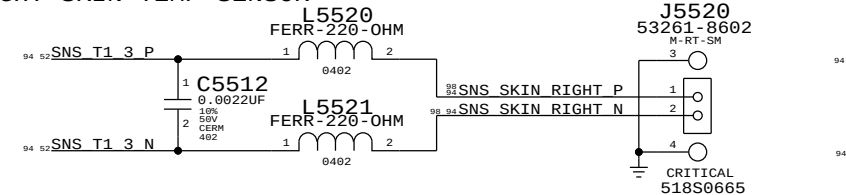
AMBIENT TEMP SENSOR



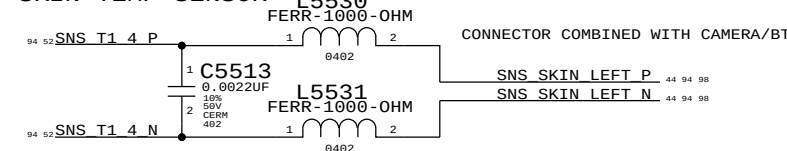
ODD TEMP SENSOR



RIGHT SKIN TEMP SENSOR

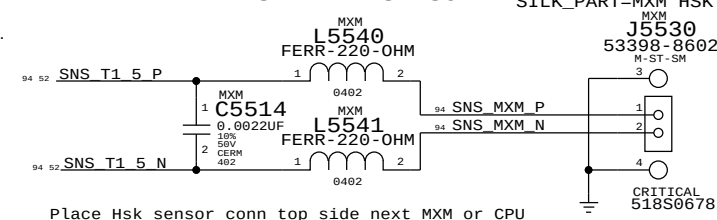


LEFT SKIN TEMP SENSOR

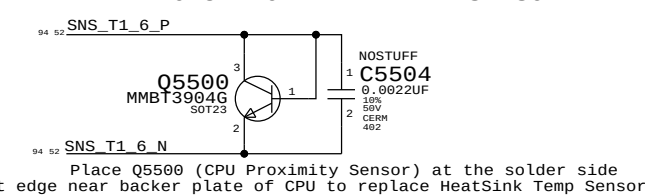


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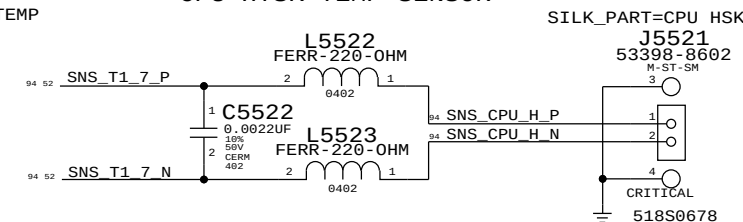
MXM HTSK TEMP SENSOR



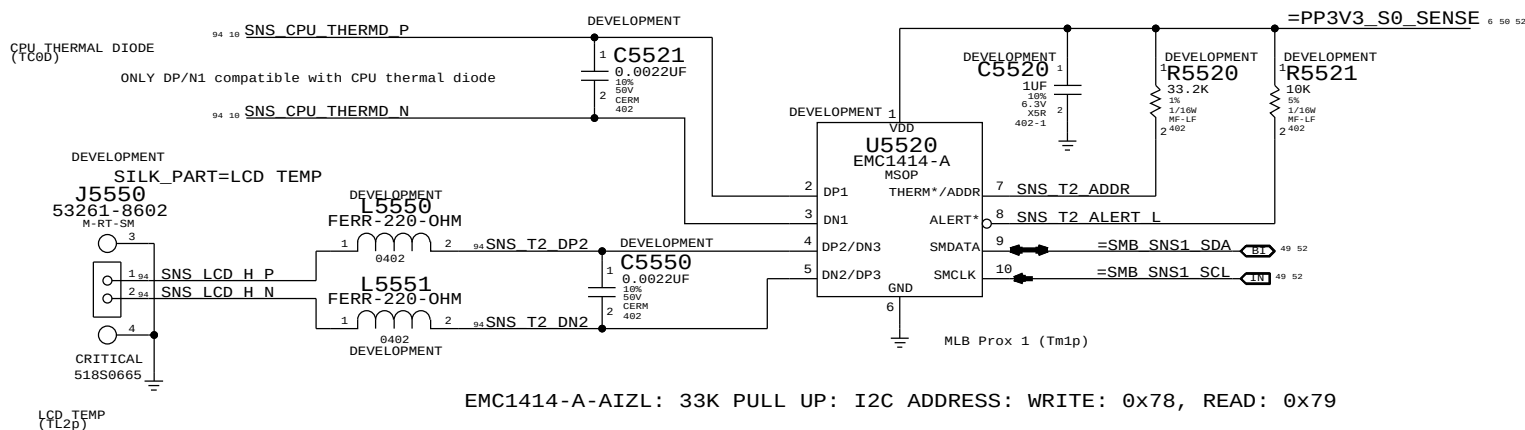
CPU PROXIMITY TEMP SENSOR



CPU HTSK TEMP SENSOR



SNS T2: DEVELOPMENT TEMP SENSOR IC



EMC1414-A-AIZL: 33K PULL UP: I2C ADDRESS: WRITE: 0x78, READ: 0x79

SYNC MASTER=K60 MARK		SYNC DATE=01/06/2011	
PAGE TITLE		TEMP SENSORS	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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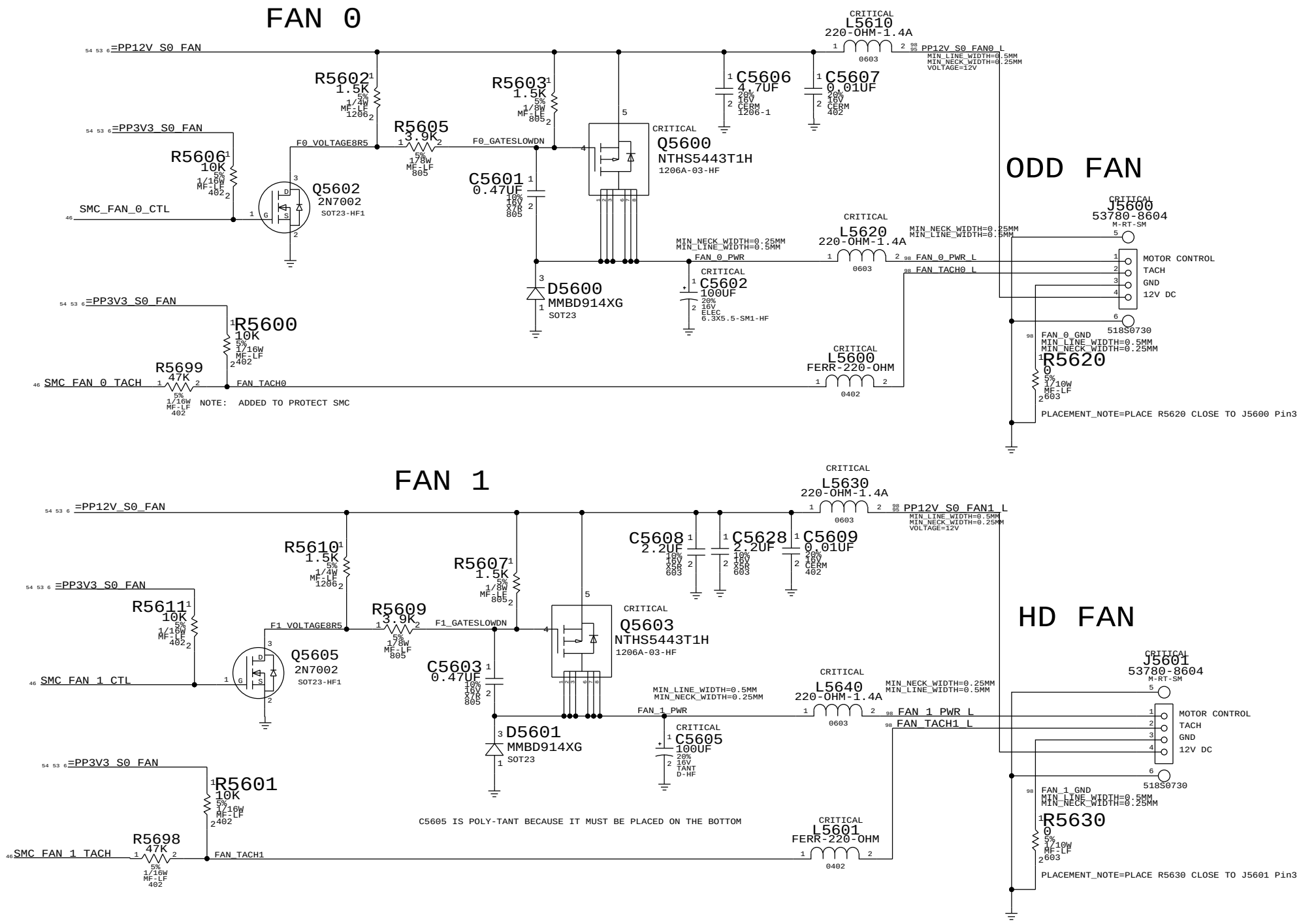
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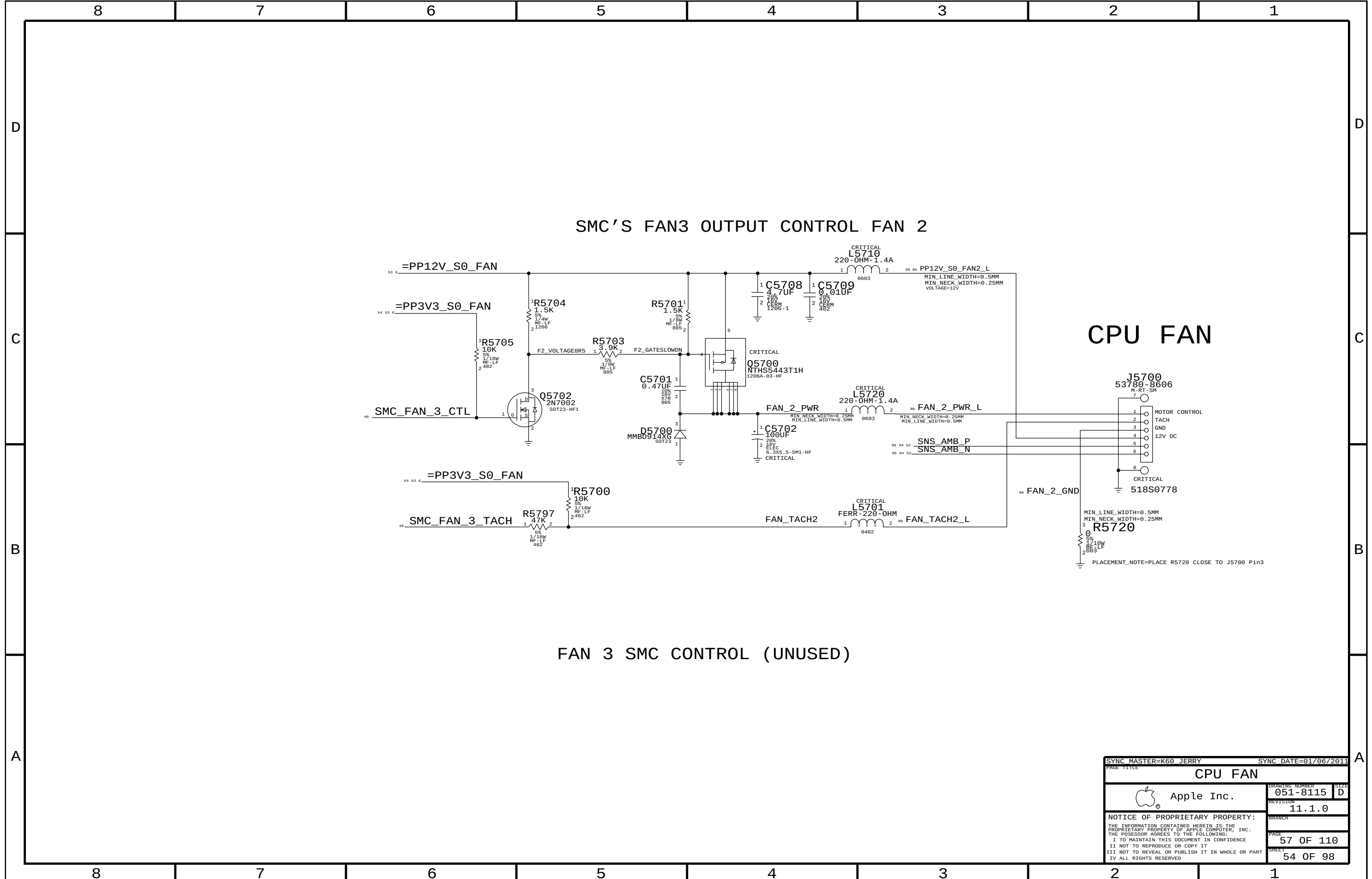
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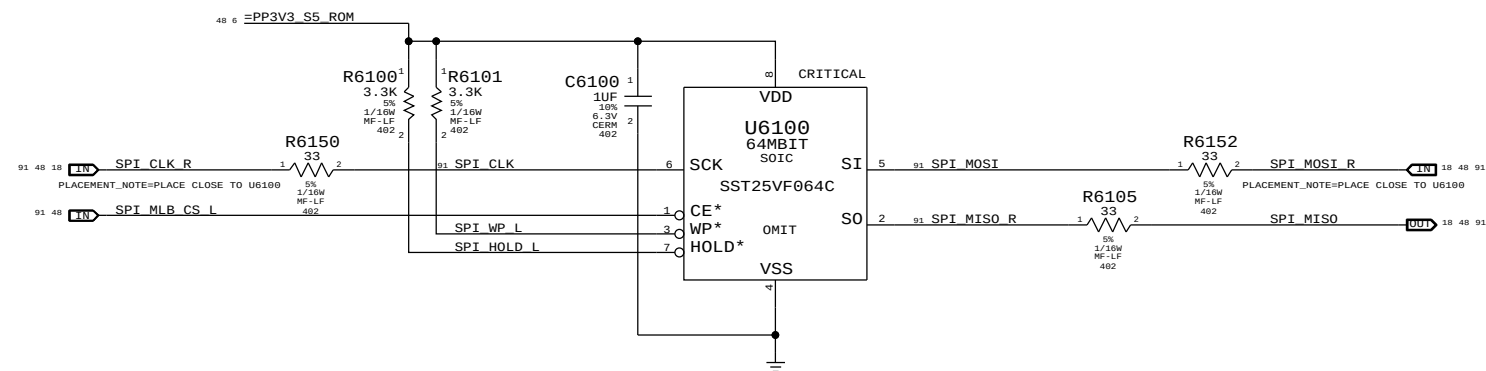
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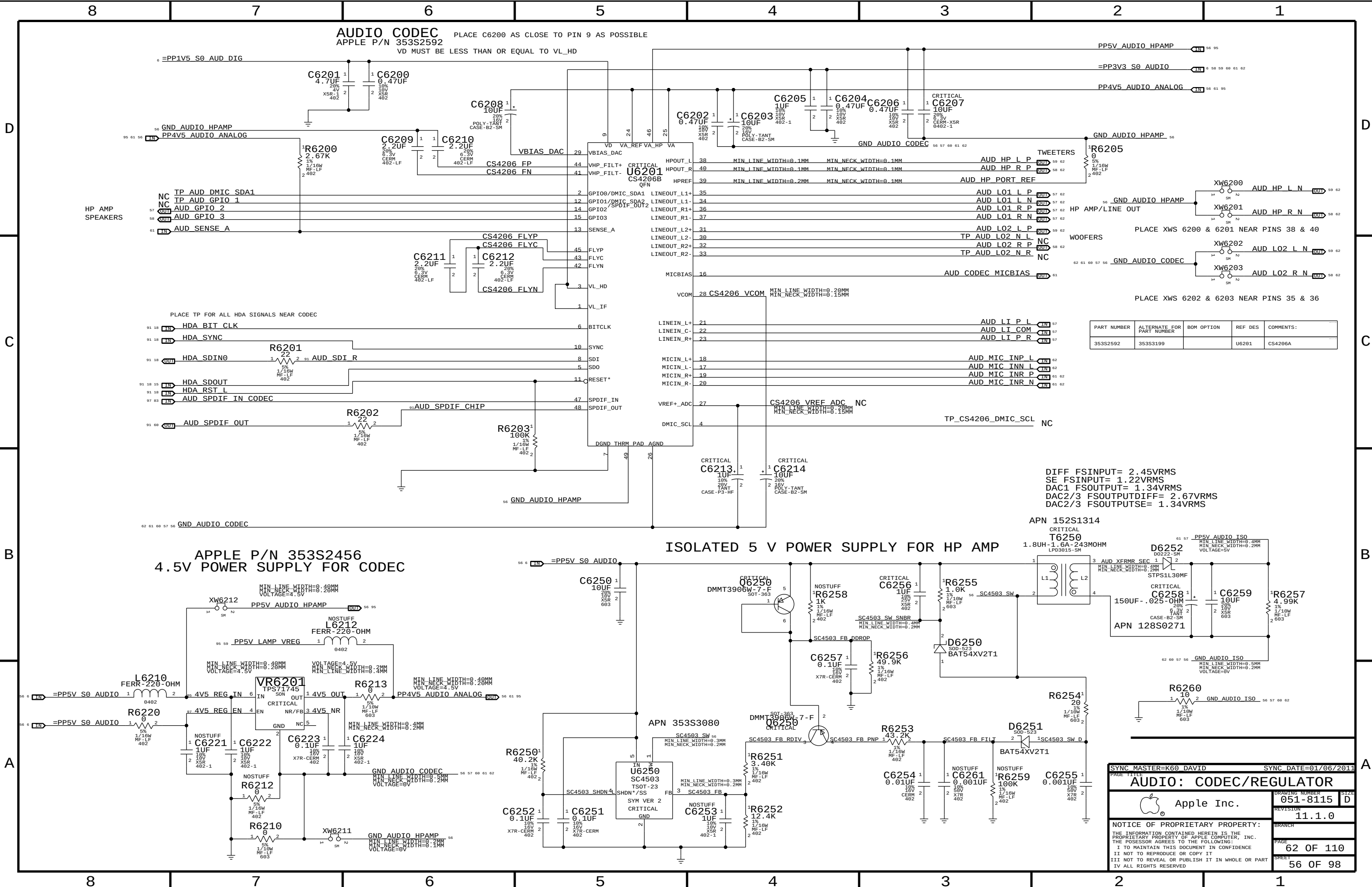
SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		HD AND OD FAN	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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SYNC MASTER=K60 JERRY		SYNC DATE=01/06/2011	
PAGE TITLE		CPU FAN	
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		PAGE	57 OF 110
		SHEET	54 OF 98



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
SPI ROM			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-8115	D
		REVISION	
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		BRANCH	
		PAGE	
		61 OF 110	
		SHEET	
		55 OF 98	



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AUDIO CODEC
APPLE P/N 353S2592
PLACE C6200 AS CLOSE TO PIN 9 AS POSSIBLE
VD MUST BE LESS THAN OR EQUAL TO VL_HD

PP5V_AUDIO_HPAMP
=PP3V3_S0_AUDIO
PP4V5_AUDIO_ANALOG

GND_AUDIO_HPAMP

AUD_HP_L_P
AUD_HP_R_P
AUD_HP_PORT_REF

AUD_L01_L_P
AUD_L01_L_N
AUD_L01_R_P
AUD_L01_R_N

AUD_L02_L_P
TP_AUD_L02_N_L
AUD_L02_R_P
TP_AUD_L02_N_R

AUD_CODEC_MICBIAS

AUD_LI_P_L
AUD_LI_COM
AUD_LI_P_R

AUD_MIC_INP_L
AUD_MIC_INN_L
AUD_MIC_INR_P
AUD_MIC_INR_N

DIFF_FSINPUT= 2.45VRMS
SE_FSINPUT= 1.22VRMS
DAC1_FSOUTPUT= 1.34VRMS
DAC2/3_FSOUTPUTDIFF= 2.67VRMS
DAC2/3_FSOUTPUTSE= 1.34VRMS

APN 152S1314
CRITICAL
T6250
1.8UH-1.6A-243MOHM
LPD3015-SM

D6252
DO222-SM
STPS1L30MF
CRITICAL
C6258
0.025-0HM
APN 128S0271
C6259
10UF
R6257
4.99K

GND_AUDIO_ISO

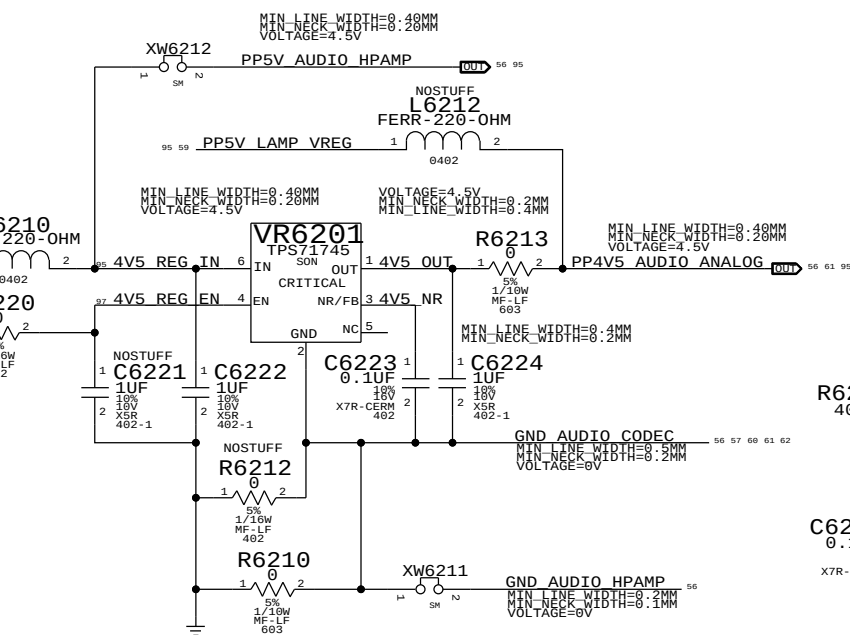
R6260

SYNC MASTER=K60 DAVID
SYNC DATE=01/06/2011

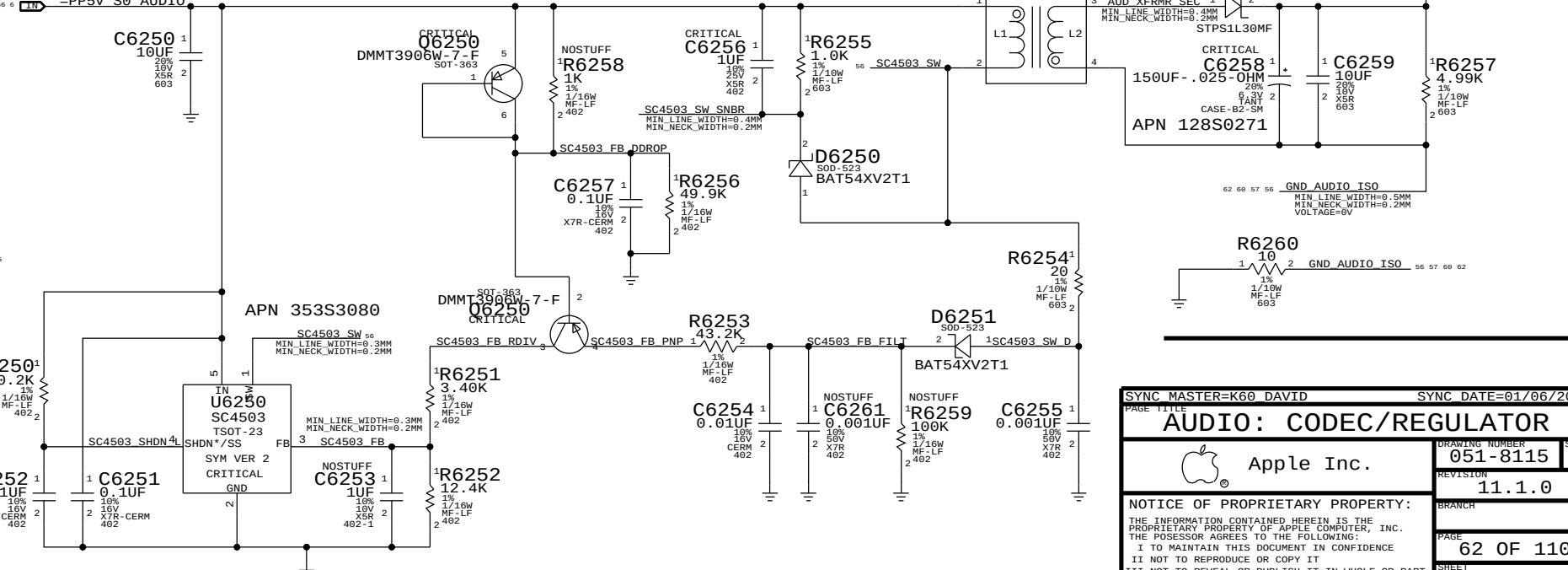
AUDIO: CODEC/REGULATOR

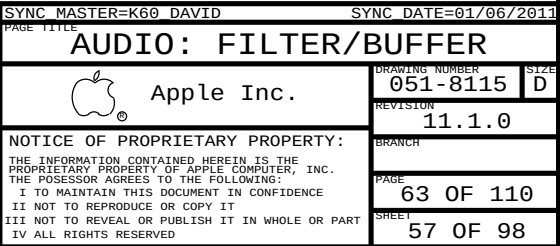
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		PAGE 62 OF 110	SHEET 56 OF 98

APPLE P/N 353S2456
4.5V POWER SUPPLY FOR CODEC



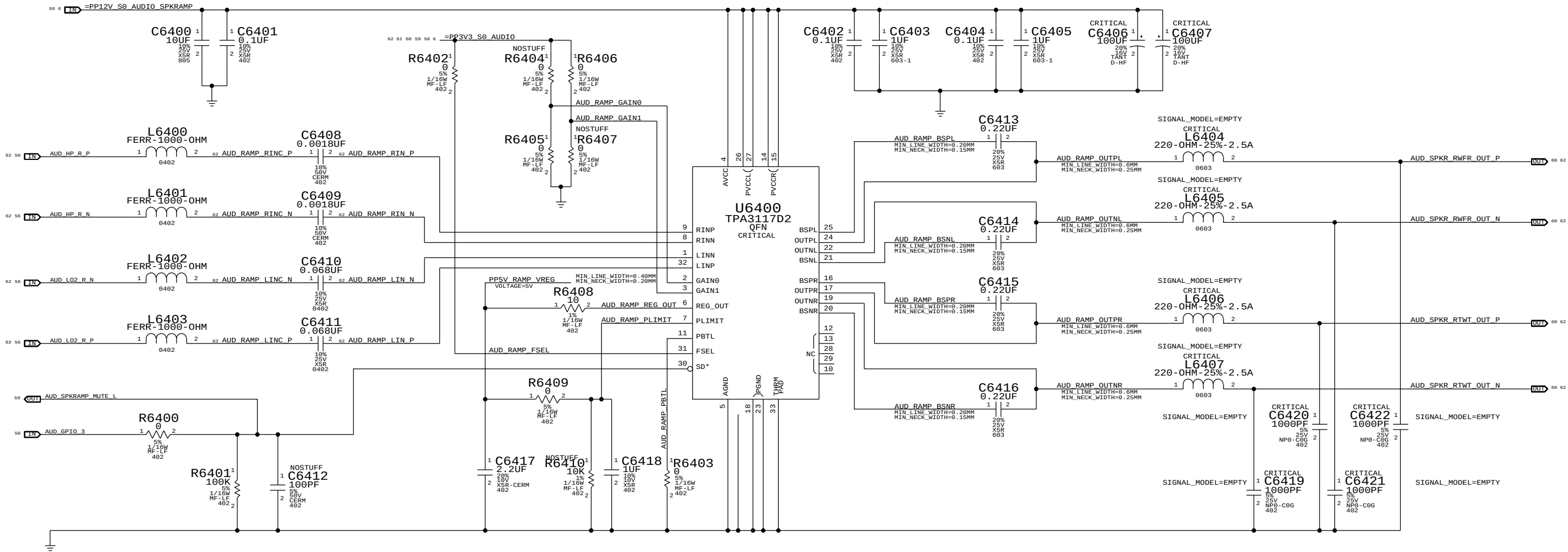
ISOLATED 5 V POWER SUPPLY FOR HP AMP





RIGHT CH SPEAKER AMP
APPLE P/N 353S3069

R6402 = HIGH = 400 KHZ
SPEAKER AMP GAIN = +15.2 DB
SPEAKER AMP RIN = 104K NOMINAL W/ +15.2 DB GAIN
FC HPF, TWEETERS = ~850 HZ (0.0018 UF)
FC HPF, WOOFERS = ~22.5 HZ (0.068 UF)



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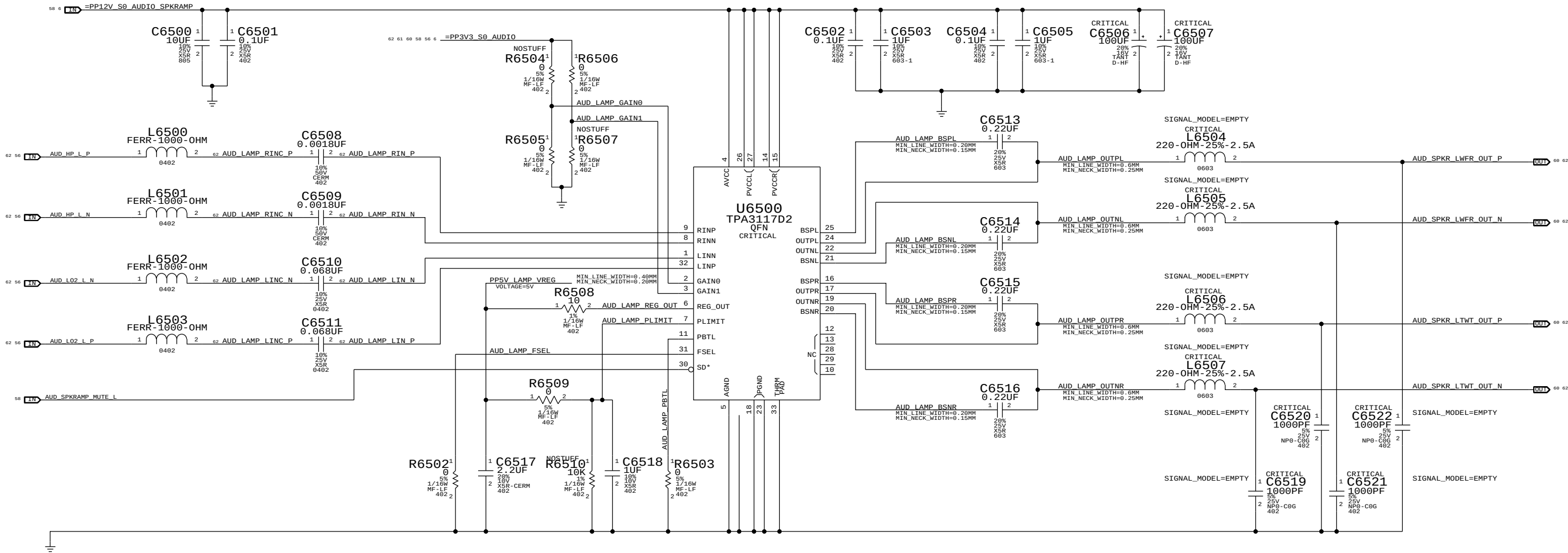
C

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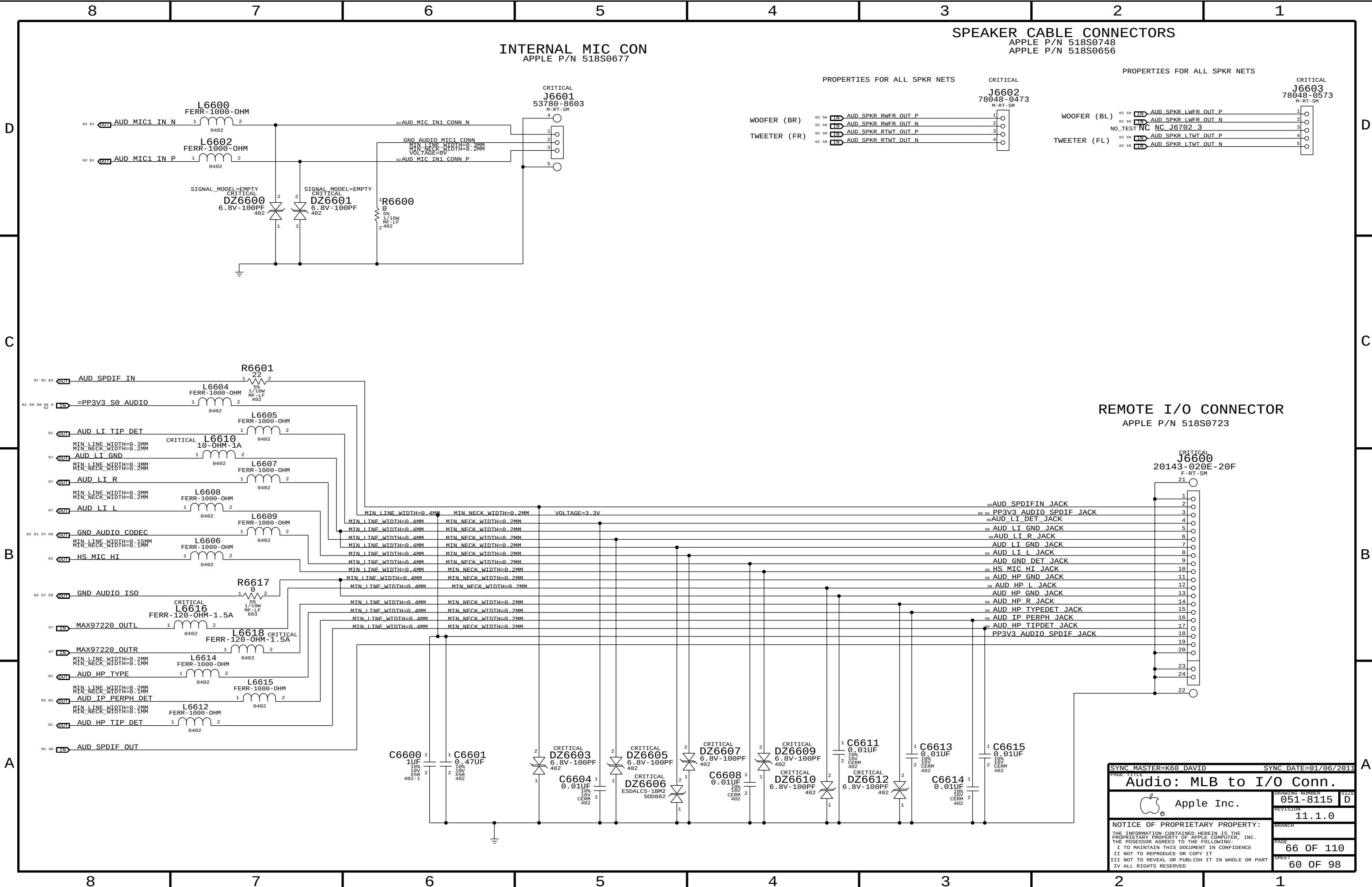
A

LEFT CH SPEAKER AMP
APPLE P/N 353S3069

R6502 = LOW = 300 KHZ
SPEAKER AMP GAIN = +15.2 DB
SPEAKER AMP RIN = 104K NOMINAL W/ +15.2 DB GAIN
FC_HPF, TWEETERS = ~850 HZ (0.0018 UF)
FC_HPF, WOOFERS = ~22.5 HZ (0.068 UF)



SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE			
AUDIO: SPEAKER AMP			
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INTERNAL MIC CON
APPLE P/N 518S0677

SPEAKER CABLE CONNECTORS

APPLE P/N 518S0748
APPLE P/N 518S0656

PROPERTIES FOR ALL SPKR NETS

CRITICAL

PROPERTIES FOR ALL SPKR NETS

CRITICAL

WOOFER (BR)
TWEETER (FR)

WOOFER (BL)
TWEETER (FL)

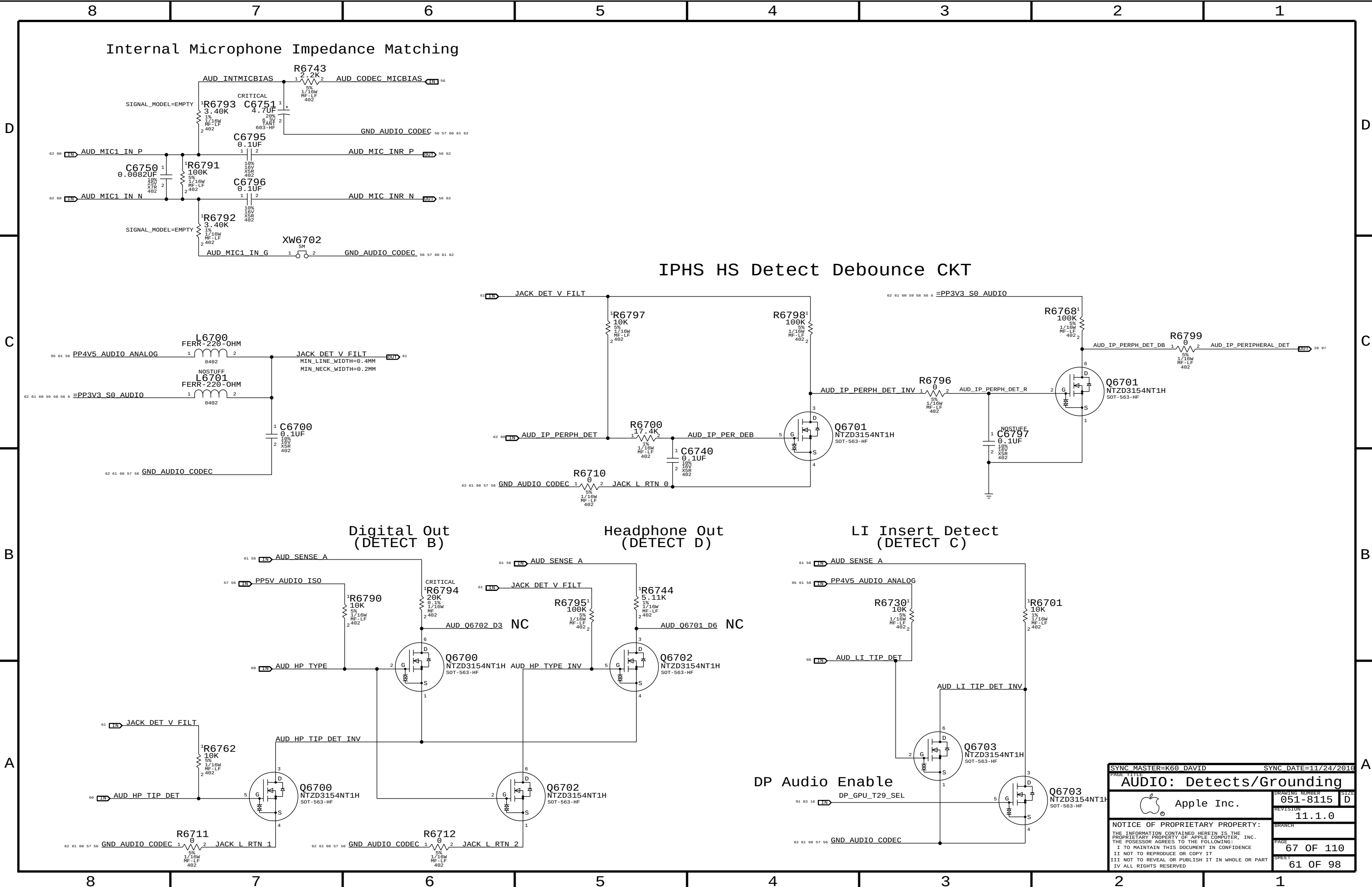
AUD_SPKR_RWER_OUT_P
AUD_SPKR_RWER_OUT_N
AUD_SPKR_RTWT_OUT_P
AUD_SPKR_RTWT_OUT_N

AUD_SPKR_LWER_OUT_P
AUD_SPKR_LWER_OUT_N
AUD_SPKR_LTWT_OUT_P
AUD_SPKR_LTWT_OUT_N

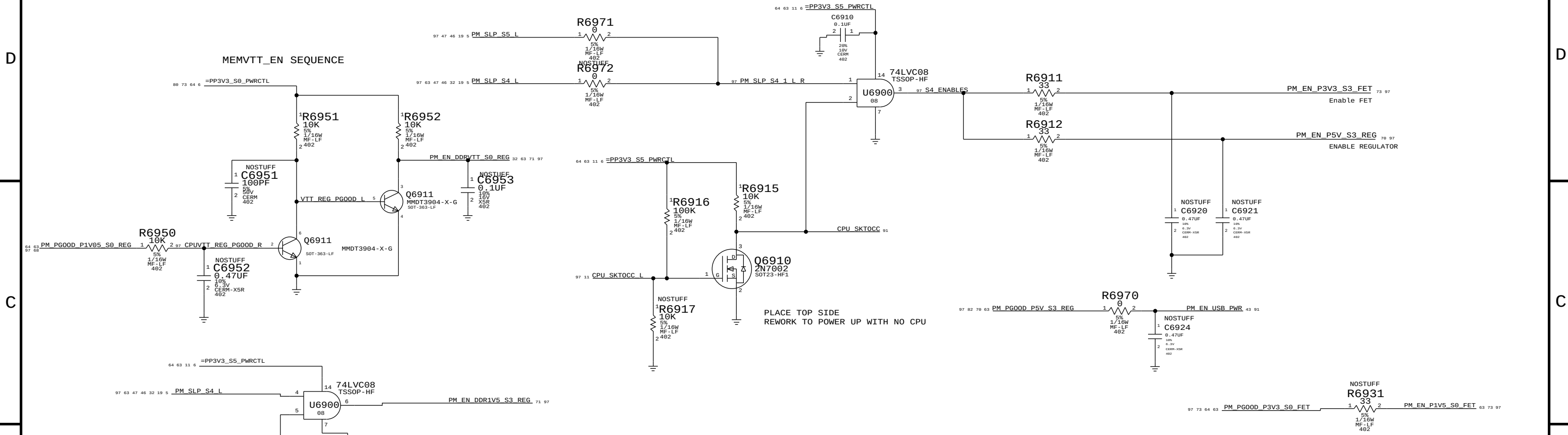
REMOTE I/O CONNECTOR

APPLE P/N 518S0723

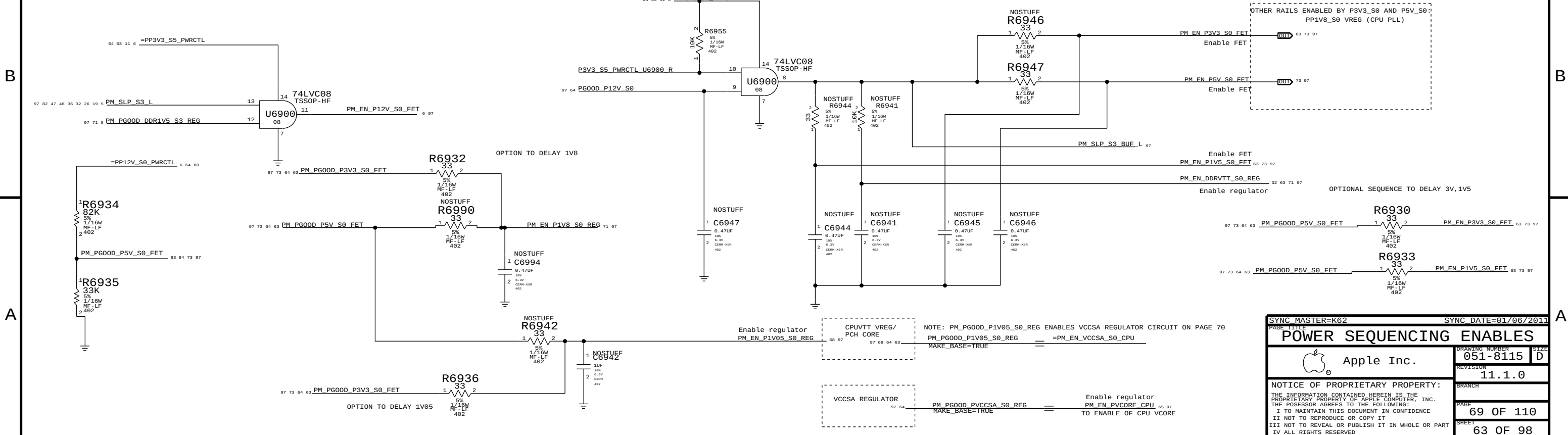
SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE		Audio: MLB to I/O Conn.	
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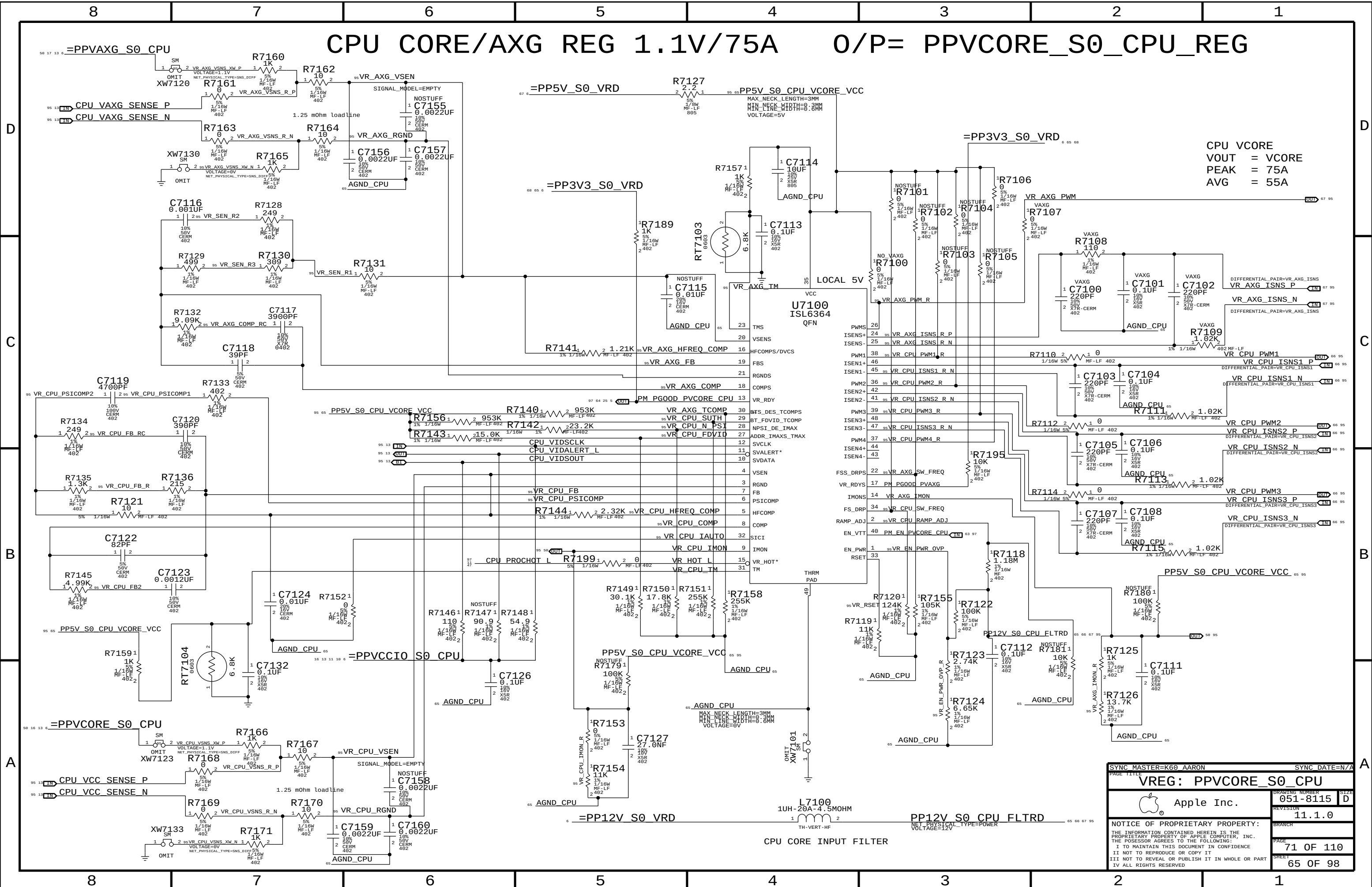
SLP_S4 ENABLES



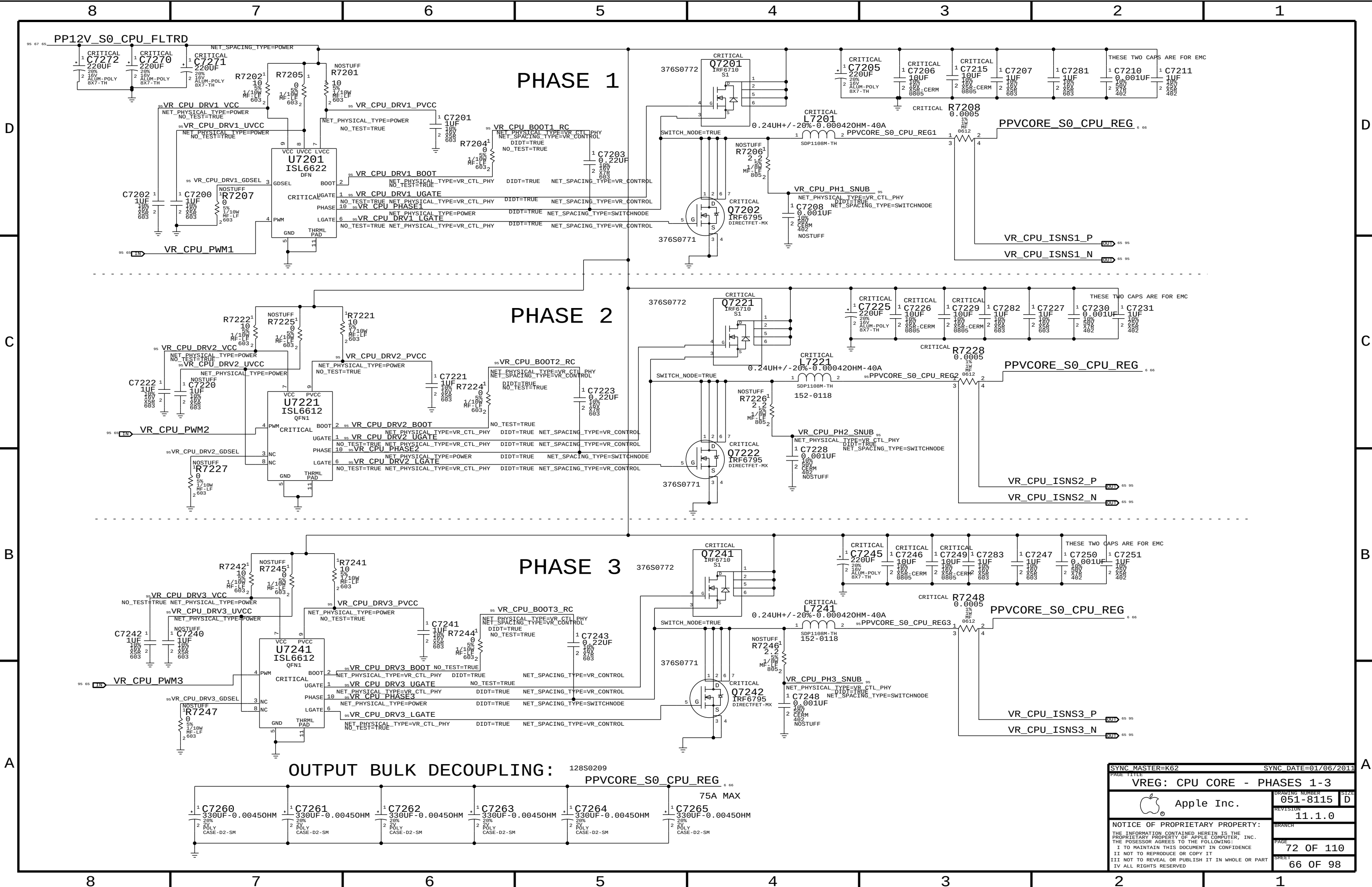
SLP_S3 ENABLES



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
POWER SEQUENCING ENABLES			
 Apple Inc.		DRAWING NUMBER	051-8115
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SYNC MASTER=K60 AARON		SYNC DATE=N/A	
PAGE TITLE			
VREG: PPVCORE_S0_CPU		DRAWING NUMBER	
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		PAGE	71 OF 110
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SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		VREG: CPU CORE - PHASES 1-3	
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		PAGE	72 OF 110
		SHEET	66 OF 98

AXG PHASE (MAX 15A)

Component Values Table:

Part	Value	Notes
R7301	5% 1/10W 0.603	
C7301	1uF 18V X5R CERM 0.603	
C7303	0.22uF 18V X7R 0.603	
C7304	1uF 18V X5R CERM 0.603	
C7305	0.001uF 28V X5R CERM 0.603	
C7306	2.2M 1/8W 0.603	
C7307	1uF 18V X5R CERM 0.603	
C7310	1uF 18V X5R CERM 0.603	
C7315	10uF 18V X5R CERM 0.603	
C7320	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7321	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7322	10uF 28V X5R CERM 0.603	
C7323	10uF 28V X5R CERM 0.603	
R7304	5% 1/10W 0.603	
R7305	5% 1/10W 0.603	
R7306	2.2M 1/8W 0.603	
R7308	0.0005 1W 0.612	

Table 1: Component Values

Part	Value	Notes
R7301	5% 1/10W 0.603	
C7301	1uF 18V X5R CERM 0.603	
C7303	0.22uF 18V X7R 0.603	
C7304	1uF 18V X5R CERM 0.603	
C7305	0.001uF 28V X5R CERM 0.603	
C7306	2.2M 1/8W 0.603	
C7307	1uF 18V X5R CERM 0.603	
C7310	1uF 18V X5R CERM 0.603	
C7315	10uF 18V X5R CERM 0.603	
C7320	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7321	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7322	10uF 28V X5R CERM 0.603	
C7323	10uF 28V X5R CERM 0.603	
R7304	5% 1/10W 0.603	
R7305	5% 1/10W 0.603	
R7306	2.2M 1/8W 0.603	
R7308	0.0005 1W 0.612	

Table 2: Component Values

Part	Value	Notes
R7301	5% 1/10W 0.603	
C7301	1uF 18V X5R CERM 0.603	
C7303	0.22uF 18V X7R 0.603	
C7304	1uF 18V X5R CERM 0.603	
C7305	0.001uF 28V X5R CERM 0.603	
C7306	2.2M 1/8W 0.603	
C7307	1uF 18V X5R CERM 0.603	
C7310	1uF 18V X5R CERM 0.603	
C7315	10uF 18V X5R CERM 0.603	
C7320	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7321	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7322	10uF 28V X5R CERM 0.603	
C7323	10uF 28V X5R CERM 0.603	
R7304	5% 1/10W 0.603	
R7305	5% 1/10W 0.603	
R7306	2.2M 1/8W 0.603	
R7308	0.0005 1W 0.612	

Table 3: Component Values

Part	Value	Notes
R7301	5% 1/10W 0.603	
C7301	1uF 18V X5R CERM 0.603	
C7303	0.22uF 18V X7R 0.603	
C7304	1uF 18V X5R CERM 0.603	
C7305	0.001uF 28V X5R CERM 0.603	
C7306	2.2M 1/8W 0.603	
C7307	1uF 18V X5R CERM 0.603	
C7310	1uF 18V X5R CERM 0.603	
C7315	10uF 18V X5R CERM 0.603	
C7320	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7321	330uF-0.00450HM 20% 2V POLY CASE-D2-SM	
C7322	10uF 28V X5R CERM 0.603	
C7323	10uF 28V X5R CERM 0.603	
R7304	5% 1/10W 0.603	
R7305	5% 1/10W 0.603	
R7306	2.2M 1/8W 0.603	
R7308	0.0005 1W 0.612	

Table 4: Component Values

Part	Value	Notes
R7301	5% 1/10W 0.603	
C7301	1uF 18V X5R CERM 0.603	
C7303	0.22uF 18V X7R 0.603	
C7304	1uF 18V X5R CERM 0.603	
C7305	0.001uF 28V X5R CERM 0.603	
C7306	2.2M 1/8W 0.603	

AXG PHASE (MAX 15A)

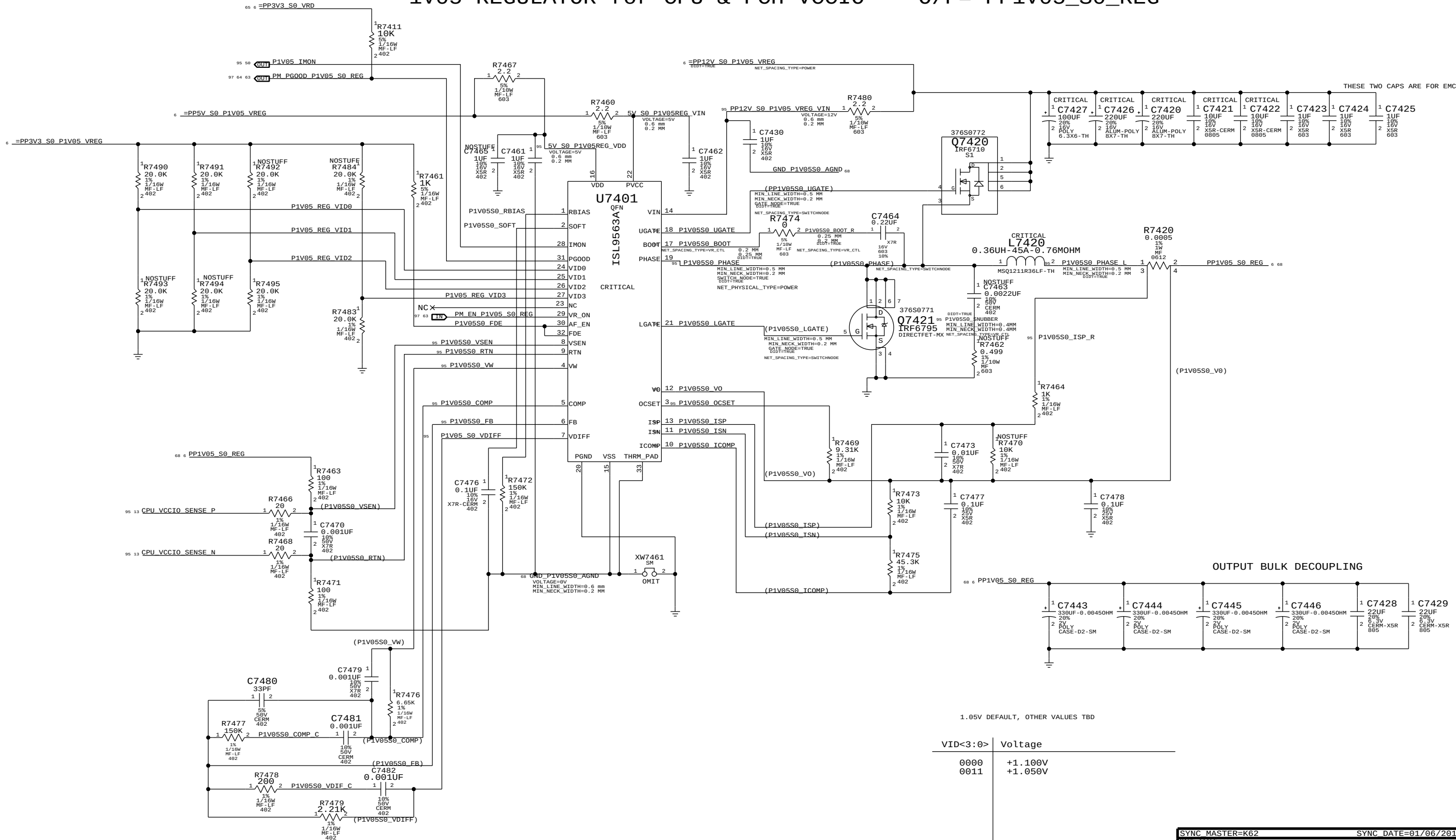
The schematic diagram illustrates the AXG PHASE (MAX 15A) power regulation circuit. Key components and connections include:

- Bootstrap Driver:** U7301 ISL6620 QFN, connected to VR_AXG_PWM, VR_AXG_DRV1_PVCC, and VR_AXG_DRV1_B00T.
- MOSFET Driver:** 07301 CSD58864Q5D SON5X6, connected to VR_AXG_DRV1_UGATE, VR_AXG_DRV1_UGATE_R, and VR_AXG_DRV1_LGATE.
- Power MOSFET:** 376S0906 CRITICAL, connected to PP12V_S0_CPU_FLTRD, VR_AXG_DRV1_UGATE, and VR_AXG_DRV1_LGATE.
- Synchronous Rectifier:** L7301 CRITICAL, connected to PPVAXG_S0_REG and VR_AXG_DRV1_LGATE.
- Capacitors:** C7301, C7303, C7304, C7305, C7306, C7307, C7310, C7315, C7320, C7321, C7322, C7323.
- Resistors:** R7301, R7304, R7305, R7306, R7308.
- Power Sources:** PP5V_S0_VRD, PP12V_S0_CPU_FLTRD.
- Output Regulation:** PPVAXG_S0_REG, VR_AXG_ISNS_P, VR_AXG_ISNS_N.

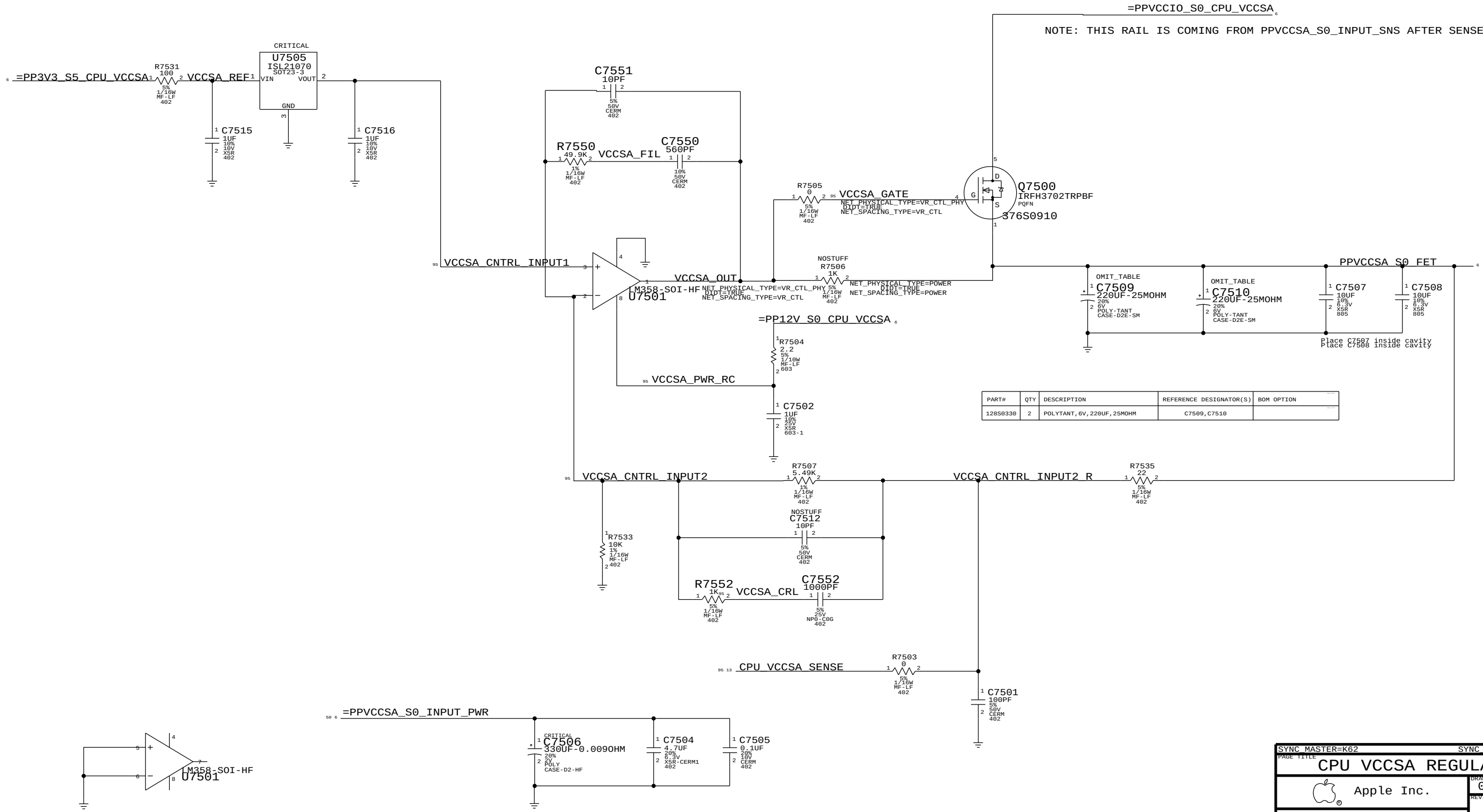
The diagram is labeled with various net names and component values, indicating a detailed and complex circuit design.

1V05 REGULATOR for CPU & PCH VCCIO

O/P= PP1V05_S0_REG



CPU VCCSA 0.925V (8.8A MAX)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0330	2	POLYTANT, 6V, 220UF, 25MOHM	C7509, C7510	

SYNC MASTER=K62		SYNC DATE=11/15/2016	
PAGE TITLE			
CPU VCCSA REGULATOR			
 Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
		BRANCH	
		PAGE	75 OF 110
		SHEET	69 OF 98
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3V3 S5 REGULATOR

5V S3 REGULATOR

Power Rating ?

5V OUTPUT

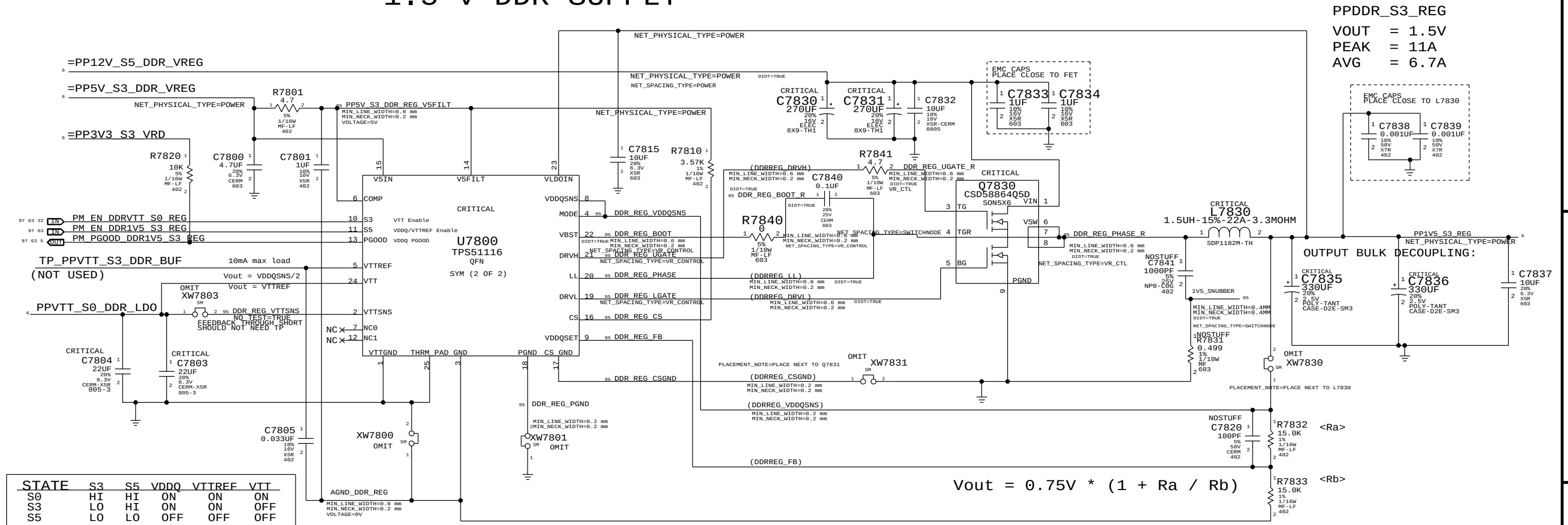
3V3 OUTPUT

OUTPUT BULK DECOUPLING:

$$V_{out} = 0.6V * (1 + R_a / R_b)$$

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		5V_S3 / 3V3_S5 VREGS	
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1.5 V DDR SUPPLY

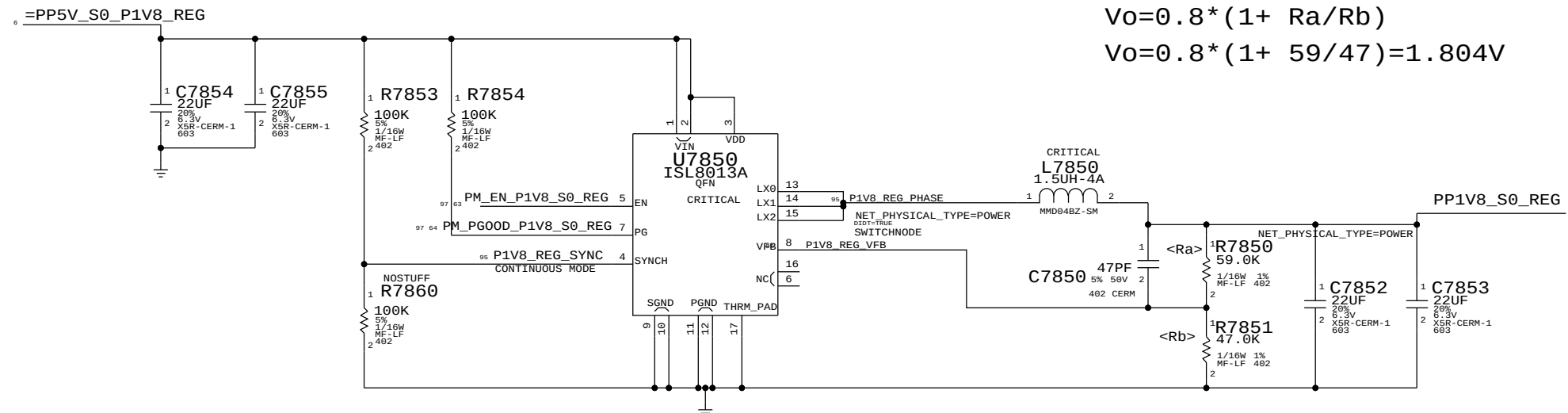


1.8 V SUPPLY

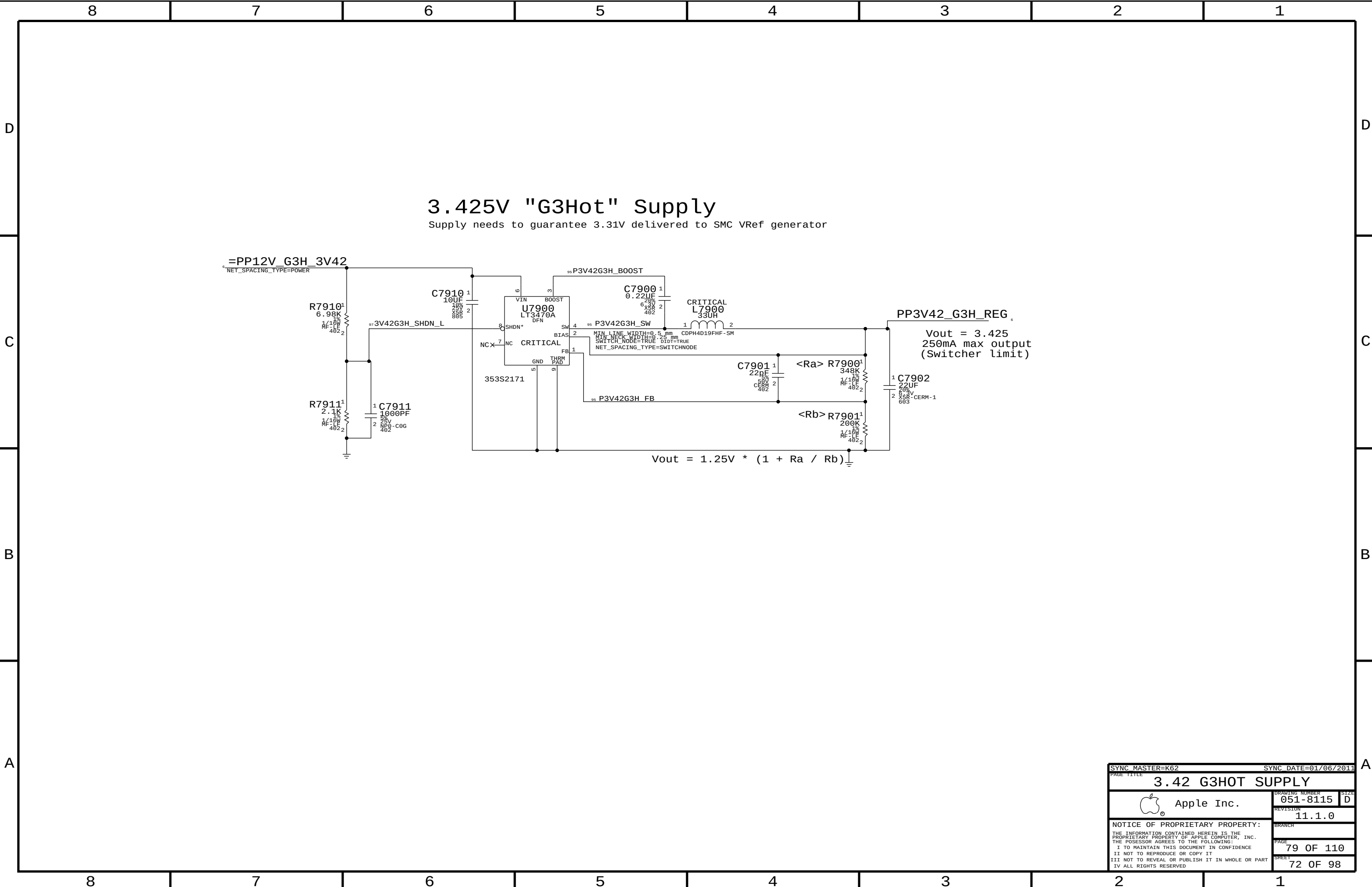
1A Average current

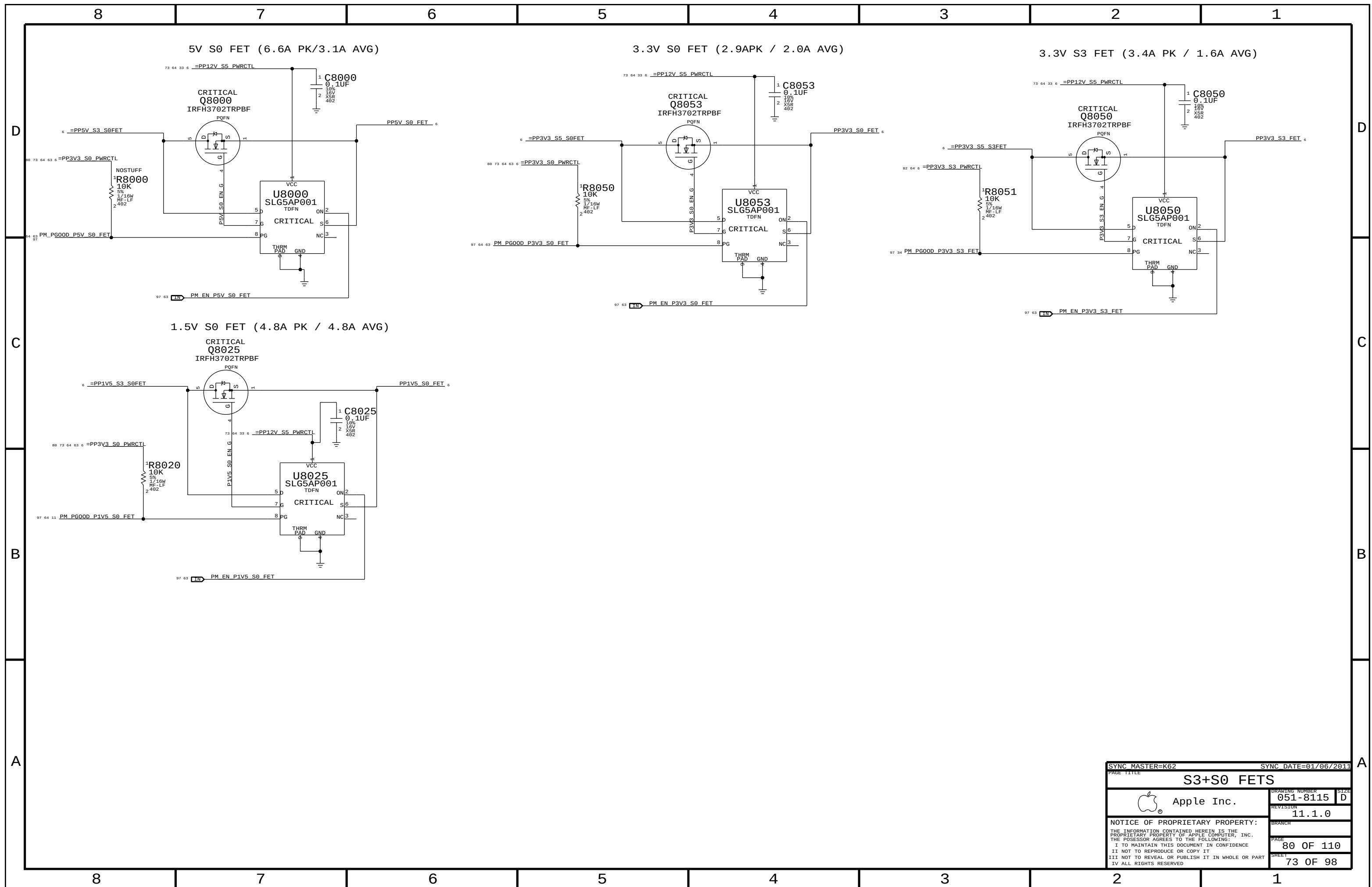
$$Vo = 0.8 * (1 + Ra/Rb)$$

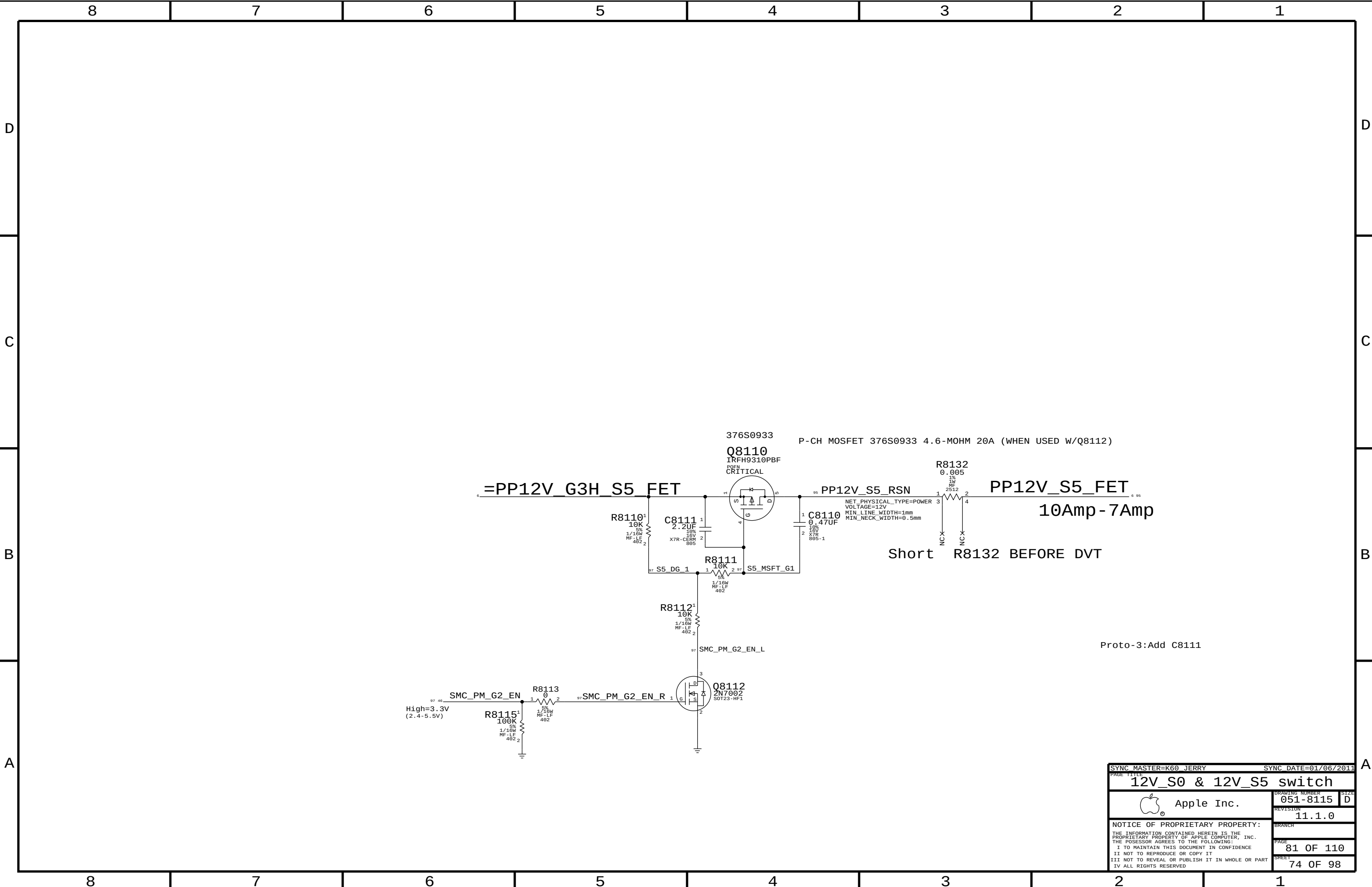
$$Vo = 0.8 * (1 + 59/47) = 1.804V$$



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		1.5V / 1.8V VREGS	
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SYNC MASTER=K60_JERRY		SYNC DATE=01/06/2011	
PAGE TITLE			
12V_S0 & 12V_S5 switch			
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		REVISION	11.1.0
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		PAGE	81 OF 110
		SHEET	74 OF 98

Power aliases required by this page:

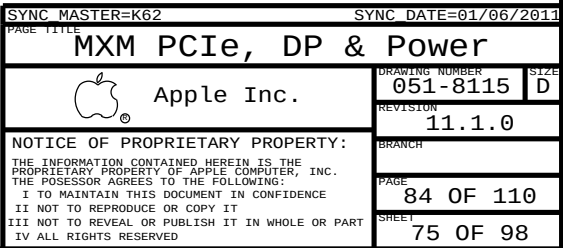
- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

- MXM



Page Notes

Power aliases required by this page:

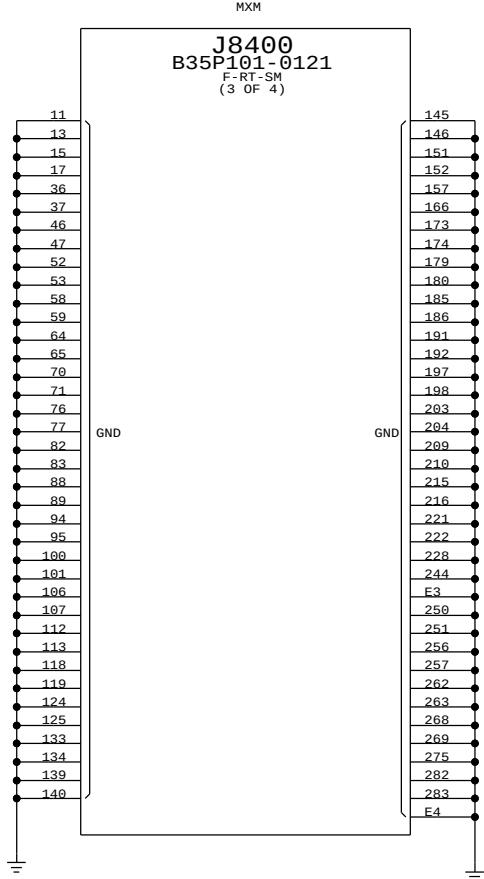
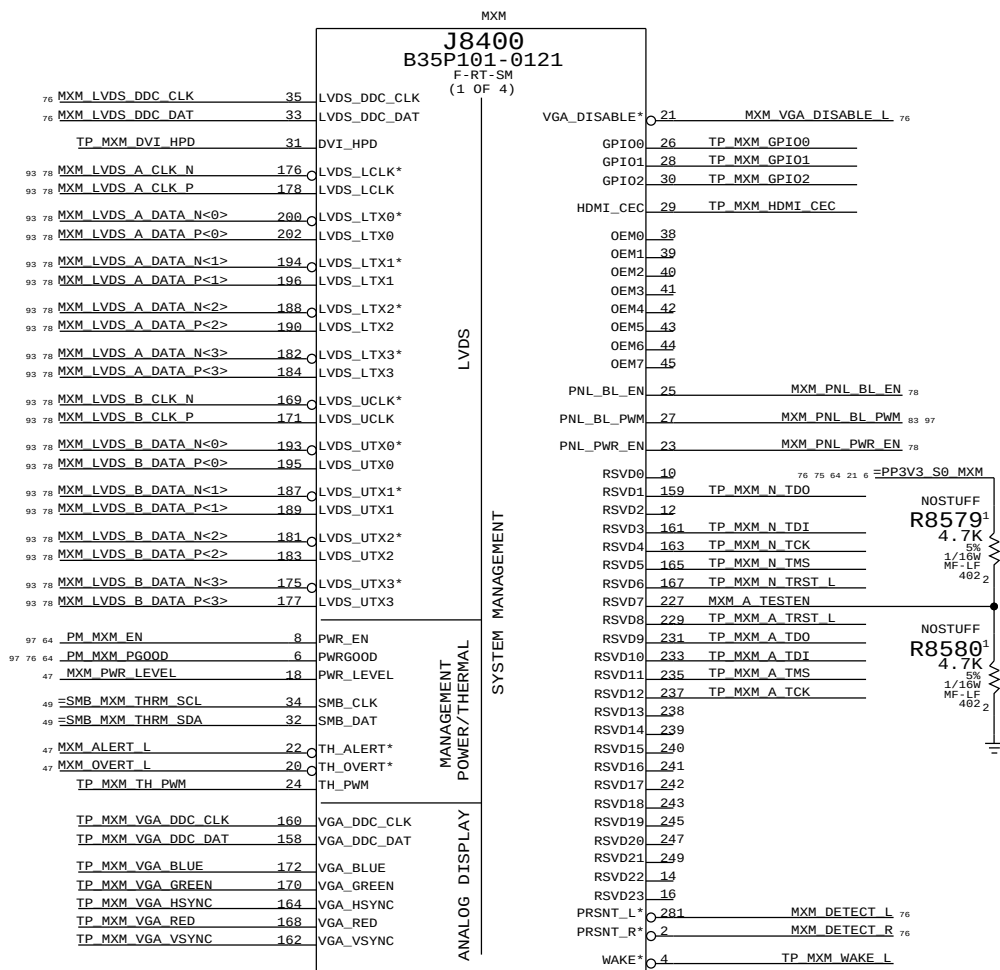
- =PP3V3_S0_MXM

Signal aliases required by this page:

- =SMB_MXM_THRM_DATA - =PM_MXM_PG00D_PULLUP
- =SMB_MXM_THRM_CLK

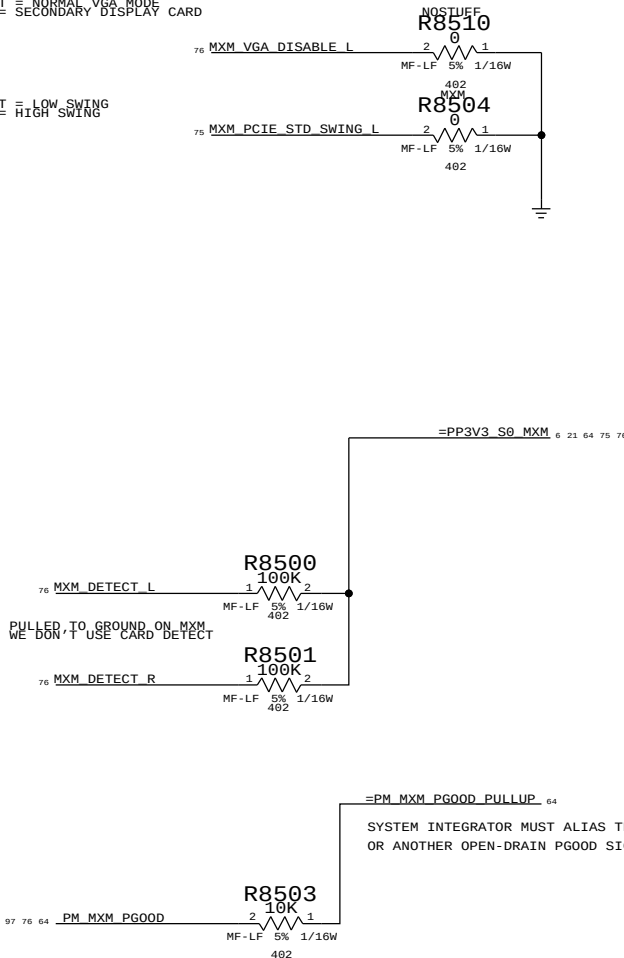
BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR



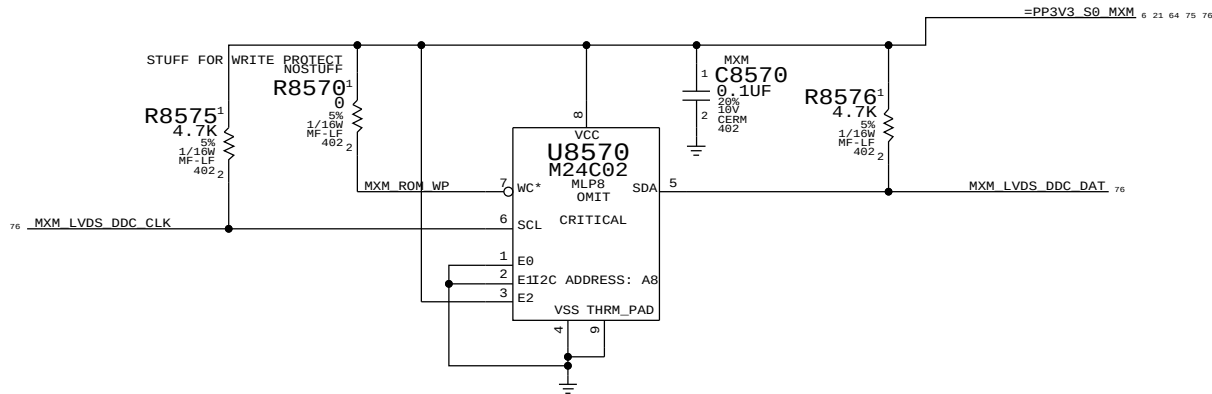
FLOAT = NORMAL VGA_MODE
GND = SECONDARY DISPLAY CARD

FLOAT = LOW SWING
GND = HIGH SWING



MXM SYSTEM INFORMATION ROM

PLACE CLOSE TO J8400



PAGE TITLE		SYNC MASTER=K60 MASTER		SYNC DATE=N/A	
MXM I/O		DRAWING NUMBER		SIZE	
Apple Inc.		051-8115		D	
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		PAGE		SHEET	
		85 OF 110		76 OF 98	

GreenCLK Implementation Notes:

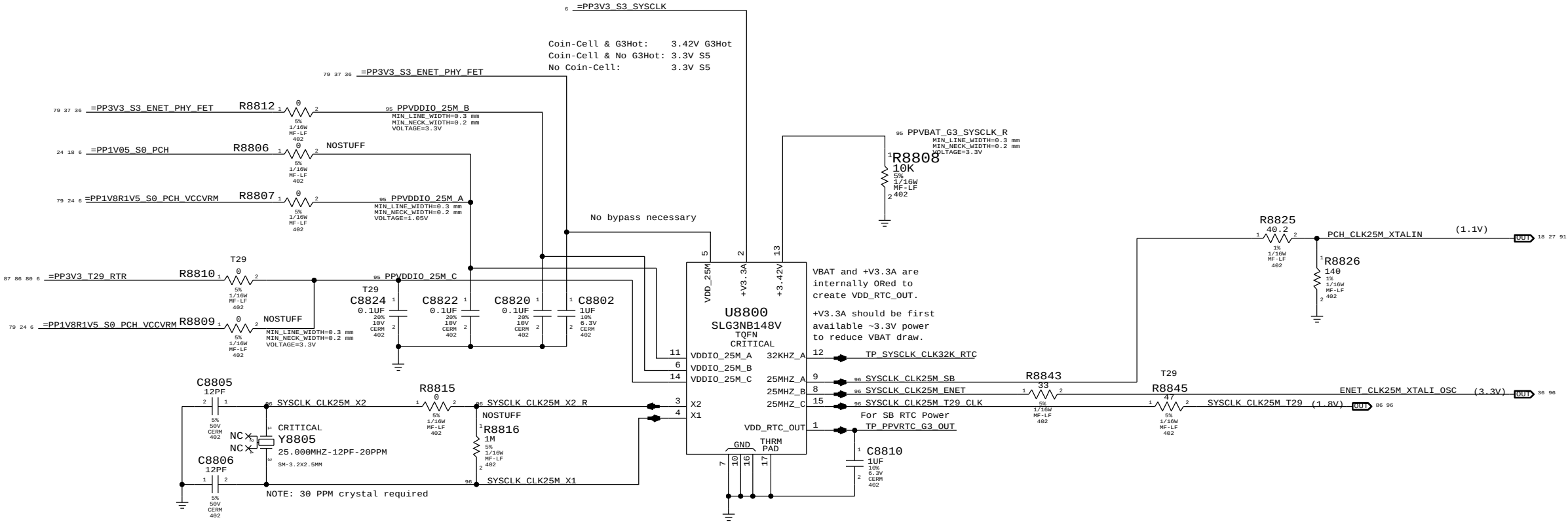
VBAT: Alias as appropriate (see note below & Desktop Example)
+V3.3A: Alias as appropriate (see note below)
VDD_25M: 3.3V matching 'highest' VDDIO power state (ENET)

VDDIO_25M_A: S8 power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.

NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.

For Cougar Point Desktop: VDDIO = VCCVRM (1.8V), Vclk = 1.1V Max, Divider: 694 / 1099
For Cougar Point Mobile: VDDIO = VCCVRM (1.5V), Vclk = 1.1V Max, Divider: 332 / 1099
For Caesar-IV (BCM57765): VDDIO = XTALVDDH (3.3V), Vclk = 3.3V Max. No Divider Necessary

System RTC Power Source & 32kHz / 25MHz Clock Generator



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
GREEN CLOCK			
 Apple Inc.		DRAWING NUMBER	051-8115
		SIZE	D
		REVISION	11.1.0
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Page Notes

Power aliases required by this page:

- =PP3V3_T29_P3V3T29FET (3.3V FET Input)
- =PP3V3_T29_FET (3.3V FET Output)
- =PP3V3_S0_T29PWRCTL
- =PP1V05_T29_P1V05T29FET (1.05V FET Input)
- =PP1V05_T29_FET (1.05V FET Output)

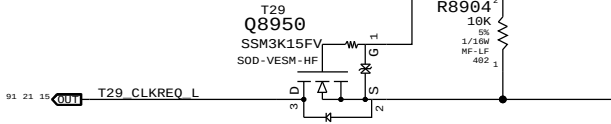
Signal aliases required by this page:

- =T29_CLKREQ_L
- T29_RESET_L

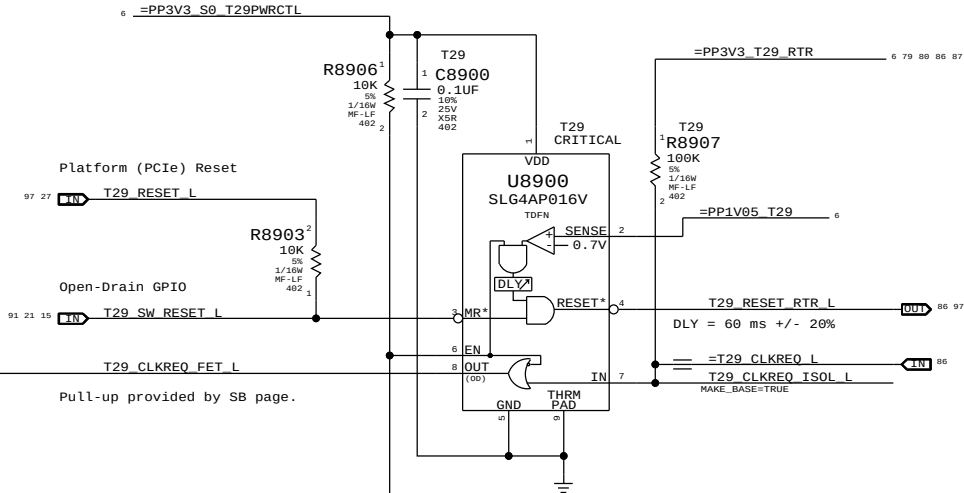
BOM options provided by this page:

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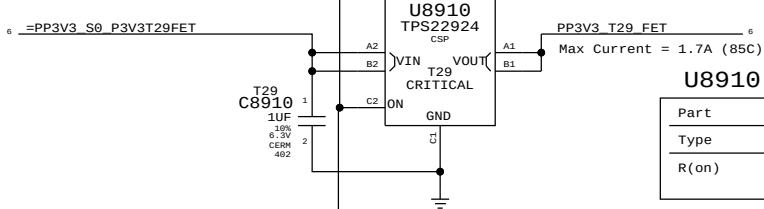
T29 CLKREQ# ISOLATION



Supervisor & CLKREQ# Isolation

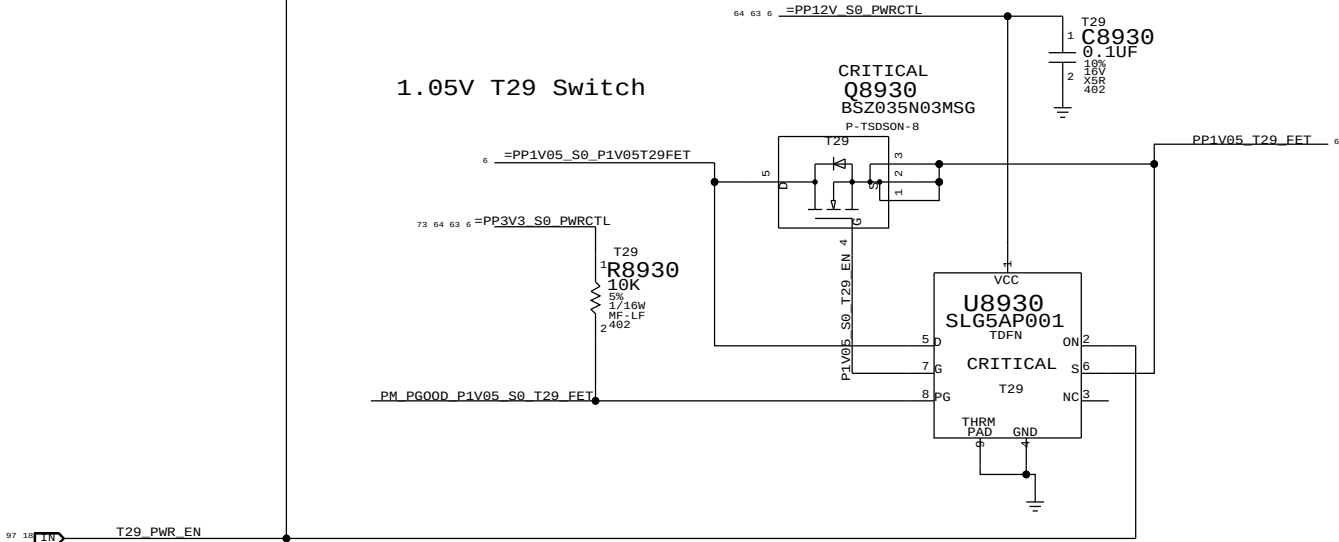


3.3V T29 Switch



Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max
Max Output: 2A	

1.05V T29 Switch



SYNC MASTER=K62		SYNC DATE=01/06/2011	
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T29 POWER			
	Apple Inc.	DRAWING NUMBER	051-8115
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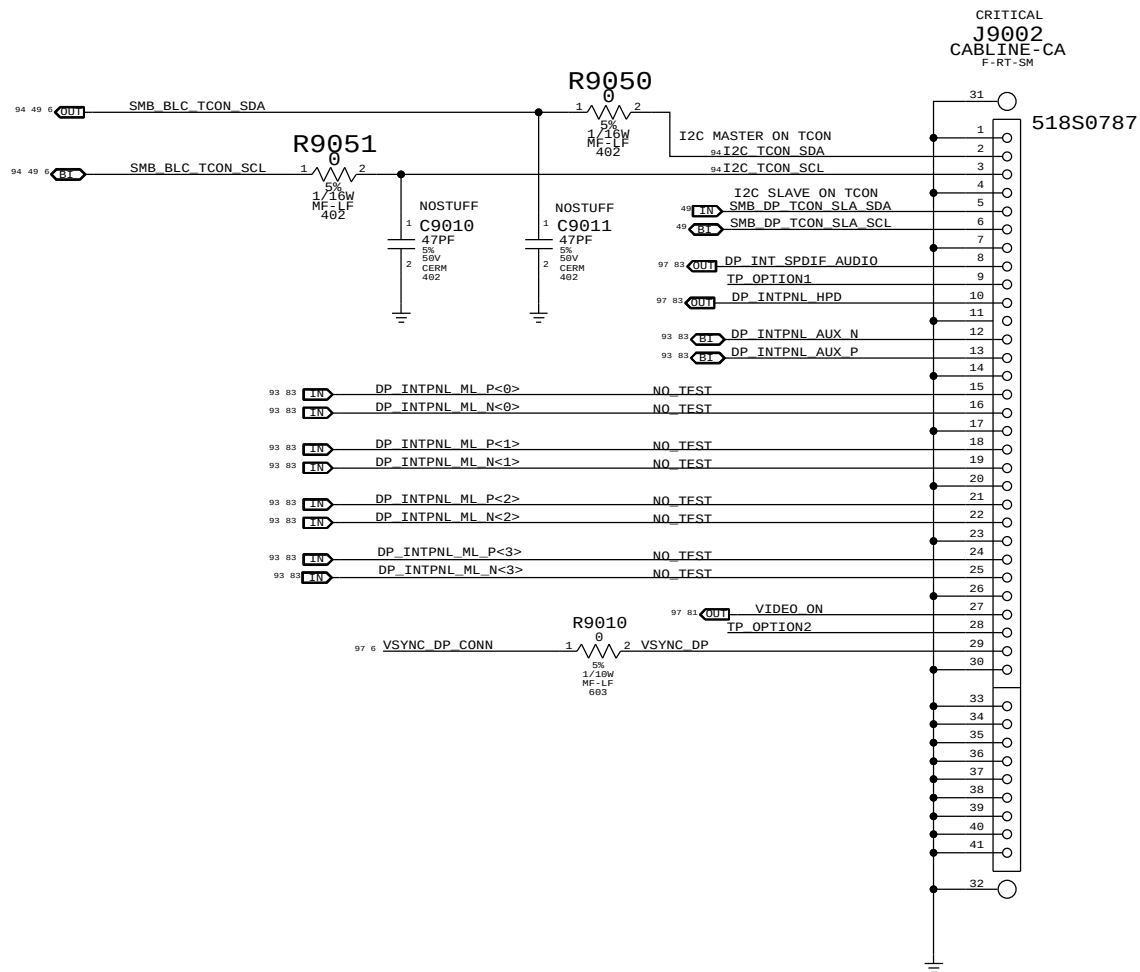
Power aliases required by this page:

- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

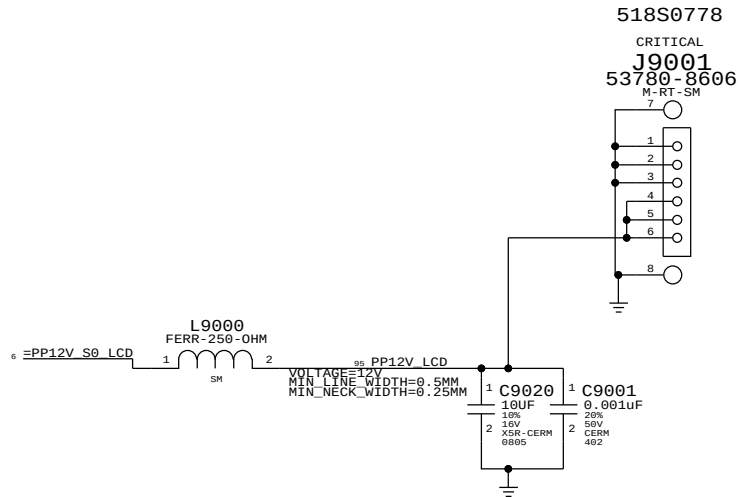
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

INTERNAL DP INTERFACE

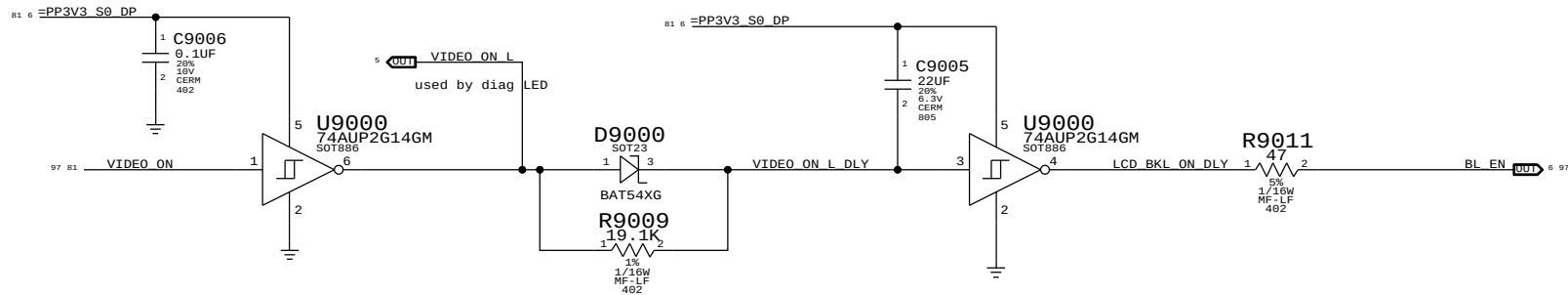


INTERNAL DP POWER



BACKLIGHT CONTROL SUPPORT

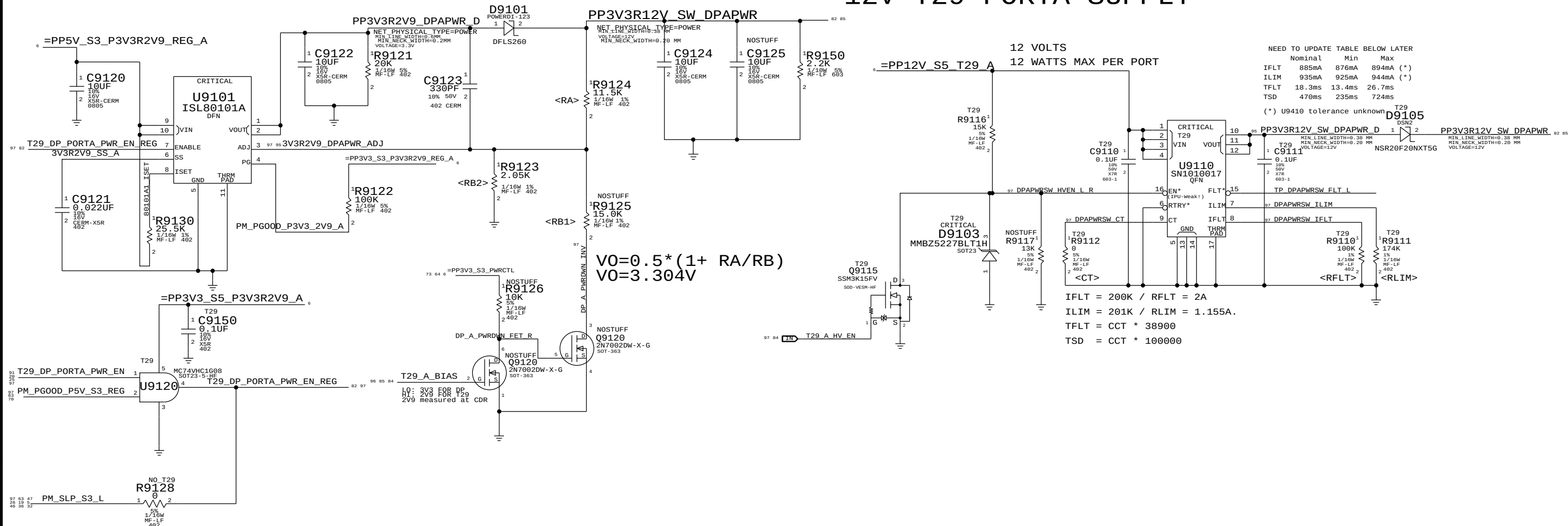
guarantee backlight is
only on when Panel has valid video

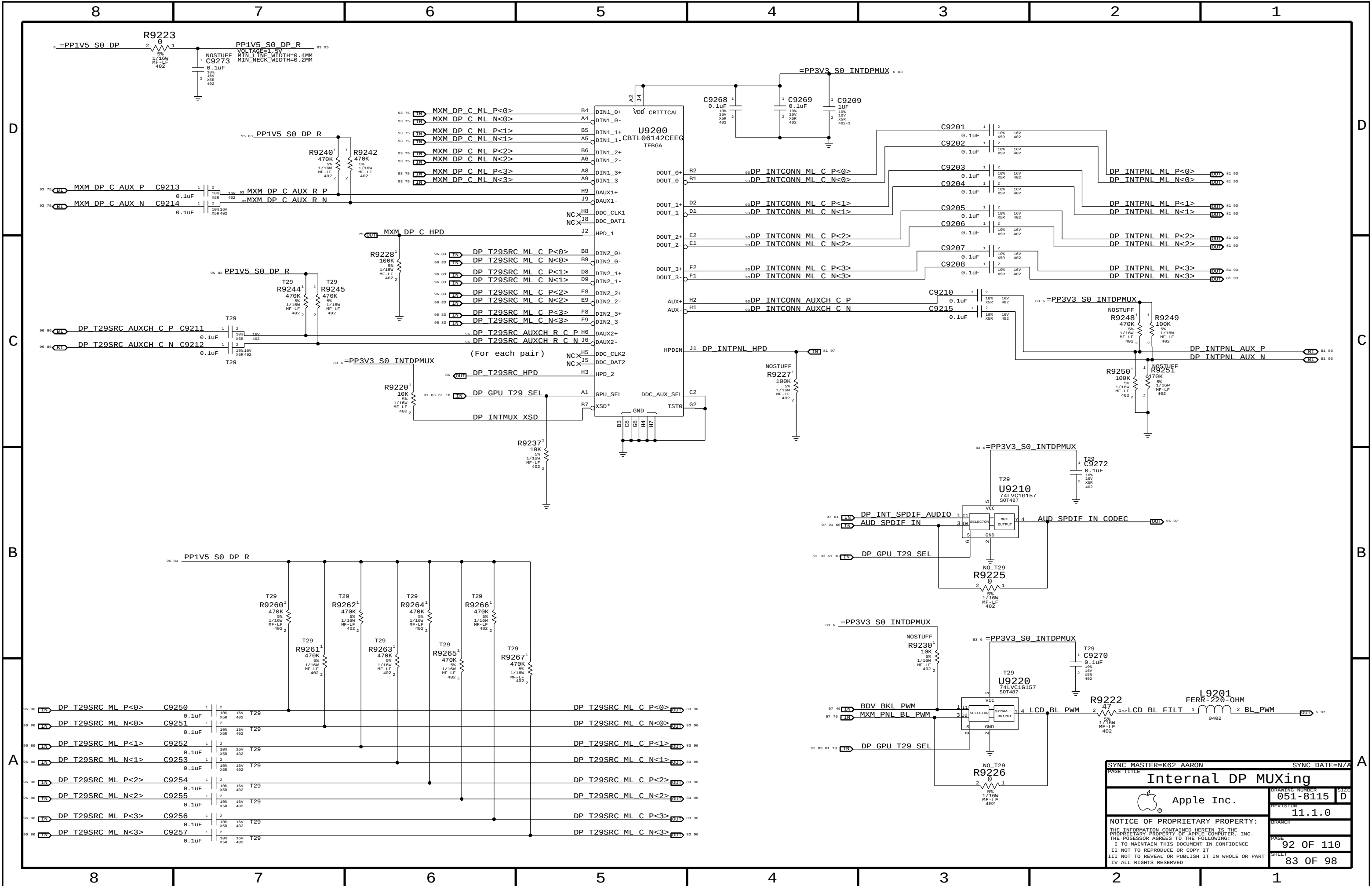


PAGE 1/111		SYNC DATE=01/06/2011	
Display: Int DP Connector		DRAWING NUMBER	051-8115
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3V3(DP)/2V9(T29) PORTA SUPPLY

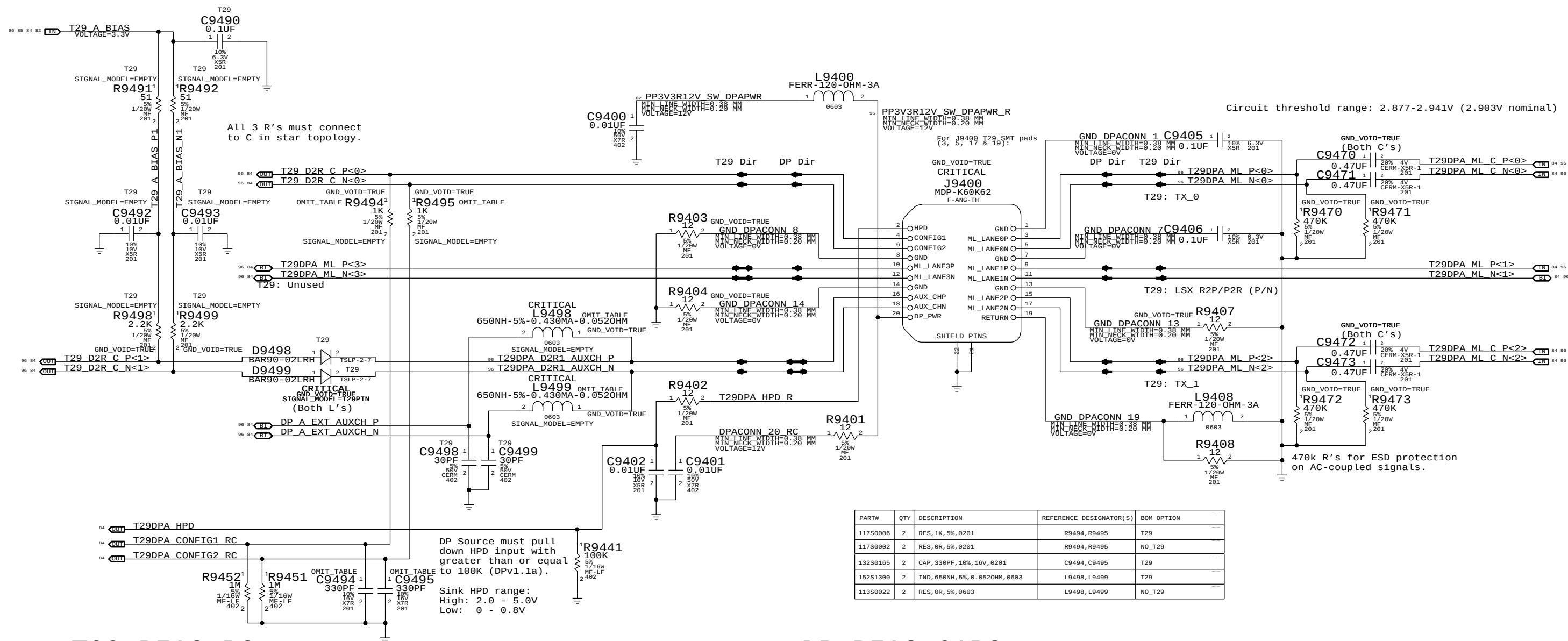
12V T29 PORTA SUPPLY



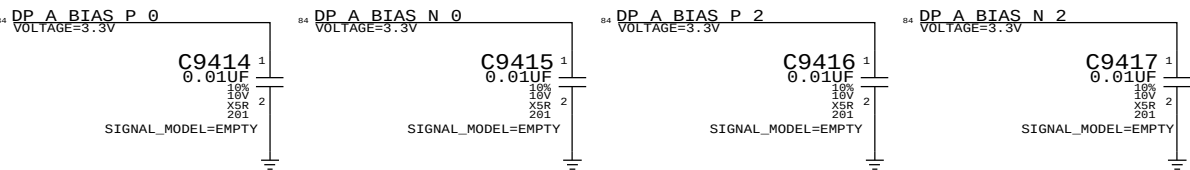


SYNC MASTER=K62 AARON		SYNC DATE=N/A	
PAGE TITLE			
Internal DP MUXing			
Apple Inc.		DRAWING NUMBER	051-8115
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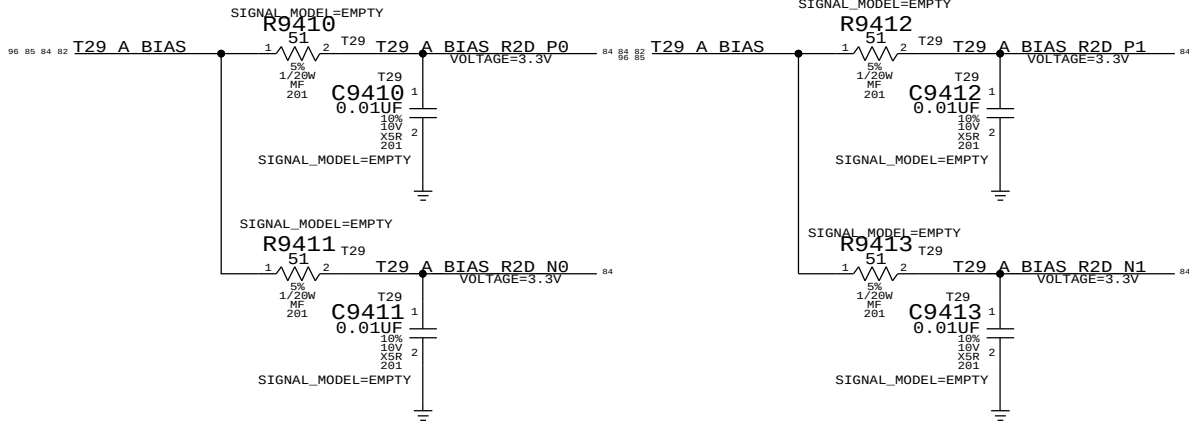
DisplayPort/T29 A Connector

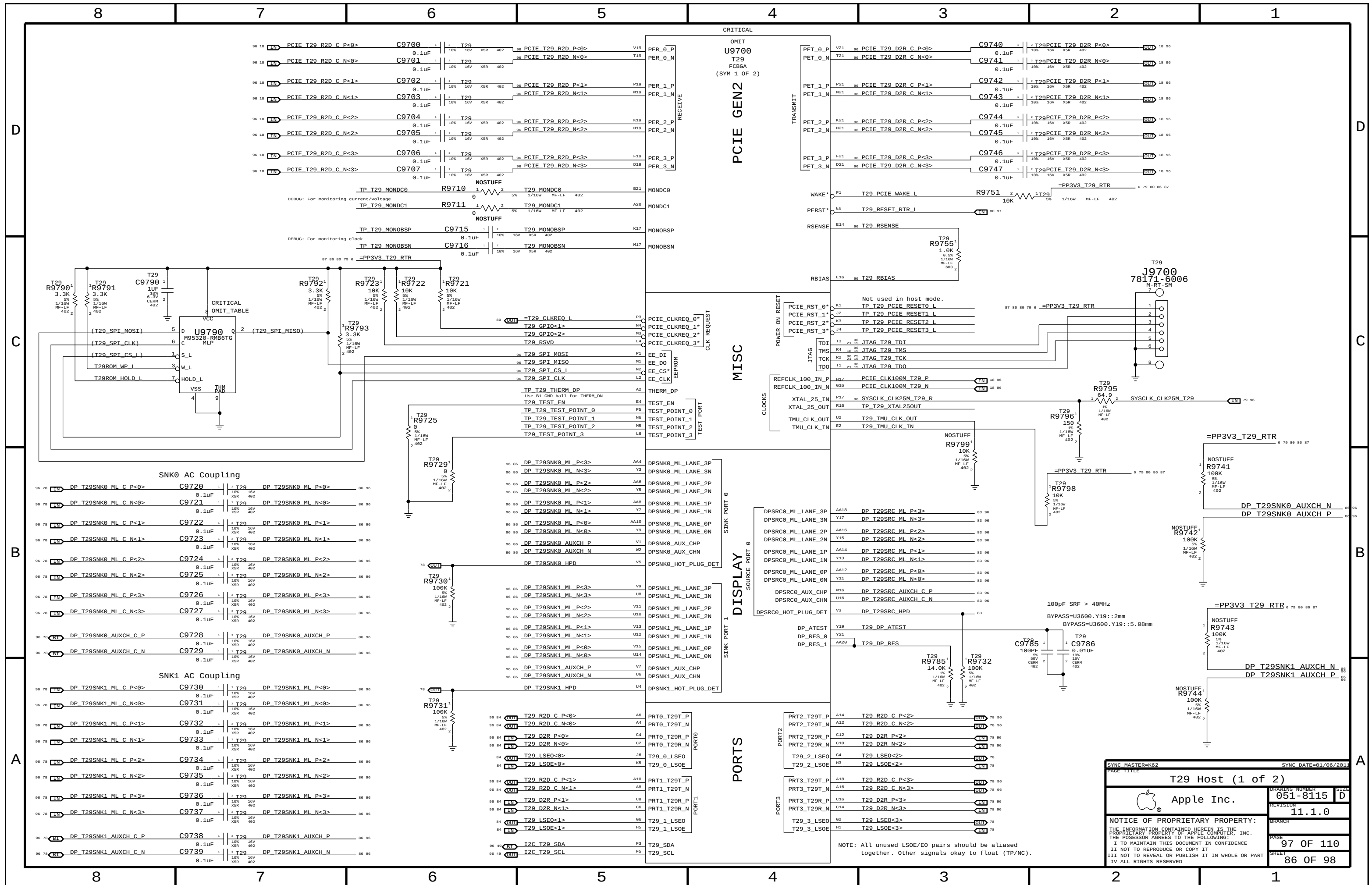


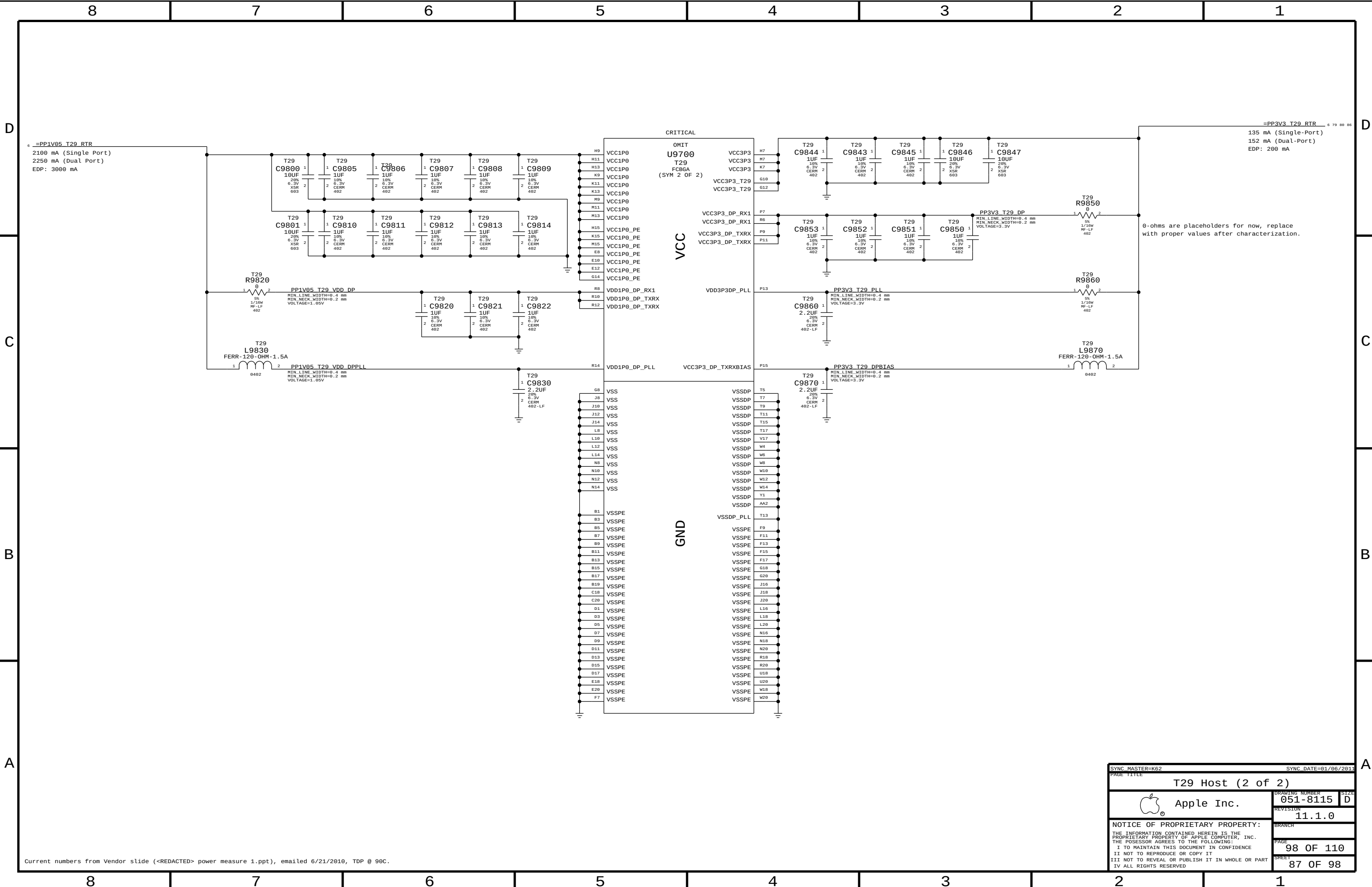
DP BIAS CAPS



T29 BIAS RC







Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
T29 Host (2 of 2)			
 Apple Inc.		DRAWING NUMBER	051-8115
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K60/62 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, BOTTOM	NO_TYPE, BGA, BGA_P1MM	MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	TOP, BOTTOM	Y	0.21 MM	0.090 MM	=STANDARD		
34_OHM_SE	*	Y	0.19 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.090 MM	=STANDARD		
39_OHM_SE	*	Y	0.159 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	TOP, BOTTOM	Y	0.151 MM	0.090 MM	=STANDARD		
42_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
68_OHM_DIFF	ISL3, ISL6	Y	0.16 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM
68_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL6	Y	0.099 MM	0.085 MM	12 MM	0.2 MM	0.1 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL6	Y	0.081 MM	0.081 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL6	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
3.5:1_SPACING	*	0.35 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PwR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P1P5MM	*	0.15 MM	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P1P5MM
CLK_PCIE	*	BGA	BGA_P1MM
CLK_LPC	*	BGA	BGA_P1MM
CLK_PCI	*	BGA	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CONTROL	*	*	SWITCHNODE
VR_CONTROL	VR_CONTROL	*	STANDARD
VR_CONTROL	SWITCHNODE	*	STANDARD
VR_CONTROL	GND	*	STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.155 MM	?
2X_DIELECTRIC	TOP, BOTTOM	0.145 MM	?
3X_DIELECTRIC	*	0.230 MM	?
3X_DIELECTRIC	TOP, BOTTOM	0.215 MM	?
4X_DIELECTRIC	*	0.305 MM	?
4X_DIELECTRIC	TOP, BOTTOM	0.285 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP, BOTTOM	0.355 MM	?
7X_DIELECTRIC	*	0.532 MM	?
7X_DIELECTRIC	TOP, BOTTOM	0.497 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

BOARD STACK-UP

TOP	HALF 0Z	SIGNAL
	0.071	PREPREG
2	TWO 0Z	GND
	0.076	PREPREG
3	ONE 0Z	SIGNAL
	0.370	PREPREG
4	TWO 0Z	POWER
	0.101	CORE
5	TWO 0Z	POWER
	0.370	PREPREG
6	ONE 0Z	SIGNAL
	0.076	PREPREG
7	TWO 0Z	GND
	0.071	PREPREG
BOTTOM	HALF 0Z	SIGNAL

BOARD THICKNESS = 62 MIL (1.5748 mm)

SYNC MASTER-K62		SYNC DATE=01/06/2011					
PAGE TITLE							
K60/K62 RULE DEFINITIONS							
 Apple Inc.		<table><tr><td>DRAWING NUMBER</td><td>SIZE</td></tr><tr><td>051-8115</td><td>D</td></tr></table>		DRAWING NUMBER	SIZE	051-8115	D
DRAWING NUMBER	SIZE						
051-8115	D						
		<table><tr><td>REVISION</td><td></td></tr><tr><td>1.1.0</td><td></td></tr></table>		REVISION		1.1.0	
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		<table><tr><td>SHEET</td><td></td></tr><tr><td>88 OF 98</td><td></td></tr></table>		SHEET		88 OF 98	
SHEET							
88 OF 98							

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
MEM_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
MEM_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
MEM_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2.5:1_SPACING	?
MEM_CTRL2MEM	*	=3:1_SPACING	?
MEM_CMD2CMD	*	=2:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DQ_SAMEBYTE	*	=3:1_SPACING	?
MEM_DQ_DIFFBYTE	*	=5:1_SPACING	?
MEM_DATA2MEM	*	=4:1_SPACING	?
MEM_DQS2MEM	*	=4:1_SPACING	?
MEM_20THER	*	=5:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE0	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE1	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE2	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE3	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE4	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE5	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE6	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE7	*	MEM_DQS2MEM
MEM_DQS	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE5	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_DQ_BYTE5	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	*	*	MEM_20OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE6	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_DQ_BYTE6	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE6	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE6	*	*	MEM_Z0THOR

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE7	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_DQ_BYTE7	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE7	*	*	MEM_20THOR

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE0	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_DQ_BYTE0	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE1	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	*	*	MEM_Z0THOR

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE1	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_DQ_BYTE1	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	*	*	MEM_20THOR

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE2	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_DQ_BYTE2	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	*	*	MEM_20THOR

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE3	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_DQ_BYTE3	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	*	*	MEM_ZOTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE4	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_DQ_BYTE4	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	*	*	MEM_20THOR

Memory Net Properties

ELECTRICAL_PAC3_SETUP		NET_TYPE		
		PHYSICAL	SPACING	
		MEM_68D	MEM_CLK	MEM A CLK P<3..0>
		MEM_68D	MEM_CLK	MEM A CLK N<3..0>
		MEM_39S	MEM_CTRL	MEM A CKE<3..0>
		MEM_39S	MEM_CTRL	MEM A CS L<3..0>
		MEM_39S	MEM_CTRL	MEM A ODT<3..0>
		MEM_34S	MEM_CMD	MEM A A<15..0>
		MEM_34S	MEM_CMD	MEM A BA<2..0>
		MEM_34S	MEM_CMD	MEM A RAS L
		MEM_34S	MEM_CMD	MEM A CAS L
		MEM_34S	MEM_CMD	MEM A WE L
		MEM_42S	MEM_DQ_BYTE0	MEM A DQ<7..0>
		MEM_42S	MEM_DQ_BYTE1	MEM A DQ<15..8>
		MEM_42S	MEM_DQ_BYTE2	MEM A DQ<23..16>
		MEM_42S	MEM_DQ_BYTE3	MEM A DQ<31..24>
		MEM_42S	MEM_DQ_BYTE4	MEM A DQ<39..32>
		MEM_42S	MEM_DQ_BYTE5	MEM A DQ<47..40>
		MEM_42S	MEM_DQ_BYTE6	MEM A DQ<55..48>
		MEM_42S	MEM_DQ_BYTE7	MEM A DQ<63..56>
		MEM_42S_D	MEM_DQS	MEM A DQS P<0>
		MEM_42S_D	MEM_DQS	MEM A DQS N<0>
		MEM_42S_D	MEM_DQS	MEM A DQS P<1>
		MEM_42S_D	MEM_DQS	MEM A DQS N<1>
		MEM_42S_D	MEM_DQS	MEM A DQS P<2>
		MEM_42S_D	MEM_DQS	MEM A DQS N<2>
		MEM_42S_D	MEM_DQS	MEM A DQS P<3>
		MEM_42S_D	MEM_DQS	MEM A DQS N<3>
		MEM_42S_D	MEM_DQS	MEM A DQS P<4>
		MEM_42S_D	MEM_DQS	MEM A DQS N<4>
		MEM_42S_D	MEM_DQS	MEM A DQS P<5>
		MEM_42S_D	MEM_DQS	MEM A DQS N<5>
		MEM_42S_D	MEM_DQS	MEM A DQS P<6>
		MEM_42S_D	MEM_DQS	MEM A DQS N<6>
		MEM_42S_D	MEM_DQS	MEM A DQS P<7>
		MEM_42S_D	MEM_DQS	MEM A DQS N<7>
LEV0		MEM_50S	PM	MEM RESET L

MEMORY MISC PROPERTIES

		NET_TYPE		
VOLTAGE	PHYSICAL	SPACING	VOLTAGE	
	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_A	
	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_B	
MEM0P	MEM_POWER_PHY	MEM_POWER	CPU_DDR_VREF	11 28
MEM0D	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMA_DACOUT	28
MEM0B	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMA_OPFB	28
MEM0Q	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMA_DQ	28
MEM0Z	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_A_SW	
MEM1A	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMB_DACOUT	28
MEM1B	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMB_OPFB	28
MEM1Q	MEM_POWER_PHY	MEM_POWER	VREFMARGIN_DIMMB_DQ	28
MEM1Z	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_B_SW	

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_POWER_WIDTH	*	Y	0.500 MM	0.250 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_POWER_PHY	*	MEM_POWER_WIDTH	MEM_POWER	*	=3:1_SPACING	?

		NET_TYPE		
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
	MEM_68D	MEM_CLK	MEM B CLK P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM B CLK N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM B CKE<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM B CS L<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM B ODT<3..0>	12 31
	MEM_34S	MEM_CMD	MEM B A<15..0>	12 31
	MEM_34S	MEM_CMD	MEM B BA<2..0>	12 31
	MEM_34S	MEM_CMD	MEM B RAS_L	12 31
	MEM_34S	MEM_CMD	MEM B CAS_L	12 31
	MEM_34S	MEM_CMD	MEM B WE_L	12 31
	MEM_42S	MEM_DQ_BYTE0	MEM B DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM B DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM B DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM B DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM B DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTES	MEM B DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM B DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM B DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS_N<7>	12 32

SYNCH MASTER=K60 ROSITA		SYNCH DATE=01/06/2011	
PAGE TITLE			
Memory Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-8115	D
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PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	0.2 MM	?
ITP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SMB	*	=2X_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
HDA	*	=2X_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

NET_TYPE			
PHYSICAL	SPACING		
18320	PM	T29_CLKREQ_L	15 21 80
18320	PM	FW_MINI_CLKREQ_L	15 18
18320	PM	BLC_GPIO	6 15 21
18320	PM	T29_SW_RESET_L	15 21 80
18320	PM	ENET_CLKREQ_L	15 18 36
18320	PM	DP_GPU_T29_SEL	18 61 83
18320	PM	T29_MCU_INT_L	20 84
18320	PM	T29_DP_PORTA_PWR_EN	20 25 82 97
18320	PM	T29_DP_PORTB_PWR_EN	20 25
18320	PM	DP_AUXCH_ISOL	15 18 25 84
18320	PM	PLT_RST_BUF_L	27
18320	PM	XDPCPU_PLTRST_L	25 27
18320	PM	PCH_PEG_CLKREQ_L	21
18320	PM	ENET_SW_RESET_L	15 21 36
18320	PM	CPU_SKTOCC	63
18320	PM	PM_EN_USB_PWR	43 63

NET_TYPE				
PHYSICAL	SPACING			
18320	PM	PCI_REQ0_L	20	
18320	PM	PCI_REQ1_L	20	
18320	PM	PCI_REQ2_L	20	
18320	PM	PCI_REQ3_L	20	
18320	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIEOUT	20 27
18320	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIEIN	18 27
18320	LPC_55S	LPC	LPC_AD<3..0>	18 46 48
18320	LPC_55S	LPC	LPC_FRAME_L	18 46 48
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	20 27
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	27 46
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	27 48
18320	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK_R	9 19 97
18320	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK	9 46 97
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 27
18320	LPC_55S	LPC	LPC_R_AD<3..0>	18
18320	LPC_55S	LPC	LPC_FRAME_R_L	18
18320	SPI_55S	SPI	SPI_CLK_1_R	18
18320	SPI_55S	SPI	SPI_MOSI_1_R	18
18320	CLK_XTAL	XTAL	USB_HUB2_XTAL1	35
18320	CLK_XTAL	XTAL	USB_HUB2_XTAL2	35
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1_R	27
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	27
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27 91
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27 91
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27 91
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27 91
18320	CLK_XTAL	XTAL	CK505_XTAL_IN	26
18320	CLK_XTAL	XTAL	CK505_XTAL_OUT	26
18320	CLK_XTAL	XTAL	CK505_XTAL_OUT_R	26
18320	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK	18 26

NET_TYPE			
PHYSICAL	SPACING		
18320	PM	ENET_RESET_LOGIC_L	36
18320	PM	ENET_RESET_FET_L	
18320	PM	ENET_CLKREQ_FET_L	36 37
18320	PM	PGOOD_5V_1V05_3V3	64 97
18320	PM	PGOOD_CPU_UNCORE	64 97
18320	PM	ALL_SYS_PWRGD	64 97
18320	PM	PGOOD_3V3_1V05	64 97
18320	PM	PGOOD_PCH_S0_R	64 97
18320	PM	AUD_IPHS_SWITCH_EN_PCH	21 25

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
[18320]	SPT_55S	SPT	SPI CLK R	18 48 55
	SPT_55S	SPT	SPI CLK	55
	SPT_55S	SPT	SPI MOSI R	18 48 55
	SPT_55S	SPT	SPI MOSI	55
	SPT_55S	SPT	SPI MISO	18 48 55
	SPT_55S	SPT	SPI MISO R	55
	SPT_55S	SPT	SPI CS0 R L	18 48
	SPT_55S	SPT	SPI CS0 L	48
	SPT_55S	SPT	SPI MLB CS L	48 55
	SPT_55S	SPT	SPI ALT CS L	48
	SPT_55S	SPT	SPIROM USE MLB	21 48
	SPT_55S	SPT	SPI ALT MOSI	48
	SPT_55S	SPT	SPI ALT MISO	48
	SPT_55S	SPT	SPI ALT CLK	48
	HDA_55S	HDA	HDA BIT CLK	18 56
	[18320]	HDA_55S	HDA	HDA BIT CLK R
HDA_55S		HDA	HDA RST L	18 56
HDA_55S		HDA	HDA RST R L	18
HDA_55S		HDA	HDA SDOOUT	15 18 56
HDA_55S		HDA	HDA SDOOUT R	18
HDA_55S		HDA	HDA SYNC	18 56
HDA_55S		HDA	HDA SYNC R	18
HDA_55S		HDA	HDA SDIN0	18 56
HDA_55S		HDA	AUD SDI R	56
		PM	AUD SPDIF IN	60 83 97
		HDA	AUD SPDIF OUT	56 60
		HDA	AUD SPDIF CHIP	56
HDA_55S		HDA	AUD SPKR OUTL01L NOUT	98
HDA_55S		HDA	AUD SPKR OUTL01L POUT	98
HDA_55S		HDA	AUD SPKR OUTL01R NOUT	98
HDA_55S		HDA	AUD SPKR OUTL01R POUT	98
[18320]	HDA_55S	HDA	AUD SPKR OUTL02L NOUT	98
	HDA_55S	HDA	AUD SPKR OUTL02L POUT	98
	HDA_55S	HDA	AUD SPKR OUTL02R NOUT	98
	HDA_55S	HDA	AUD SPKR OUTL02R POUT	98
	CLK_XTAL	XTAL	PCH CLK25M XTALOUT R	27
	CLK_XTAL	XTAL	PCH CLK25M XTALIN R	27
	CLK_XTAL	XTAL	PCH CLK25M XTALOUT	18 27
	CLK_XTAL	XTAL	PCH CLK25M XTALIN	18 27 70
	PCH_55S	COMP_PCH	PCH USB_RBIAS	20
	PCH_55S	COMP_PCH	PCH SATA3COMP	18
	PCH_55S	COMP_PCH	PCH XCLK_RCOMP	18
	PCH_55S	COMP_PCH	PCH DMI_COMP	19
	PCH_55S	COMP_PCH	PCH SATA1COMP	18
	CLK_XTAL	XTAL	USB HUB1_XTAL1	34
	CLK_XTAL	XTAL	USB HUB1_XTAL2	34
	PCH_55S	COMP_PCH	USB HUB1_RBIAS	34
[18320]	PCH_55S	ITP_PCH	XDP PCH TCK	18 25
	PCH_55S	ITP_PCH	XDP PCH TMS	18 25
	PCH_55S	ITP_PCH	XDP PCH TDI	18 25
	PCH_55S	ITP_PCH	XDP PCH TDO	18 25
	PCH_55S	COMP_PCH	PCH DMI2RBIAS	19
	PCH_55S	COMP_PCH	PCH SATA3RBIAS	18
	PCH_55S	COMP_PCH	USB HUB2_RBIAS	35

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SYNC_DATE=01/06/2011

IBEX PEAK CONSTRAINTS

Apple Inc.

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GRAPHICS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_850	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_rule_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

USE 5X_DIELECTRIC IN K62

PAIRS SHOULD BE WITHIN 100 MILS OF CLOCK LENGTH.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.

ELECTRICAL_CONSTRAINT_SET
ASSIGNED IN CONT. MGR. NET_TYPE

PHYSICAL		SPACING			
DP_85D	DISPLAYPORT	DP_INTCNN ML C P<3..0>			83
DP_85D	DISPLAYPORT	DP_INTCNN ML C N<3..0>			83
DP_85D	DISPLAYPORT	DP_INTCNN AUXCH C P			83
DP_85D	DISPLAYPORT	DP_INTCNN AUXCH C N			83
DP_85D	DISPLAYPORT	DP_INTPNL ML P<3..0>			81
DP_85D	DISPLAYPORT	DP_INTPNL ML N<3..0>			81
DP_85D	DISPLAYPORT	DP_INTPNL AUX P			81
DP_85D	DISPLAYPORT	DP_INTPNL AUX N			81
DP_85D	DISPLAYPORT	DP_EXTM ML P<3..0>			84
DP_85D	DISPLAYPORT	DP_EXTM ML N<3..0>			84
DP_85D	DISPLAYPORT	DP_EXTM AUXCH N			84
DP_85D	DISPLAYPORT	DP_EXTM ML C P<3..0>			78
DP_85D	DISPLAYPORT	DP_EXTM ML C N<3..0>			78
DP_85D	DISPLAYPORT	DP_EXTM AUXCH C P			78
DP_85D	DISPLAYPORT	DP_EXTM AUXCH C N			78
DP_85D	DISPLAYPORT	DP_EXTB ML P<3..0>			75
DP_85D	DISPLAYPORT	DP_EXTB ML N<3..0>			75
DP_85D	DISPLAYPORT	DP_EXTB AUXCH P			75
DP_85D	DISPLAYPORT	DP_EXTB AUXCH N			75
DP_85D	DISPLAYPORT	DP_EXTB ML C P<3..0>			75
DP_85D	DISPLAYPORT	DP_EXTB ML C N<3..0>			75
DP_85D	DISPLAYPORT	DP_EXTB AUXCH C P			75
DP_85D	DISPLAYPORT	DP_EXTB AUXCH C N			75
DP_85D	DISPLAYPORT	MXM DP B ML P<3..0>			75
DP_85D	DISPLAYPORT	MXM DP B ML N<3..0>			75
DP_85D	DISPLAYPORT	MXM DP B AUX P			75
DP_85D	DISPLAYPORT	MXM DP B AUX N			75
DP_85D	DISPLAYPORT	MXM DP C ML P<3..0>			75
DP_85D	DISPLAYPORT	MXM DP C ML N<3..0>			75
DP_85D	DISPLAYPORT	MXM DP C AUX P			75
DP_85D	DISPLAYPORT	MXM DP C AUX N			75
DP_85D	DISPLAYPORT	MXM DP C AUX R P			83
DP_85D	DISPLAYPORT	MXM DP C AUX R N			83
DP_85D	DISPLAYPORT	MXM DP D ML P<3..0>			75
DP_85D	DISPLAYPORT	MXM DP D ML N<3..0>			75
DP_85D	DISPLAYPORT	MXM DP D AUX P			75
DP_85D	DISPLAYPORT	MXM DP D AUX N			75

UNUSED VIDEO NET PHYSICAL CONSTRAINTS

DP_85D	DP_85D	DISPLAYPORT	MXM LVDS A CLK P	76	78
	DP_85D	DISPLAYPORT	MXM LVDS A CLK N	76	78
DP_85D	DP_85D	DISPLAYPORT	MXM LVDS B CLK P	76	78
	DP_85D	DISPLAYPORT	MXM LVDS B CLK N	76	78
DP_85D	DP_85D	DISPLAYPORT	MXM LVDS A DATA P<3..0>	76	78
	DP_85D	DISPLAYPORT	MXM LVDS A DATA N<3..0>	76	78
DP_85D	DP_85D	DISPLAYPORT	MXM LVDS B DATA P<3..0>	76	78
	DP_85D	DISPLAYPORT	MXM LVDS B DATA N<3..0>	76	78

D

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SMC Constraints			
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ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
	THERM_DIFF	THERMAL	SNS I MXM P	50
	THERM_DIFF	THERMAL	SNS I MXM N	50
	THERM_DIFF	THERMAL	SNS DIMM 1V5 P	50
	THERM_DIFF	THERMAL	SNS DIMM 1V5 N	50
	SNS_DIFF	THERMAL	VR ISNS VCORE P	50
	SNS_DIFF	THERMAL	VR ISNS VCORE N	50
	SNS_DIFF	THERMAL	VR ISNS VAXG_P	50
	SNS_DIFF	THERMAL	VR ISNS VAXG_N	50
	SNS_DIFF	THERMAL	VR ISNS 1V05_P	95
	SNS_DIFF	THERMAL	VR ISNS 1V05_N	95
	THERM_DIFF	THERMAL	SNS CPU 1V5_P	50
	THERM_DIFF	THERMAL	SNS CPU 1V5_N	50
	THERM_DIFF	THERMAL	SNS VCCSA_P	50
	THERM_DIFF	THERMAL	SNS VCCSA_N	50
	THERM_DIFF	THERMAL	SNS 1V05_PCH_P	50
	THERM_DIFF	THERMAL	SNS 1V05_PCH_N	50
		THERMAL	GND_SMC_AVSS	46 47
		THERMAL	SMC_CPU_1V5_ISENSE	46 50
		THERMAL	SMC_CPU_1V5_ISENSE_R	50
		THERMAL	SMC_CPU_1V5_VSENSE	46 50
		THERMAL	GND_SMC_AVSS	46 47
		THERMAL	SMC_DIMM_ISENSE	46 50
		THERMAL	SMC_DIMM_1V5_R	50
		THERMAL	SMC_DIMM_VSENSE	46 50
		THERMAL	GND_SMC_AVSS	46 47
		THERMAL	SMC_VCCSA_ISENSE	46 50
		THERMAL	SMC_VCCSA_ISENSE_R	50
		THERMAL	SMC_VCCSA_VSENSE	46 50
		THERMAL	GND_SMC_AVSS	46 47
		THERMAL	SMC_PCH_1V05_ISENSE	46 50
		THERMAL	SMC_VAXG_VSENSE	46 50
		THERMAL	SMC_PCH_1V05_VSENSE	46 50
		THERMAL	GND_SMC_AVSS	46 47
		THERMAL	SMC_1V05_ISENSE	46 50
		THERMAL	SMC_VAXG_ISENSE	46 50
		THERMAL	SMC_1V05_VSENSE	46 50
		THERMAL	SMC_GPU_ISENSE	46 50
		THERMAL	SMC_GPU_VSENSE	46 50
		THERMAL	SMC_VCORE_ISENSE	46 50
		THERMAL	SMC_VCORE_VSENSE	46 50
		THERMAL	SMC_CPU_VSENSE	

ELECTRICAL CONSTRAINT SET	NET_TYPE		
	PHYSICAL	SPACING	
	THERM_DIEF	THERMAL	SNS T1 1 P 52
	THERM_DIEF	THERMAL	SNS T1 1 N 52
	THERM_DIEF	THERMAL	SNS T2 DP2 52
	THERM_DIEF	THERMAL	SNS T2 DN2 52
1P200	THERM_DIEF	THERMAL	SNS T1 2 P 52
1P200	THERM_DIEF	THERMAL	SNS T1 2 N 52
1P200	THERM_DIEF	THERMAL	SNS T1 3 P 52
1P200	THERM_DIEF	THERMAL	SNS T1 3 N 52
1P200	THERM_DIEF	THERMAL	SNS T1 4 P 52
1P200	THERM_DIEF	THERMAL	SNS T1 4 N 52
1P200	THERM_DIEF	THERMAL	SNS T1 5 P 52
1P200	THERM_DIEF	THERMAL	SNS T1 5 N 52
1P200	THERM_DIEF	THERMAL	SNS T1 6 P 52
1P200	THERM_DIEF	THERMAL	SNS T1 6 N 52
1P200	THERM_DIEF	THERMAL	SNS T1 7 P 52
1P200	THERM_DIEF	THERMAL	SNS T1 7 N 52
1P200	THERM_DIEF	THERMAL	SNS CPU THERMD P 10 52
1P200	THERM_DIEF	THERMAL	SNS CPU THERMD N 10 52
1P200	THERM_DIEF	THERMAL	SNS LCD H P 52
1P200	THERM_DIEF	THERMAL	SNS LCD H N 52
	THERM_DIEF	THERMAL	SNS_ODD_P 52 98
	THERM_DIEF	THERMAL	SNS_ODD_N 52 98
	THERM_DIEF	THERMAL	SNS_CPU_H_P 52
	THERM_DIEF	THERMAL	SNS_CPU_H_N 52
1P200	THERM_DIEF	THERMAL	SNS_SKIN_RIGHT_P 52 98
1P200	THERM_DIEF	THERMAL	SNS_SKIN_RIGHT_N 52 98
1P200	THERM_DIEF	THERMAL	SNS_SKIN_LEFT_P 44 52 98
1P200	THERM_DIEF	THERMAL	SNS_SKIN_LEFT_N 44 52 98
	THERM_DIEF	THERMAL	SNS_AMB_P 52 54 98
	THERM_DIEF	THERMAL	SNS_AMB_N 52 54 98
	THERM_DIEF	THERMAL	SNS_MXM_P 52
	THERM_DIEF	THERMAL	SNS_MXM_N 52
		THERMAL	HDD_O0B_TEMP_FILT 42 51 98
		THERMAL	HDD_O0B_TEMP_FB 42
		THERMAL	HDD_O0B_TEMP_R 51
		THERMAL	SMC_HDD_O0B_TEMP 46 51

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29	*	=5X_DIELECTRIC	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	0.2 MM	?

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_XTAL	*	=4X_DIELECTRIC	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SMB	*	=2x_DIELECTRIC	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_COMP	*	0.2 MM	?

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	NO_TEST=TRUE	T29_960	T29	T29 R2D C P<3..0> 78 84 85
	NO_TEST=TRUE	T29_960	T29	T29 R2D C N<3..0> 78 84 85
	NO_TEST=TRUE	T29_960	T29	T29 D2R C P<3..0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29 D2R C N<3..0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29 R2D P<3..0> 84
	NO_TEST=TRUE	T29_960	T29	T29 R2D N<3..0> 84
	NO_TEST=TRUE	T29_960	T29	T29 D2R P<3..0> 78 84 85
	NO_TEST=TRUE	T29_960	T29	T29 D2R N<3..0> 78 84 85
	NO_TEST=TRUE	T29_960	T29	T29 R2D C F P<3..0> 84
	NO_TEST=TRUE	T29_960	T29	T29 R2D C F N<3..0> 84
	NO_TEST=TRUE	T29_960	T29	T29DPA ML P<3..0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPA ML N<3..0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPB ML P<3..0> 84
	NO_TEST=TRUE	T29_960	T29	T29DPB ML N<3..0> 84
	NO_TEST=TRUE	T29_960	T29	DP A EXT AUXCH P 84 85
	NO_TEST=TRUE	T29_960	T29	DP A EXT AUXCH N 84 85
	NO_TEST=TRUE	T29_960	T29	DP SDRVA AUXCH C P 84
	NO_TEST=TRUE	T29_960	T29	DP SDRVA AUXCH C N 84
	NO_TEST=TRUE	T29_960	T29	DP SDRVB AUXCH C N
	NO_TEST=TRUE	T29_960	T29	DP B EXT AUXCH P
	NO_TEST=TRUE	T29_960	T29	DP B EXT AUXCH N
	NO_TEST=TRUE	T29_960	T29	T29DPA D2R1 AUXCH P 85
	NO_TEST=TRUE	T29_960	T29	T29DPA D2R1 AUXCH N 85
	NO_TEST=TRUE	T29_960	T29	T29DPB D2R3 AUXCH P
	NO_TEST=TRUE	T29_960	T29	T29DPB D2R3 AUXCH N
	NO_TEST=TRUE	T29_960	T29	T29DPA ML C N<0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPA ML C P<0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPA ML C N<2> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPA ML C P<2> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPB ML C N<0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPB ML C P<0> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPB ML C N<2> 84 85
	NO_TEST=TRUE	T29_960	T29	T29DPB ML C P<2> 84 85

NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_P<3..0>	86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_N<3..0>	86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_C_P<3..0>	86 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_C_N<3..0>	86 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_P<3..0>	86 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_N<3..0>	86 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_C_P<3..0>	86 86
NO_TEST=TRUE	CLK_PCIE_98D	CLK_PCIE	PCIE_CLK100M_T29_P	86 86
NO_TEST=TRUE	CLK_PCIE_98D	CLK_PCIE	PCIE_CLK100M_T29_N	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_C_P<3..0>	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_C_N<3..0>	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_P<3..0>	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_N<3..0>	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_P<3..0>	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_N<3..0>	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_P	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_N	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_C_P<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_C_N<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_P	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_N	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_P<3..0>	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_N<3..0>	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_P	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_N	86 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_C_P<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_C_N<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_C	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_N	78 86

NO_TEST=TRUE	DP_850	DISPLAYPORT	DP T29SRC AUXCH R C P	83
NO_TEST=TRUE	DP_850	DISPLAYPORT	DP T29SRC AUXCH R C N	83
NO_TEST=TRUE	DP_850	DISPLAYPORT	DP T29SRC AUXCH C P	83 86
NO_TEST=TRUE	DP_850	DISPLAYPORT	DP T29SRC AUXCH C N	83 86

NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_AUXCH P	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_AUXCH N	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_AUXCH P	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_AUXCH N	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_ML C P<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_ML C N<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_ML P<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_ML R<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_ML R P<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVA_ML R C P<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_ML C N<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_ML P<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_ML N<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_ML R P<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	DP_SDRVB_ML R N<3...0>	04
NO TEST=TRUE	T29_980	DISPLAYPORT	T29_A_RSVD N	04
NO TEST=TRUE	T29_980	DISPLAYPORT	T29_B_RSVD N	04
NO TEST=TRUE	T29_980	DISPLAYPORT	T29_B_RSVD P	04

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
		T29_SPT_55S	T29_SPT	JTAG T29 TDI	15 21 86
		T29_SPT_55S	T29_SPT	JTAG T29 TMS	15 18 86
		T29_SPT_55S	T29_SPT	JTAG T29 TCK	15 21 25
		T29_SPT_55S	T29_SPT	JTAG T29 TDO	86 15 21 86
IE89		T29_SPT_55S	T29_SPT	T29_SPT_MOSI	86
IE90		T29_SPT_55S	T29_SPT	T29_SPT_MISO	86
IE90		T29_SPT_55S	T29_SPT	T29_SPT_CS_L	86
IE90		T29_SPT_55S	T29_SPT	T29_SPT_CLK	86
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	79 86 96
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	86 96
		T29_SMB_55S	T29_SMB	I2C_T29_SDA	49 86
		T29_SMB_55S	T29_SMB	I2C_T29_SCL	49 86
IE91		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB	79
IE99		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET	79
IE99		CLK_25M_55S	CLK_25M	ENET_CLK25M_XTALI_OSC	36 79
IE99		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_CLK	79
IE99		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	79 86 96
IE99		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	86 96
IE99		T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X2	79
IE99		T29_VTAL_100D	T29_VTAL	SYSCLK_CLK25M_X2_R	79
IE99		T29_VTAL_100D	T29_VTAL	SYSCLK_CLK25M_X1	79
IE99		T29_55S	T29_COMP	T29_RSENSE	86
IE99		T29_55S	T29_COMP	T29_RBIAS	86
IE99		T29_COMP	T29_COMP	T29_A_BIAS	82 84 85
IE99		T29_COMP	T29_COMP	T29_B_BIAS	84
IE99		T29_COMP	T29_COMP	DP_A_BIAS	84

